Sequential Contests and the Later-/Earlier-Mover Advantage*

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Abstract

This paper provides a general study of a sequential contest modeled as a multi-player incomplete-information contest that allows for a rich set of timing architecture. The contest proceeds in multiple periods. Players arrive and exert efforts to compete for a prize. They observe the efforts made by their earlier opponents, but not those of their contemporaneous and future rivals. We establish the existence and uniqueness of a symmetric Perfect Bayesian equilibrium (PBE) and fully characterize the equilibrium. The equilibrium results further allows us to compare players' ex ante expected payoffs with respect to their timing positions. We demonstrate that a player always secures a larger payoff when he is assigned to a later period. This later-mover advantage is robust to (i) alternative tie-breaking rules, (ii) the information environment, and (iii) the payment rule for losers. Finally, we show that players' payoff comparison sensitively depends on winner's payment rule: Drastically contrasting predictions arise when the winner pays the second-highest bid, i.e., in a war of attrition.

Keywords: Sequential Contest; Later-/Earlier-mover Advantage; All-pay Auction; War of Attrition

JEL Classification Codes: C72, D43, D44, D82, L13.

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1 Introduction

Many competitive activities resemble a contest, in which contenders strive to leapfrog while their efforts are nonrefundable regardless of win or loss. Such phenomena are widespread in socioeconomic contexts, ranging from electoral campaigns (Snyder, 1989), lobbying (Che and Gale, 1998), and internal labor markets inside firms (Lazear and Rosen, 1981), to R&D races (Loury, 1979; Lee and Wilde, 1980). A voluminous economics literature has been developed to illuminate the strategic essence of contest games.

Many contest-like competitions in practice are inherently sequential, in that contenders enter and act in succession that allows a later entrant to observe the history of earlier actions. Political campaigns, for instance, take place in series (see, e.g., Morgan, 2003; Fu, 2006). Firms may enter a race for an innovative technology successively. As Hinnosaar (2021a) and Fischer, Güth, Kaplan, and Zultan (2021) point out, firms' actions are often subject to disclosure requirement or leaked to competitors. For instance, EU countries typically require mandatory disclosure of firms' R&D activities (La Rosa and Liberatore, 2014). In the United States, the Honest Leadership and Open Government Act of 2007 in the United States amended the Lobbying Disclosure Act of 1995, which elevated public disclosure requirements regarding lobbying activity and funding. On the leading crowdsourcing platform Taskcn, a participant is given access to earlier submissions (Liu, Yang, Adamic, and Chen, 2014).

Dynamic interactions arise in such scenarios. Earlier movers condition their actions on prior moves; while an earlier mover shapes their strategies in anticipation of future opponents' reactions, which complicates a strategical analysis of sequential contests. The complexity can be further compounded when the contest involves richer timing architectures: For instance, multiple players can enter in a single period and act simultaneously; they observe the actions taken by earlier opponents but not those of the contemporaneous ones. A full-fledged analysis involves substantial analytical difficulty, which confines the majority of previous studies to limited settings. Dixit (1987), Baik and Shogren (1992), and Hoffmann and Rota-Graziosi (2012), for instance, assume a two-player two-period structure.

In this paper, we investigate a sequential contest game modeled as an all-pay auction. Analogous to standard static models—e.g., Moldovanu and Sela (2001) and Moldovanu, Sela, and Shi (2007)—ex ante symmetric players strive for a commonly valued prize, and the highest bidder wins; players' private types (abilities) are independently and identically distributed, and a higher ability yield a lower marginal effort cost. In contrast, our model accommodates a full spectrum of timing architectures: The contest proceeds in multiple periods, while multiple players can be clustered in a single period; all contemporaneous within each period take actions simultaneously, while they observe the moves of their earlier opponents. A fully sequential contest emerges when each period consists of a single player, while a standard simultaneous contest results when all players are concentrated in a single

period. The literature has yet to provide an equilibrium analysis of this general sequential contest game. Segev and Sela (2014) and Jian, Li, and Liu (2017) allow for multiple players but assume a fully sequential structure. Hinnosaar (2021a) provides a remarkable exception to the literature that allows for an unrestricted timing architecture but assumes a lottery contest, which differs from our setting. Our paper fills the gap.

The dynamic nature of the sequential contest game further compels us to explore the natural question about players' payoff comparison with respect to their timing positions, i.e., the earlier- vs. later-mover advantages—or the classical trade-off between commitment and flexibility—in the literature (see, e.g., Gal-Or, 1987; Dowrick, 1986; Dixit, 1987). The economics literature has long recognized that players' decisions and payoffs in oligopolistic rivalry depend sensitively on the timing of their moves (e.g., Gal-Or, 1985, 1987; Dowrick, 1986). We aim to complement this extensive literature in two main dimensions.

First, the vast majority of the literature—with few notable exceptions such as Hinnosaar (2021a)—has typically adopted duopolistic settings to analyze the classical Cournot/Stackelberg trade-off. The conventional wisdom obtained in duopolistic settings deserve to be examined in broader context. Shinkai (2000), for instance, demonstrates the challenge caused by more general sequential structures: He considers a three-firm three-period model, which generalizes the two-firm Stackelberg setting in Gal-Or (1987); it is shown that players' payoffs can be nonmonotone along the sequence. Our setting imposes no restrictions on the number of players and the prevailing timing architecture.

Second, the previous studies generally hold that the strategic substitutability/complementarity between players' strategic actions—or equivalently, the slopes of their best response curves—plays the decisive role in payoff comparisons. In contrast to quantity/price setting games, an all-pay auction does not yield a well-behaved best response correspondence. Let us consider a two-player contest and a simple tie-breaking rule such that a later mover beats an earlier one when their efforts tie. The later mover simply matches earlier effort insofar as it costs less than the prize value; as a result, his best response would increase with earlier effort—which yields an upward-sloping best response curve—but drops to zero when it is sufficiently high, i.e., when matching it is rendered unprofitable. Players' best responses would further be obscured when they are embedded in more complex timing architectures, which, as will be discussed below, enormously complicates their effort decision. This calls for an alternative perspective to understanding the nature of earlier-/later-mover advantage. Hinnosaar (2021a) analyzes a general sequential lottery contest, in which case a continuous best response function arises and a rationale of strategic substitutability remains. Our setting thus departs from this strand of literature.

Findings and Implications: Summary We first conduct a comprehensive equilibrium analysis of the sequential contest game described above. To meet the analytical challenges posed by the dynamic interactions involved in the game, we take advantage of the recursive property of the payoff structure and convert the game into one that resembles a simultaneous-move all-pay auction with an *endogenously* determined prize. The pseudo prize value is shaped by players' ability distribution function and depends on the maximum effort of one's earlier opponents. Our setup allows for a broad class of ability distribution functions, without imposing specific requirement of the shape of the distribution. This may cause irregularity to players' payoff functions and therefore discontinuity in players' bidding strategies with respect to their abilities. Despite the nuance, we establish that there exists a unique symmetric Perfect Bayesian equilibrium (PBE) in the game and provide a complete equilibrium characterization.

The equilibrium result allows us to formally explore the following question: Ceteris paribus, would a player be ex ante better off if he were assigned to a later vis-à-vis an earlier timing position? The answer is straightforward in a classical two-player setting, while allowing for multiple players adds substantial complexity to the analysis. We next briefly elaborate on the nuance.

Let us begin with a two-player sequential-move contest with a simple tie-breaking rule as described above. The second mover, upon observing the first mover's effort, either simply matches the earlier bid, or stays inactive—i.e., exerting zero effort—if his ability is too low, in which case matching the earlier bid incurs an excessive cost. Evidently, in the equilibrium, the first mover exerts a larger amount of ex ante expected effort than the second mover because the latter player never outbids the first mover and stays inactive with a positive probability. Further, the first mover is ex ante less likely to win: He wins only if his late opponent's ability substantially falls below his own, such that the late opponent chooses to stay inactive.[1] We can thus conclude a clear-cut second-mover advantage in the setting.

Next, imagine a simple case of three players and fully sequential moves. The second mover cannot simply match the earlier effort, which allows him to defeat the first mover but may not be optimal given the threat from the third mover. He may stay inactive, match the earlier effort, or exert a strictly higher effort, depending on his expectation of the future mover's type and reaction. As a result, the second mover may contribute the highest ex ante expected effort, as he may attempt to preempt future competition: The monotonicity in duopolistic settings may break down, which would nullify the simple rationale laid out above for duopoly. It deserves to note that this rationale does not fully reveal the complex nature of the problem. Recall that our model allows multiple players to be clustered in the same

[1] In a simultaneous-move setting, the two ex ante symmetric players win with an equal probability. However, with sequential moves, the second mover can win the contest even if his type is lower than the first.

period. Consider a simple situation in which multiple players move simultaneously in the last period: Matching the highest earlier bid is never optimal for a player, as he, in order to win the contest, has to outbid not only the earlier opponents but also all contemporaneous players. It is thus unclear a priori whether a later timing position is ex ante preferred to an earlier one.

Under plausible conditions, we demonstrate that as conjectured above, the monotonicity in effort, in general, does not persist beyond a two-player setting: In a fully-sequential contest, the *penultimate* mover may exert the highest ex ante expected effort, which reveals the complexity in the general setting. Despite the nuance, however, we establish a payoff monotonicity arises in the general setting: Regardless of the prevailing contest architecture, a player ends up with a strictly higher ex ante expected payoff in later timing position vis-à-vis an earlier one. Our result thus provides a formal argument for an unambiguous later-mover advantage in the context of multi-player contests.

We then examine three variations to our baseline model to test the boundary of our results. First, we consider an alternative (symmetric) tie-breaking rule, such that a later mover secures a lead if and only if his effort strictly exceeds that of his earlier opponents. Second, we introduce an alternative information structure: Players' abilities are fully revealed prior to the contest. As a result, under a given timing architecture, every contest is a subgame with a complete-information sequential all-pay auction, and we provide a complete equilibrium characterization. Third, we infuse a winner-pay component to the model, such that the winner pays the full cost of his effort, while a loser pays only a portion of that; the model, in the extreme case, boils down to a first-price auction. We demonstrate that the later-mover advantage established in our baseline setting is robust to all these variations.

Finally, we consider a fundamental variation to our model. Namely, we allow for a second-prize winner-payment rule, such that the winner bears the cost of the second-highest effort, which leads to a model that resembles a war-of-attrition game à la Krishna and Morgan (1997); Bulow and Klemperer (1999); Hafer (2006); and Bergemann, Brooks, and Morris (2019) but is accommodates a sequential structure. In contrast to our first-price baseline model, an earlier mover can be placed in a favorable position: One can exert an effort more than the prize value that deters future competition, since he does not pay the cost of his own effort but only that of the second highest. We characterize the equilibrium and demonstrate that players' payoffs are in general nonmonotone along the sequence.

The contrast to our baseline first-price model sheds light on the strategic nature of sequential contest games and the later mover advantage established in all-pay auctions. The prevailing winner-payment rule plays a critical role. In a first-price model, an earlier player takes into account future competitions and thus tends to underbid (relative to his effective prize valuation) to avert a loss. A later player, however, faces less uncertainty, as

he faces fewer future competitors. These facts render a later-mover advantage. In contrast, a second-prize model gives rise to a dichotomous bidding strategy: A player may either underbid—e.g., simply staying inactive—or overbid to preempt. He is tempted to overbid because he does not bear his full effort cost, and he does so when he is sufficiently confident of his eventual win, i.e., when he is of a high-ability type. The heightened competition poses a threat to low-ability earlier players and may lead them to exit. As a result, an unambiguous payoff ranking with respect to the order of moves does not exist in general. These observations offer novel insights into the nature of earlier-/later-mover advantage in sequential games.

Link to the Literature This paper belongs to the small but burgeoning literature on sequential contests. Dixit (1987), Baik and Shogren (1992), Morgan (2003), and Hoffmann and Rota-Graziosi (2012) all consider Tullock contests in which two players move sequentially. Morgan and Várdy (2007) adopt a similar framework but assume that the follower has to bear a small cost to observe the leader's effort. Glazer and Hassin (2000), in contrast, allow for three-period sequential plays. Our paper is more closely related to Segev and Sela (2014) and Jian, Li, and Liu (2017). Both adopts incomplete-information all-pay auction frameworks and assumes a fully sequential structure. Segev and Sela (2014) study fully-sequential all-pay auctions with multiple ex ante heterogeneous players and examine the effect of the number and ability distributions on the expected highest effort. Jian, Li, and Liu (2017) assume symmetric players and compares players' winning probabilities with respect to the order of moves. We also assume symmetric players but impose no restrictions on timing architectures. Konrad and Leininger (2007) consider multi-player complete-information all-pay auctions, in which players' marginal effort costs are commonly known. Players are split into an early group and an later group. It is shown that as in simultaneous-move contests, only the player with the lowest cost ends up with a positive expected payoff, while the payoff depends on his own timing position vis-à-vis those of the others. Further, the player with the lowest cost prefers the late timing, while all the others are indifferent. Our analysis in Section 5.2 generalizes Konrad and Leininger (2007), as it examines complete-information all-pay auctions with a general timing architecture.

Hinnosaar (2021a) examines a complete-information lottery contest with multiple symmetric players. He develops a novel technique to analyze a complex multi-player (partially) sequential contest, in which standard backward induction comes to no avail. The technique further enables Hinnosaar (2021b) to examine the boundary for Stackelberg independence to hold in multi-firm capacity-choice games. Hinnosaar (2021a) remarkably generalizes and formalizes Dixit's thesis that the efforts (and expected payoffs) of earlier players are strictly higher than those of later players, which results from the strategic substitutability of efforts

in a symmetric sequential lottery contest. Our paper examines a drastically contrasting game theoretical context. All-pay auctions does not generate a continuous and well-behaved best-response correspondence as a lottery contest, and players' strategic concerns sharply differ. Our results also diverge from that of Hinnosaar (2021a). He shows that a later mover exerts a lower effort and an earlier mover acquires an advantage. We nevertheless demonstrate nonmonotonicity in equilibrium efforts and establish a later-mover advantage.

This paper contributes to the extensive literature on players' payoff comparison with respect to their timing positions in sequential-move games—such as Gal-Or (1985, 1987), Dixit (1987), Dowrick (1986), Daughety (1990), Deneckere and Kovenock (1992), Amir and Grilo (1999), Van Damme and Hurkens (1999, 2004), Amir and Stepanova (2006), and von Stengel (2010), among many others—in various contexts, ranging from quantity/capacity- to price-setting competitions.[2,3] As stated above, this strand of literature typically focuses on duopolistic rivalry. Shinkai (2000) extends the framework to a three-firm three-period setting and illuminates the nuance caused in the more extensive sequence. To the best of our knowledge, our paper and Hinnosaar (2021a) provide the few exceptions in the literature that examine earlier-/later-mover advantage in a general framework that affords unrestricted timing architecture. As stated above, all-pay auctions do not yield well-behaved best response correspondence, which stands in contrast to the frameworks adopted in these studies. Further, we demonstrate the nuances caused by the winner-payment rule, which we believe provides a novel perspective to understanding earlier-/later-mover advantage in sequential games.[4]

The rest of the article is organized as follows. Section 2 sets up the model. Section 3 characterizes the equilibrium, and discusses the nuance of a multi-player model. Section 4 compares players' ex ante expected payoffs with respect to their timing positions and establishes the result of later-mover advantage. Section 5 provides further discussion within the first-price framework. Section 6 examines an extension of second-price all-pay auctions. Section 7 concludes. Proofs of the results are collected in the Appendix.

[2]Mailath (1993) considers this issue from a signaling perspective. An informed firm is allowed to delay its quantity decision, and the timing choice itself conveys private information to the market. He shows that the only stable outcome is that the informed firm moves first.

[3]Kempf and Rota-Graziosi (2010) consider a setting in which two jurisdictions set tax rates and endogenize leadership in tax competition.

[4]Jian, Li, and Liu (2017) establish a later-mover advantage in terms of players' equilibrium winning probabilities in a sequential all-pay auction, but they consider only first-price settings and a fully-sequential architecture.

2 The Model

A contest involves $N \geq 2$ ex-ante identical risk-neutral players, indexed by $i \in \mathcal{N} \equiv \{1, \ldots, N\}$. The players arrive sequentially and each exert effort upon arrival to compete for a prize of a unity value. The contest proceeds in $T \geq 2$ periods, and the players are, accordingly, partitioned into T groups. Denote by $\mathcal{N}_t$ the set of players who arrive in period t, and let $n_t := |\mathcal{N}_t| \geq 1$ indicate the number of players in this period. A player who acts in period $t \in \mathcal{T} \equiv \{1, \ldots, T\}$ observes the efforts sunk by earlier players but not those in contemporaneous and future periods. The architecture of the contest is then fully described by a vector $\boldsymbol{n} := (n_1, \ldots, n_T)$, with $N = \sum_{t=1}^{T} n_t$. The contest is fully sequential if it has an architecture $\boldsymbol{n} := (1, \ldots, 1)$, while it degenerates to a fully simultaneous one with $\boldsymbol{n} := (N, 0, \ldots, 0)$.

A player i, when exerting an effort (or bid) $b^i \geq 0$, incurs a cost $c(b^i) = b^i/a^i$, where $a^i > 0$ can be interpreted as a measure of one's ability and is privately known to the player.[5] Abilities are drawn independently from an interval $[0, 1]$ according to a common distribution function $F(\cdot)$. We assume that $F(\cdot)$ admits a positive and smooth density $f(\cdot) \equiv F'(\cdot)$ and is piecewise analytic.

The competition is modeled as an all-pay auction. The player with the highest effort wins the contest. Specifically, a period-t player $i \in \mathcal{N}_t$, when exerting an effort $b^i \geq 0$, is the *sole winner* of the prize if and only if (i) his effort is greater than or equal to those in previous periods—i.e., $b^i \geq b^j$ for $j \in \cup_{k=1}^{t-1} \mathcal{N}_k$—and (ii) his effort is strictly larger than those in contemporaneous and future periods—i.e., $b^i > b^j$ for $j \in \cup_{k=t}^{T} \mathcal{N}_k \setminus \{i\}$. In the event that multiple players in period t place the same highest bid, the prize is randomly distributed among these players. To put it more formally, fixing a set of effort entries $\boldsymbol{b} \equiv (b^1, \ldots, b^N)$, contestant i's probability of winning the contest is given by

$$p^i(\boldsymbol{b}) := \begin{cases} 1, & \text{if } b^i \geq \max_{j \in \cup_{k=1}^{t-1} \mathcal{N}_k} \{b^j\} \text{ and } b^i > \max_{j \in \cup_{k=t}^{T} \mathcal{N}_k \setminus \{i\}} \{b^j\}, \\ 1/m, & \begin{array}{l} \text{if } b^i \geq \max_{j \in \cup_{k=1}^{t-1} \mathcal{N}_k} \{b^j\},\ b^i > \max_{j \in \cup_{k=t+1}^{T} \mathcal{N}_k} \{b^j\}, \\ \text{and } b^j \text{ is among the } m \text{ highest of } \{b^j\}_{j \in \mathcal{N}_t} \text{ with a tie,} \end{array} \\ 0, & \text{if } b^i < \max_{j \in \cup_{k=1}^{t} \mathcal{N}_k \setminus \{i\}} \{b^j\} \text{ or } b^i \leq \max_{j \in \cup_{k=t+1}^{T} \mathcal{N}_k} \{b^j\}, \end{cases} \tag{1}$$

and his ex post expected payoff, for a given ability level a^i, is

$$p^i(\boldsymbol{b}) - b^i/a^i, \text{for all } i \in \mathcal{N}. \tag{2}$$

[5] We follow the tradition in the contest literature and accommodate player heterogeneity in their cost functions (e.g., Moldovanu and Sela, 2001, 2006; Moldovanu, Sela, and Shi, 2007; Brown and Minor, 2014). It is noteworthy that the model is isomorphic to an alternative setting in which effort is interpreted as bid in the auction literature: Players value prize differently but bear the same effort costs. All our results remain qualitatively unchanged under this model specification.

We consider the solution concept of perfect Bayesian equilibrium (PBE) for the sequential contest game throughout the paper except for the case studied in Section 5.1, and focus on the symmetric equilibrium in which all players who move in the same period adopt the same bidding strategy.

Before we proceed to characterize the equilibrium of the game, the following remark elaborates on the tie-breaking rule adopted in (1).

Remark 1 ***(Tie-breaking Rule)*** *Cautious readers may have noted that the tie-breaking rule assumed in* (1) *is asymmetric, which "favors" a later-mover at first glance, in that the latter can gain a lead by simply matching previous efforts. This tie-breaking rule is commonly adopted in the literature, as it ensures the existence of a PBE.*[6] *A player's best response might not be well-defined if a symmetric tie-breaking rule is adopted—i.e., letting the prize be randomly awarded to one of players with the highest bid regardless of players' arrival times.*[7] *Section 5.1 allows for a symmetric tie-breaking rule and shows that our results largely remain intact under an alternative equilibrium concept, the ϵ-equilibrium.*

3 Equilibrium Analysis

In this section, we first lay out the fundamentals of the analysis, and then formally characterize the equilibrium of the game. We close this section with a brief discussion of the nuance involved in a multi-player sequential all-pay auction.

3.1 Preliminaries of Equilibrium Analysis

The profile of equilibrium bidding strategies can be derived by backward induction. Fix a bid profile $(b^j)_{j\in\cup_{k=1}^{t-1}\mathcal{N}_k}$, define $\gamma_t := \max_{j\in\cup_{k=1}^{t-1}\mathcal{N}_k}\{b^j\}$ for $t \in \{2,\ldots,T\}$, and let $\gamma_1 \equiv 0$. In words, γ_t is the maximum effort in the contest prior to period t. A symmetric PBE is denoted by $\{b_t^*(a;\gamma_t)\}_{t=1}^T$, where a is a player's ability and $b_t^*(a;\gamma_t)$ is his equilibrium bidding strategy when he arrives in period t.[8]

[6] See, e.g., Andreoni, Che, and Kim (2007), Simon and Zame (1990), and Maskin and Riley (2000).

[7] Similarly, consider an asymmetric Bertrand duopoly game. A pure-strategy Nash equilibrium exists if ties are broken in favor of the lower-cost firm, whereas a pure-strategy equilibrium does not exist with an "equal sharing" tie-breaking rule.

[8] In principle, a period-t player's information set is summarized by the bid profile $(b^j)_{j\in\cup_{k=1}^{t-1}\mathcal{N}_k}$ from all previous periods rather than the maximum bid $\gamma_t \equiv \max_{j\in\cup_{k=1}^{t-1}\mathcal{N}_k}\{b^j\}$ and we need to specify a belief system when defining a PBE. However, in our setting, a player's payoff depends only on his own ability and whether his bid is the highest among all contestants; there is no need for the player to infer his rivals' types from the bidding profile. Therefore, a period-t player's bidding behaviors are the same across all information sets that lead to the same maximum bid γ_t, and it is unnecessary to specify a belief system.

Before characterizing the equilibrium, we introduce several notations. Fixing a contest architecture $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$, define a sequence of functions $\{H_t(b), a_t^*(\gamma), \tilde{\pi}_t(b,a)\}_{t=1}^T$, recursively, as follows:

$$H_T(b) \equiv 1, H_{t-1}(b) := H_t(b) F^{n_t}\left(a_t^*(b)\right), \forall b \in [0,1], \tag{3}$$

$$a_t^*(\gamma) := \max\left\{0 < a \leq 1 : \tilde{\pi}_t(b,a) \leq 0, \forall b \in [\gamma, 1]\right\}, \tag{4}$$

$$\tilde{\pi}_t(b,a) := H_t(b) F^{n_t - 1}(a) - b/a. \tag{5}$$

It can be verified from the above definition that $a_t^*(\gamma)$ and $H_t(b)$ are well-defined and satisfy the following properties:

(a) $a_t^*(\gamma)$ is continuous, piecewise differentiable, and weakly increasing on $[0,1]$, satisfying $a_t^*(\gamma) \geq \gamma$ for all $t \in \mathcal{T}$; and

(b) $H_t(b)$ is continuous, piecewise differentiable, and strictly increasing on $[0,1]$, with $H_t(0) = 0$ and $H_t(1) = 1$ for all $t \in \mathcal{T} \setminus \{T\}$.[9]

The sequence of functions $\{H_t(b), a_t^*(\gamma), \tilde{\pi}_t(b,a)\}_{t=1}^T$ is key to the equilibrium characterization of the sequential contest game, and will be intuitively interpreted below as our analysis unfolds. We hereby present four lemmata that pave the way for our equilibrium result. Lemma 1 delineates useful properties of players' bidding strategy in a hypothetical symmetric PBE.

Lemma 1 ***(Properties of Equilibrium Bidding Strategy)*** *Consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ and suppose that there exists a symmetric PBE. A period-t player's equilibrium bidding strategy $b_t^*(a; \gamma_t)$ satisfies the following properties:*

(i) $b_t^*(a; \gamma_t)$ is increasing in a on $(0,1)$;

(ii) $b_t^*(a; \gamma_t) = 0$ for $a \leq a_t^*(\gamma_t)$ and $b_t^*(a; \gamma_t) \geq \gamma_t$ for $a > a_t^*(\gamma_t)$;

(iii) $b_t^*(a; \gamma_t)$ strictly increases with a on $\left(a_t^*(\gamma_t), 1\right)$ if $n_t \geq 2$.

Lemma 1(i) is intuitive: A stronger player tends to bid more aggressively. Lemma 1(ii) reveals the nature of $a_t^*(\cdot)$ and establishes the role it plays in the equilibrium bidding strategy $b_t^*(a; \gamma_t)$: $a_t^*(\gamma_t)$ is the threshold ability level above (below) which a period-t player would remain active (inactive) in equilibrium. Recall that $a_t^*(\cdot)$ increases with its arguments and γ_t is the maximum effort prior to period-t, which implies that higher earlier effort elevates the threshold for active bidding, thereby discouraging future competition. By Lemma 1(iii),

[9] See the proof of Lemma 5 in the appendix for more details.

when a period t involves two or more players, one's effort strictly increases with his ability provided that he is willing to place a positive bid, i.e., $a > a_t^*(\gamma_t)$. Note that the strict monotonicity does not necessarily hold in the case with $n_t = 1$. To see this, recall that in a two-player sequential contest $\boldsymbol{n} = (1, 1)$, the later mover simply matches the first mover's effort γ_2 irrespective of his own ability, provided that it exceeds γ_2, i.e., $a_2^*(\gamma_2) = \gamma_2$ and $b_2^*(a; \gamma_2) = \gamma_2$ for $a > a_2^*(\gamma_2)$.

Let $b \geq 0$ be the realized highest effort by the end of a period $t \in T$, which leads to an eventual win if and only if it exceeds all subsequent efforts from period $t + 1$. Lemma 1(ii) allows us to derive the equilibrium probability of this event. By Lemma 1, b ends up as the eventual winning effort if and only if all subsequent players stay inactive—i.e., a period-ℓ player's ability falls below the threshold $a_\ell^*(b)$, $\forall \ell \in \{t + 1, \ldots, T\}$—which occurs with a probability of $\Pi_{\ell=t+1}^{T} F^{n_\ell}\left(a_\ell^*(b)\right)$: Otherwise, at least one player in later periods would stay active and exert an effort above b. Notably, this probability can be expressed recursively, which boils down to the function $H_t(b)$ as specified in (3). We formally state this fact in the following lemma.

Lemma 2 ***(Equilibrium Winning Probability of a Provisional Winner)*** *Consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ and suppose that there exists a symmetric PBE. Let $b \geq 0$ be the realized highest effort by the end of period $t \in \mathcal{T}$. Then $H_t(b)$ gives the probability of the effort b's exceeding all subsequent efforts in equilibrium.*

Lemma 2 enables us to exploit the recursive nature of the sequential contest game, which simplifies the equilibrium analysis. Consider a period-t player with ability $a > 0$. In a symmetric PBE, by exerting an arbitrary effort $b \geq \gamma_t$, he earns an expected payoff[10]

$$\pi_t(b, a; \gamma_t) := H_t(b) F^{n_t - 1}\left((b_t^*)^{-1}(b; \gamma_t)\right) - b/a. \tag{6}$$

In a given symmetric PBE, he wins with a probability $H_t(b) F^{n_t-1}\left((b_t^*)^{-1}(b; \gamma_t)\right)$: As stated above, the effort b allows him to beat future opponents with a probability $H_t(b)$, and prevail over his contemporaneous competitors with a probability $F^{n_t-1}\left((b_t^*)^{-1}(b; \gamma_t)\right)$.

The strategic interactions between a period-t player and those from subsequent periods are encapsulated in the provisional winning probability function $H_t(\cdot)$: His equilibrium bidding strategy can be solved for as if he competed in a static contest for a prize of a value $H_t(b)$, which technically dissolves the *dynamic* linkages between contestants across different periods. Despite the analogy, the pseudo prize value, $H_t(b)$, endogenously depends on the player's own bid b, so the equilibrium bidding strategy fundamentally differs from that in a standard static contest.

[10] The inverse function $(b_t^*)^{-1}(b; \gamma_t)$ is well-defined by Lemma 1(iii) for $b \geq \gamma_t$ and $n_t \geq 2$. Note that there is no need to specify $(b_t^*)^{-1}(b; \gamma_t)$ for the case of $n_t = 1$ given that $F^{n_t-1}(a) = 1$ for all $a \in [0, 1]$

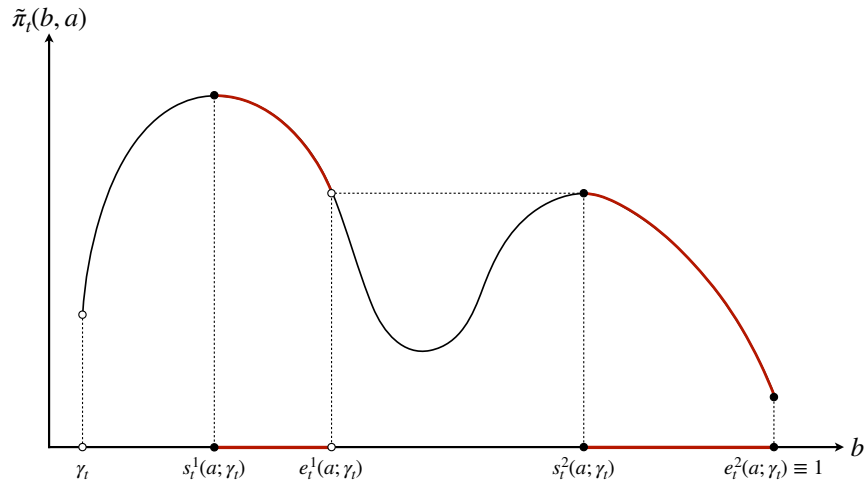


Figure 1: Illustration of $S_t(a;\gamma_t)$.

Next, we narrow the set of possible equilibrium bids. Consider a period-t player with ability $a > 0$ who faces contemporaneous competition, i.e., $n_t \geq 2$. Recall the function $\tilde{\pi}_t(b,a)$ defined in (5):

$$\tilde{\pi}_t(b,a) \equiv H_t(b)F^{n_t-1}(a) - b/a,$$

Fixing $\gamma_t \geq 0$, we define

$$S_t(a;\gamma_t) := \left\{b \in [\gamma_t, 1] : \tilde{\pi}_t(b,a) > \tilde{\pi}_t(b',a), \forall b' \in (b,1]\right\}.$$

The set $S_t(a;\gamma_t)$ is type-dependent and shrinks as ability a increases, i.e., $S_t(a';\gamma_t) \subseteq S_t(a;\gamma_t)$ for $a < a'$. Further, it follows from the piecewise analyticity of the ability distribution $F(\cdot)$ that $H_t(b)$ and thus $\tilde{\pi}_t(b,a)$ are continuous and piecewise analytic with respect to b on $[\gamma_t, 1]$; this in turn implies that $S_t(a;\gamma)$ can be expressed as the union of finitely many disjoint intervals. For ease of exposition, we denote by $m_t(a;\gamma_t) \in \mathbb{N}_+$ the number of disjoint intervals included in the set $S_t(a;\gamma)$. Then $S_t(a;\gamma)$ can be written as

$$S_t(a;\gamma_t) = \left[s_t^1(a;\gamma_t), e_t^1(a;\gamma_t)\right)\bigcup\left[s_t^2(a;\gamma_t), e_t^2(a;\gamma_t)\right)\bigcup\cdots\bigcup\left[s_t^{m_t(a;\gamma_t)}(a;\gamma_t), e_t^{m_t(a;\gamma_t)}(a;\gamma_t)\right],$$

with $e_t^m(a;\gamma_t) < s_t^{m+1}(a;\gamma_t)$ for $1 \leq m \leq m_t(a;\gamma_t) - 1$ and $e_t^{m_t(a;\gamma_t)}(a;\gamma_t) \equiv 1$. Figure 1 graphically illustrates the set $S_t(a;\gamma_t)$, in which case $m_t(a;\gamma_t) = 2$.

Now we are ready to show that a period-t player's equilibrium effort must be contained within the above-defined set, provided that it exceeds γ_t. To put this formally, $b_t^*(a,\gamma_t) \in S_t(a;\gamma_t)$ if $b_t^*(a,\gamma_t) \geq \gamma_t$ is continuous in the neighborhood of a. To see this, suppose, to the contrary, that $b_t^*(a,\gamma_t) \notin S_t(a;\gamma_t)$. By definition, there exists some effort $b' > b_t^*(a,\gamma_t)$ such that $\tilde{\pi}_t(b',a) \geq \tilde{\pi}_t(b_t^*(a,\gamma_t),a)$. We now claim that the player's payoff would strictly

exceed the value of the constructed function $\tilde{\pi}_t(b', a)$ when he deviates from $b_t^*(a, \gamma_t)$ to b'. This is caused by the fact that the higher effort b' increases not only the probability of outbidding his future opponents but also that of beating the contemporaneous ones. To put this more specifically, let a' be the maximum ability such that $b_t^*(a, \gamma_t) \leq b'$. Because the equilibrium bidding strategy $b_t^*(a, \gamma_t)$ is weakly increasing in and continuous at a, we can conclude that $a' > a$. In a symmetric PBE, the player's actual expected payoff from the deviation ends up as $H_t(b')F^{n_t-1}(a') - b'/a$: In other words, he behaves as if he were of an ability a', which allows him to defeat his contemporaneous opponents with a probability $F^{n_t-1}(a')$. This payoff strictly exceeds $\tilde{\pi}_t(b', a) \equiv H_t(b')F^{n_t-1}(a) - b'/a$ and thus overshadows the equilibrium payoff $\tilde{\pi}_t(b_t^*(a, \gamma_t), a)$. Contradiction ensues.

The next lemma demonstrates that the smallest element in $S_t(a; \gamma_t)$ is indeed the bid that a player of the threshold ability $a = a_t^*(\gamma_t)$ tends to place in equilibrium.

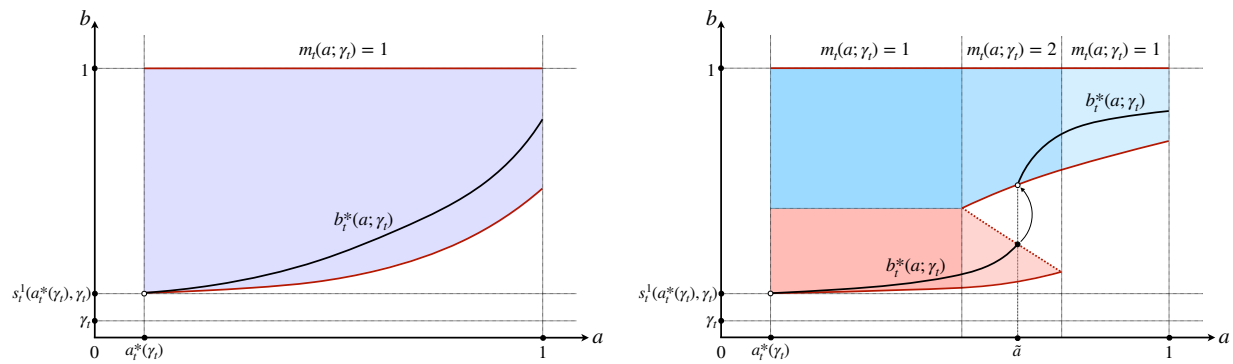
Lemma 3 ***(Equilibrium Bid at $a_t^*(\gamma_t)$)*** *Consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ and suppose that there exists a symmetric PBE. If $n_t \geq 2$ and $a_t^*(\gamma_t) < 1$, then $\lim_{a \searrow a_t^*(\gamma_t)} b_t^*(a; \gamma_t) = s_t^1(a_t^*(\gamma_t); \gamma_t)$.*

For notational convenience, we use $b_t^*(a - 0; \gamma_t)$ and $b_t^*(a + 0; \gamma_t)$ to denote the left and right limits of $b_t^*(a; \gamma_t)$, respectively. The following lemma can be obtained.

Lemma 4 ***(Potential Discontinuity of Players' Bidding Strategy)*** *Consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ and suppose that there exists a symmetric PBE. Fix a period t with $n_t \geq 2$ and player's ability $\tilde{a} \in (a_t^*(\gamma_t), 1]$. If $b_t^*(\tilde{a} - 0; \gamma_t) = e_t^m(\tilde{a}; \gamma_t)$ for some $1 \leq m \leq m_t(\tilde{a}; \gamma_t) - 1$, then $b_t^*(\tilde{a} + 0; \gamma_t) = s_t^{m+1}(\tilde{a}; \gamma_t)$.*

Lemma 4, as well as Lemma 3, alludes to the potential discontinuity in players' equilibrium bidding strategy. by Lemma 1, a period-t player, for a given γ_t, would increase his bid as his ability ascends. When the bid is in the interior of the set $S_t(a; \gamma_t)$, the player's bidding strategy would gradually increase with a for $a > a_t^*(\gamma_t)$, as Figure 2(a) illustrates. Recall that the set of eligible efforts $S_t(a; \gamma_t)$ shrinks as a ascends (see Figure 2). When the player's effort reaches the end of some interval in the set $S_t(a; \gamma_t)$—i.e. $e_t^m(a; \gamma_t)$—he would refrain from exerting an effort in the "undesirable" region $\left[e_t^m(a; \gamma_t), s_t^m(a; \gamma_t)\right)$; his effort jumps directly to the lowest bid in the next adjacent interval in $S_t(a; \gamma_t)$, i.e., $s_t^{m+1}(a; \gamma_t)$. This scenario is depicted in Figure 2(b).

To understand why the boundary of the set $S_t(a; \gamma_t)$ can be played in the equilibrium, it is useful to further inspect the constructed function (5) and the equilibrium payoff function (6). Consider a symmetric PBE. When a player's effort increases, it increases the probability of his outperforming his contemporaneous opponents: Such an *equilibrium effect* is nevertheless omitted in the expression of (5). Suppose that all other players' bidding strategy



(a) Continuous Equilibrium Bidding Strategy (b) Discontinuous Equilibrium Bidding Strategy

Figure 2: Illustration of Equilibrium Bidding Strategy $b_t^*(a;\gamma_t)$.

contains a jump from $b^\sharp$ to $b^{\sharp\sharp}$ when one's ability increases. All bids between $b^\sharp$ and $b^{\sharp\sharp}$ would yield the same probability of winning the contemporaneous competition. As a result, the aforementioned equilibrium effect dissolves around the jump.

The jump predicted in Lemma 4 and Figure 2(b) is impossible in a Bayesian Nash equilibrium of a *static* all-pay auction. This discontinuity, if it exists, largely stem from the dynamic nature of the sequential contest game, which is captured by the provisional winning probability function $H_t(\cdot)$ in the equilibrium interim expected payoff (3). Recall that $H_t(\cdot)$ is derived from the ability distribution $F(\cdot)$. This fact inspires us to conjecture that the continuity/discontinuity in players' equilibrium bidding strategies depends on the properties of the distribution function $F(\cdot)$. Our analysis in Section 3.3 verifies this conjecture and unpacks the underlying logic.

3.2 Equilibrium Result: Existence, Uniqueness, and Characterization

We are now ready to verify the existence and uniqueness of a symmetric PBE in the sequential contest game under an arbitrary contest architecture $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ and fully characterize it. Let $h_t(b) := H_t'(b)$. Denote by $G_n(\cdot)$ the inverse function of $aF^{n-1}(a)$ for an arbitrary positive integer $n \in \mathbb{N}_+$.[11]

Theorem 1 ***(Equilibrium of Sequential All-pay Contests)*** *Consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$. There exists a unique symmetric PBE $\{b_t^*(a;\gamma_t)\}_{t=1}^T$ of the contest game, which is fully characterized as follows:*[12]

[11] It is straightforward to verify that $G_n(\cdot)$ is well-defined and is strictly increasing on $[0,1]$, with $G_n(0) = 0$ and $G_n(1) = 1$.

[12] The symmetric PBE is unique in the sense that if there exist two symmetric PBE of the contest game—

(i) If $n_t = 1$, then

$$b_t^*(a;\gamma_t)\begin{cases} = 0, & \text{if } a \le a_t^*(\gamma_t), \\ = \gamma_t, & \text{if } a_t^*(\gamma_t) < a \le a_t^{**}(\gamma_t), \\ \in \arg\max_{b>\gamma_t}[H_t(b) - b/a], & \text{if } a > a_t^{**}(\gamma_t), \end{cases} \tag{7}$$

where $a_t^*(\gamma_t)$ is defined in (4) and can be simplified as $a_t^*(\gamma_t) = \min_{b\ge\gamma_t} b/H_t(b)$, and $a_t^{**}(\gamma_t) := \sup_{a_t^*(\gamma_t)\le a\le 1}\big\{a : H_t(\gamma_t) - \gamma_t/a > H_t(b') - b'/a, \forall b' \in (\gamma_t, 1]\big\}$.

(ii) If $n_t \ge 2$, then $b_t^*(a;\gamma_t) = 0$ for $a \le a_t^*(\gamma_t)$. For $a > a_t^*(\gamma_t)$, $b_t^*(a;\gamma_t)$ increases continuously and is governed by the following differential equation:

$$(n_t-1)aF^{n_t-2}(a)f(a)H_t\big(b_t^*(a;\gamma_t)\big) + aF^{n_t-1}(a)h_t\big(b_t^*(a;\gamma_t)\big)(b_t^*)'(a;\gamma_t) - (b_t^*)'(a;\gamma_t) = 0, \tag{8}$$

with the initial condition $b_t^*\big(a_t^*(\gamma_t)+0;\gamma_t\big) = s_t^1\big(a_t^*(\gamma_t);\gamma_t\big)$. When $b_t^*(\tilde{a};\gamma_t) = e_t^m(\tilde{a};\gamma_t)$ for some $\tilde{a} \in (0,1)$ and $1 \le m \le m_t(\tilde{a};\gamma_t) - 1$, $b_t^*(a;\gamma_t)$ jumps to $s_t^{m+1}(\tilde{a};\gamma_t)$ at $a = \tilde{a}$ and then increases continuously from $\tilde{a}$ again according to (8), with the initial condition $b_t^*\big(\tilde{a}+0;\gamma_t\big) = s_t^{m+1}(\tilde{a};\gamma_t)$.

Theorem 1 establishes the existence and uniqueness of a symmetric PBE in sequential contests. A player's equilibrium bidding strategy depends on the number of contemporaneous opponents. Theorem 1(i) considers the bidding strategy of a player who is alone in a period, while Theorem 1(ii) addresses the scenario in which multiple players are clustered. It is straightforward to observe that a player would remain inactive if he is of low ability—i.e., $a \le a_t^*(\gamma_t)$—in either scenario, as predicted in Lemma 1(ii). The predictions diverge across the two scenarios when the player's ability is sufficiently high. With $n_t = 1$, the player matches the highest prior bid γ_t when his ability remains in an intermediate range—i.e., $a_t^*(\gamma_t) < a < a_t^{**}(\gamma_t)$, while he strictly outbids γ_t if his ability exceeds the cutoff $a_t^{**}(\gamma_t)$. In contrast, with $n_t \ge 2$, he strictly outbids γ_t whenever his ability exceeds $a_t^*(\gamma_t)$: Contemporaneous competition compels him to step up effort to avoid a tie.

A closer look at Lemma 1 and Theorem 1 sheds light on players' bidding behavior with respect to their timing positions, allowing us to identify the set of players who constantly stay inactive. Recall from Lemma 1(ii) that a period-t player with ability below $a^*(\gamma_t)$ would remain inactive. Thus a player would always exert zero effort if $a^*(\gamma_t) = 1$ for all $\gamma_t \in [0,1]$, which is equivalent to $\tilde{\pi}_t(b,1) \le 0$ for all $b \in [0,1]$ by (4). This condition, together with (5) and $F(1) = 1$, implies $H_t(b) \le b$ for all $b \in [0,1]$. Let $\mathcal{T}_0 := \{t \in \mathcal{T} : H_t(b) \le b, \forall b \in [0,1]\}$.

denoted by $\{b_t^*(a;\gamma_t)\}_{t=1}^T$ and $\{b_t^{**}(a;\gamma_t)\}_{t=1}^T$—then for all $t \in \mathcal{T}$ and $\gamma_t \ge 0$, the collection of ability a such that $b_t^*(a;\gamma_t) = b_t^{**}(a;\gamma_t)$ has F-measure one.

It can be verified that $t' \in \mathcal{T}_0$ if $t \in \mathcal{T}_0$ and $t' < t$. Define $t_0 := \max \mathcal{T}_0$, and it is obvious to infer $0 \leq t_0 \leq T - 1$.[13] The following result naturally ensues.

Proposition 1 ***(Players Who Always Remain Inactive)*** *Consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$. In the unique symmetric PBE $\{b_t^*(a; \gamma_t)\}_{t=1}^T$ of the contest game, all players in periods 1 through t_0 choose to stay inactive regardless of his ability and the previous maximum bid, i.e., $b_t^*(a; \gamma_t) = 0$ for all $a \in (0, 1]$, $\gamma_t \in [0, 1]$, and $t \in \mathcal{T}_0$.*

Proposition 1 states that players that arrive in early periods—i.e., $t \leq t_0$—always stay inactive, regardless of their own types. These players obviously receive zero expected payoff in equilibrium, which alludes to a disadvantage of being earlier movers in a contest.

3.3 Discussion: Concavity/Convexity of Ability Distribution

In this part, we examine how the players' equilibrium bidding strategy depends on the ability distribution $F(\cdot)$.

3.3.1 Concave Distribution Function

We begin with the following case.

Assumption 1 ***(Concave Ability Distribution)*** *The ability distribution function $F(\cdot)$ is continuous, twice differentiable, strictly concave, and satisfies $\lim_{a \searrow 0}[f(a)a] = 0$.*

Assumption 1 requires a concave CDF of players' ability. The same assumption is made in Segev and Sela (2014), which characterizes the unique PBE of a fully sequential all-pay auction.[14] The following result—in parallel with Theorem 1—can be obtained.

Proposition 2 ***(Concave Ability Distribution and Equilibrium Bidding Strategy)*** *Suppose that Assumption 1 is satisfied and consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$. The unique symmetric PBE of the contest game, $\{b_t^*(a; \gamma_t)\}_{t=1}^T$, is characterized as follows:*

(i) If $n_t = 1$, then

$$b_t^*(a; \gamma_t) = \begin{cases} 0, & \text{if } a \leq a_t^*(\gamma_t), \\ \gamma_t, & \text{if } a_t^*(\gamma_t) < a \leq a_t^{**}(\gamma_t), \\ h_t^{-1}\left(\frac{1}{a}\right), & \text{if } a > a_t^{**}(\gamma_t), \end{cases}$$

where $a_t^*(\gamma_t) = \gamma_t / H_t(\gamma_t)$, and $a_t^{**}(\gamma_t) = 1/h_t(\gamma_t)$.

[13] In the case that $\mathcal{T}_0$ is an empty set, we let $t_0 = 0$.

[14] Proposition 2 extends their results to allow multiple players to move simultaneously.

(ii) If $n_t \geq 2$, then for $a \leq a_t^*(\gamma_t)$, $b_t^*(a;\gamma_t) = 0$; for $a > a_t^*(\gamma_t)$, $b_t^*(a;\gamma_t)$ solves the differential equation (8) *with the initial condition $b_t^*(a_t^*(\gamma_t)+0;\gamma_t) = \gamma_t$.*

Two remarks are in order regarding the case with concave distribution function. First, recall, by Proposition 1, early players—i.e., those who arrive in from periods 1 to t_0—stay inactive regardless. In the case with a concave distribution, it can be verified that $H_t(b) > b$ for all $b \in (0,1)$, which in turn implies that $t_0 = 0$; that is, all players exert positive efforts in equilibrium with positive probabilities. To put this more formally, there always exists some pair (a, γ_t) such that a period-t player would choose to actively compete in the contest. This observation can largely be attributed to the concavity of the ability distribution function $F(\cdot)$. Intuitively, with a concave distribution, an early player would expect a relatively *mild* competition ahead, as the future competitors are more likely to be of low ability, which encourages him to strive for a win as long as his ability is sufficiently high.

Second, there is no jump in a player's equilibrium bidding strategy once he decides to exert a positive effort: It is straightforward to see that the set $S_t(a;\gamma_t)$ degenerates to a single interval for all $a \in (a_t^*(\gamma_t), 1]$, as Figure 2(a) depicts. To understand the logic, let us consider a two-player sequential contest $(1,1)$ and focus on the first mover. The second mover would simply match the first mover's effort as long as his ability is above the early effort and remain inactive otherwise. Thus, the first mover's expected payoff of placing a bid b is $F(b) - b/a$. A concave ability distribution ensures that his expected payoff function is concave in the effort, which implies that he will *gradually* increase his effort as his ability increases and ensures the continuity.

Players' equilibrium bidding strategy sensitively depends on the concavity/curvature of the distribution function $F(\cdot)$, which is further evidenced by the case with convex ability distribution.

3.3.2 Convex Distribution

We now analyze the case with convex ability distribution.

Assumption 2 ***(Convex Ability Distribution)*** *The ability distribution function $F(\cdot)$ is continuous, twice differentiable, and weakly convex.*

Recall that $G_n(\cdot)$ is the inverse function of $aF^{n-1}(a)$. The following result is obtained, which characterizes the symmetric PBE in the sequential contest.

Proposition 3 ***(Convex Ability Distribution and Equilibrium Bidding Strategy)*** *Suppose that Assumption 2 is satisfied and consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$. In*

the unique symmetric PBE, $b_t^*(a;\gamma_t) = 0$ *for all* $a \in (0,1]$, $\gamma_t \in [0,1]$ *and* $t \in \{1,\ldots,T-1\}$, *and*

$$b_T^*(a;\gamma_T) = \begin{cases} 0, & \text{if } a \le G_{n_T}(\gamma_T), \\ \gamma_T + \int_{G_{n_T}(\gamma_T)}^{a} (n_T - 1) s f(s) F^{n_T - 2}(s) ds, & \text{if } a > G_{n_T}(\gamma_T). \end{cases}$$

With a (weakly) convex ability distribution, $H_t(b) \le b$ for all $b \in [0,1]$ and $t \in \{1,\ldots,T-1\}$, which indicates $t_0 = T-1$.[15] As a result, all players that arrive prior to period T would stay inactive in equilibrium regardless of their own actual abilities. Period-T players behave as if they participate in a simultaneous all-pay auction with n_T players.[16]

This observation stands in sharp contrast to that in the case with a concave ability distribution function, which reaffirms the fundamental role played by the curvature of distribution function in shaping equilibrium behavior . We continue with the two-player sequential contest $(1,1)$ to elaborate on the intuition. Recall that the first mover's expected payoff of placing a bid b is $F(b) - b/a$, which is convex with a convex ability distribution. Thus, the payoff can be maximized by either a zero or a sufficiently large effort. Further, a convex ability distribution implies more intense future competition, as high-ability players are likely to emerge in later periods. This discourages players' investment to avoid futile effort and the resultant loss. As a result, the players except those from the last period choose to drop out of the competition.

Analogous to the case with a concave distribution, however, discontinuity does not arise in equilibrium either, which is due to the strong discouragement effect caused by a convex ability distribution and resultant expectation of intense future competition.

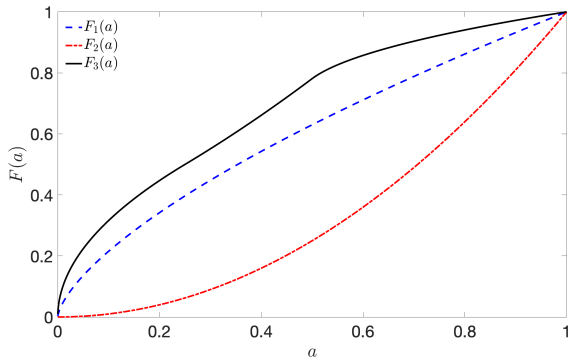
3.3.3 Curvature of Ability Distribution Function: Further Discussion

The sharp contrast between the two cases presented above testifies to the significant role played by the curvature of the distribution function. However, neither case yields the aforementioned discontinuity in equilibrium bidding strategies, and we interpret the observations based on the implications of concave/convex distribution functions. This naturally inspires us to conjecture that discontinuity in equilibrium bidding strategy may arise with a distribution that lacks such regularity, e.g., one to be concave in some regions and convex in the others. We provide an example to confirm this conjecture.

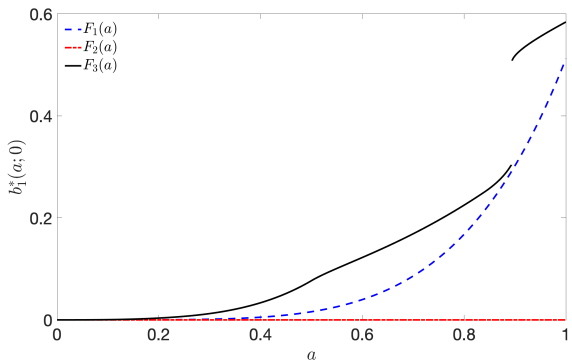
Example 1 ***(Equilibrium Bidding Strategy in Two-period Contests)*** *Consider the*

[15] The convexity of players' ability distribution assumed in Assumption 2 is indeed a sufficient condition; it can verified that Proposition 3 holds if the ability distribution satisfies $F(a) \le a$ for all $a \in [0,1]$.

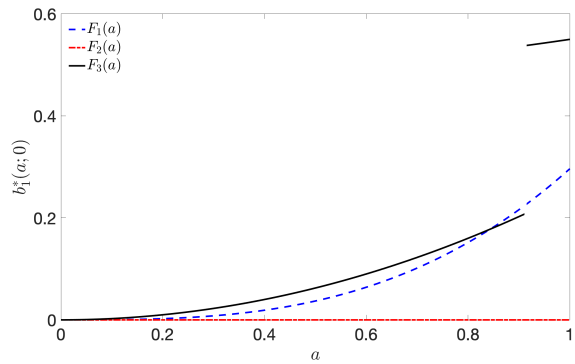
[16] If $n_T = 1$, the last player would bid the reserve price if his ability is greater than the reserve price and remains inactive otherwise in the equilibrium.



(a) CDF



(b) First mover's bidding strategy in contest $(1,1)$



(c) First mover's bidding strategy in contest $(2,1)$

Figure 3: First Mover's Bidding Strategy in Two-Period Sequential Contests under Different Ability Distributions.

following three ability distributions: (i) $F_1(a) = a^{2/3}$*; (ii)* $F_2(a) = a^2$*; and (iii)*

$$F_3(a) = \begin{cases} \sqrt{a}, & a < \frac{1}{4}, \\ \frac{(a+0.75)^2}{2}, & \frac{1}{4} \leq a \leq \frac{1}{2}, \\ 0.3837\sqrt{a - 0.4764} + 0.7224, & a > \frac{1}{2}. \end{cases}$$

Figure 3(a) depicts these CDFs. Note that $F_3(\cdot)$ *is convex in* a *on* $[\frac{1}{4}, \frac{1}{2}]$ *and concave on* $(0, \frac{1}{4})$ *and* $(\frac{1}{2}, 1]$*. Figure 3(b) illustrates the first mover's equilibrium bidding strategy a two-player sequential contest* $(n_1, n_2) = (1, 1)$ *under each distribution. A jump in the bidding function with respect to the first mover's ability,* a*, arises under* $F_3(\cdot)$*. Further, the discontinuity persists when an additional player is added into the first period, which yields contemporaneous competition* $(n_1 = 2)$*, as is evidenced by Figure 3(c).*

4 Later-Mover Advantage

In this section, we invoke the equilibrium results of this contest game and rank players' ex ante expected payoffs with respect to their timing positions. We first demonstrate a second-mover advantage in a simple two-player setting $(1, 1)$. We then briefly elaborate on the nuance and complexity when generalizing the model to a multi-player setting, and establish a general later-mover advantage under an arbitrary contest architecture.

4.1 Benchmark: Two-player Sequential Competition

Consider a two-player sequential-move contest with $\boldsymbol{n} = (1, 1)$. With slight abuse of notation, denote by by a_1 and a_2 the first and second movers' abilities, respectively. Similarly, denote the first and second movers' efforts by b_1 and b_2, respectively. By Theorem 1, the second mover, upon observing the first mover's effort b_1, either simply matches the earlier bid—i.e., $b_2 = b_1$—or stays inactive—i.e., $b_2 = 0$—if his ability is too low, in which case matching the earlier bid incurs excessive cost. Obviously, in the equilibrium, the first mover exerts an ex ante higher expected effort than the second mover because the latter player would not outbid the former and stays inactive with a positive probability.

Moreover, the first mover is ex ante less likely to win: He wins if and only if his late opponent's ability substantially falls below his own, such that the late opponent chooses to stay inactive. To see this, note that the first mover must bid below his own ability, i.e., $b_1 < a_1$; he otherwise ends up with negative expected payoff. Further, recall that the second mover would match b_1 and wins the competition as long as his ability exceeds the first mover's effort, i.e., $a_2 \geq b_1$. As a result, the second mover can win the contest even if his ability falls below that of his first-moving opponent, i.e., in the event of $a_2 \in [b_1, a_1]$. In contrast, he would lose the competition in a simultaneous-move contest: In a symmetric Bayesian Nash equilibrium, ex ante symmetric players adopts the same bidding strategy and win with an equal probability.

This rationale allows us to conclude a second-mover advantage in a two-player setting.

Remark 2 ***(Two-player Sequential All-pay Auctions)*** *Consider a two-player sequential all-pay auction with $\boldsymbol{n} = (1, 1)$. The expected equilibrium effort of the first mover is weakly higher than that of the second mover. Meanwhile, the expected winning probability of the first mover is strictly less than that of the second mover. As a result, the ex ante expected payoff of the first mover is strictly less than that of the second mover in the unique PBE.*

4.2 General Case: Multi-player Sequential Competition

In this part, we first show in an example that the clear-cut effort comparison obtained in the two-player setting does not extend to a multi-player setting in general, which nullifies the rationale laid out above for payoff comparison. Despite the complexity, we establish that ex ante expected payoff remains monotone with respect to players' timing position in a general setting.

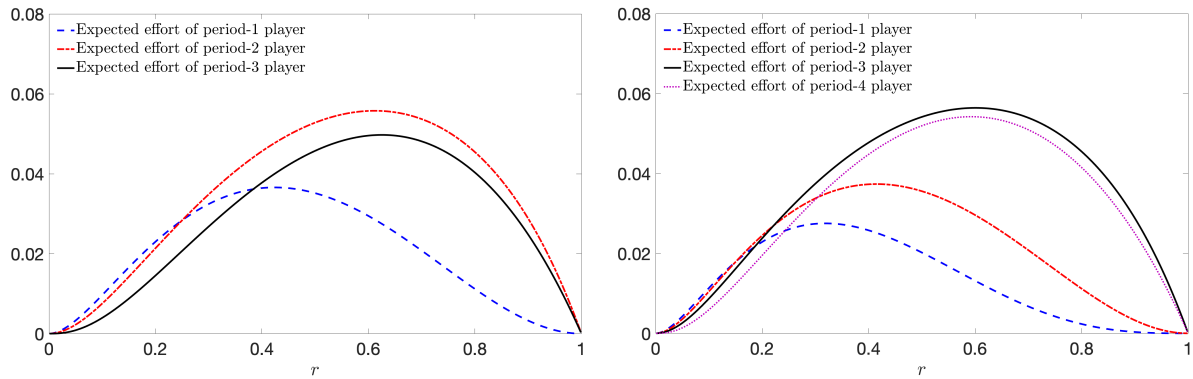
Ambiguous Effort Ranking Although Theorem 1 fully characterizes the equilibrium, a closed-form solution to the equilibrium bidding strategy is unavailable in general. We assume a distribution function of the form $F(a) = a^r$—with $r \in (0,1)$—and a fully sequential architecture with $\boldsymbol{n} = (1, \ldots, 1)$. This case enables a handy solution, which suffices to demonstrate the nuance caused by a multi-player contest setting. Note that a larger r implies a more *favorable* distribution, as it causes the probability mass to concentrate more on the upper end of support and competent players are more likely to realize. Let b_t^* be a period-t player's bid in equilibrium and $\mathbb{E}[b_t^*]$ be its expectation.

Example 2 ***(Ranking Players' Efforts in Fully Sequential Contests)*** *Suppose $F(a) = a^r$, with $r \in (0,1)$. Consider an N-player fully sequential contest, with $\boldsymbol{n} = (1, \ldots, 1)$ and $T = N$. The following statements hold:*

(i) If $N = 2$, then $\mathbb{E}[b_1^*] > \mathbb{E}[b_2^*]$ for all $r \in (0,1)$.

(ii) If $N \geq 3$, then $\mathbb{E}[b_1^*] > \mathbb{E}[b_2^*] > \cdots > \mathbb{E}[b_T^*]$ when r is sufficiently close to 0, and $\mathbb{E}[b_{T-1}^*] > \mathbb{E}[b_T^*] > \mathbb{E}[b_{T-2}^*] > \cdots > \mathbb{E}[b_1^*]$ when r is sufficiently close to 1.

Example 2(i) reiterates the observation of Remark 2 in this parameterized setting: The first mover is expected to exert more effort than the second mover in a two-player sequential contest $(1,1)$. Example 2(ii) demonstrates that the monotonicity of equilibrium efforts with respect to the order of players' moves may not persist when more than two players are involved in the contest. Specifically, when r is close to 1, the *penultimate* mover exerts the highest expected effort. To understand the logic, let us consider a three-player fully sequential contest $(1,1,1)$ and focus on the second mover. Fix an arbitrary period-1 effort b_1^*, the second mover has no incentive to exert an effort strictly higher than b_1^* in the absence of a third mover. However, anticipating a third player in the final period, matching b_1^* is no longer necessarily optimal even if a_2 exceeds b_1^*; by Proposition 2(i), he may exert a strictly higher effort than b_1^* once his ability a_2 surpasses the cutoff $a_2^{**}(b_1)$ to avoid being overtaken by the third player. It is thus straightforward to infer that the case of $\mathbb{E}[b_2^*] > \mathbb{E}[b_1^*]$ is more likely as r ascends, which implies that high-ability players are more likely to emerge and



(a) Contest $(1,1,1)$: $T = N = 3$ (b) Contest $(1,1,1,1)$: $T = N = 4$

Figure 4: Players' Expected Efforts in Fully Sequential All-pay Auctions.

compels him to step up his effort. It is also intuitive to observe $\mathbb{E}[b_2^*] > \mathbb{E}[b_3^*]$. The last player behaves in essentially the same way as the second mover in the two-player case: He either match $\gamma_3 = \max\{b_1^*, b_2^*\}$ or stay inactive, since no future competition is ahead.

Figure 4 depicts players' individual efforts for all values of $r \in (0,1)$, assuming $N = 3$ or 4. By Figure 4(a), three patterns arise as r ascends: (i) $\mathbb{E}[b_1^*] > \mathbb{E}[b_2^*] > \mathbb{E}[b_3^*]$; (ii) $\mathbb{E}[b_2^*] > \mathbb{E}[b_1^*] > \mathbb{E}[b_3^*]$; and (iii) $\mathbb{E}[b_2^*] > \mathbb{E}[b_3^*] > \mathbb{E}[b_1^*]$. The comparison is further complicated when the number of players increases from three to four, as Figure 4(b) illustrates. Six patterns emerge as r increases: (i) $\mathbb{E}[b_1^*] > \mathbb{E}[b_2^*] > \mathbb{E}[b_3^*] > \mathbb{E}[b_4^*]$; (ii) $\mathbb{E}[b_2^*] > \mathbb{E}[b_1^*] > \mathbb{E}[b_3^*] > \mathbb{E}[b_4^*]$; (iii) $\mathbb{E}[b_2^*] > \mathbb{E}[b_3^*] > \mathbb{E}[b_1^*] > \mathbb{E}[b_4^*]$; (iv) $\mathbb{E}[b_3^*] > \mathbb{E}[b_2^*] > \mathbb{E}[b_1^*] > \mathbb{E}[b_4^*]$; (v) $\mathbb{E}[b_3^*] > \mathbb{E}[b_2^*] > \mathbb{E}[b_4^*] > \mathbb{E}[b_1^*]$; and (vi) $\mathbb{E}[b_3^*] > \mathbb{E}[b_4^*] > \mathbb{E}[b_2^*] > \mathbb{E}[b_1^*]$.

Before closing this part, it is noteworthy that the sequential contest involves additional subtlety when there exist multiple players acting simultaneously in certain period. Effort ranking across periods would be substantially complicated in a partially sequential all-pay auction.[17]

Unambiguous Payoff Ranking We now formally address the primary research question of this section: Does a player benefit from being an early/later mover? An answer to this question requires that we compare players' expected payoffs with respect to the order of moves. Despite the complexity in effort comparison shown in Example 2, we verify that players' equilibrium expected payoffs can be *unambiguously* and *monotonically* ranked.[18]

Let Π_t^* denote a period-t player's equilibrium expected payoff, with $t \in \mathcal{T}$, in a sequential

[17]Numerical result for four-player three-period sequential contests–i.e., $(N,T) = (4,3)$—are available from the authors upon request.

[18]It is useful to point out that the payoff comparison in a sequential complete-information Tullock lottery contest must coincide with the effort comparison, as stated in Hinnosaar (2021a): Player i's equilibrium payoff $x_i^*(1/X^* - 1)$—where x_i^* player i's equilibrium effort and X^* is the equilibrium aggregate effort—is proportional to his individual effort.

contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$. Recall that $\mathcal{T}_0 \equiv \{t \in \mathcal{T} : H_t(b) \leq b, \forall b \in [0,1]\}$ indicates the set of the periods in which players always choose to remain inactive in equilibrium, with $t_0 \equiv \max \mathcal{T}_0$. The following result can be obtained.

Theorem 2 ***(Later-mover Advantage in Sequential All-pay Auctions)*** *Consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$. A player's expected payoff is higher than those of all earlier movers in the unique symmetric PBE. Formally,* $0 = \Pi_1^* = \cdots = \Pi_{t_0}^* < \Pi_{t_0+1}^* < \cdots < \Pi_T^*$.

Recall that $t_0 = 0$ under Assumption 1—i.e., a concave ability distribution—and $t_0 = T - 1$ under Assumption 2, i.e., a convex ability distribution. The following result can immediately be obtained.

Corollary 1 ***(Later-mover Advantage with Concave/Convex Ability Distributions)*** *Consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$. The following statements hold in the unique symmetric PBE:*

(i) Suppose that Assumption 1 is satisfied. Then $0 < \Pi_1^* < \cdots < \Pi_T^*$.

(ii) Suppose that Assumption 2 is satisfied. Then $\Pi_1^* = \cdots = \Pi_{T-1}^* = 0 < \Pi_T^*$.

Theorem 2 and Corollary 1 formally establish later-mover advantage in a multi-player sequential all-pay auction. We sketch the proof as follows. For ease of exposition, let us consider a fully sequential contest, with $N = T$. Recall that the profile of equilibrium bidding strategies is denoted by $\boldsymbol{b}^* := \left\{b_1^*(a;\gamma_1), \ldots, b_T^*(a;\gamma_T)\right\}$. Fix an arbitrary period $\tau \in \{t_0 + 1, \ldots, T-1\}$, we conduct the following thought experiment. We modify period-$(\tau+1)$ player's bidding strategy from $b_{\tau+1}^*(a;\gamma_{\tau+1})$ to

$$b_{\tau+1}^\dagger(a;\gamma_{\tau+1}) := b_\tau^*(a;\gamma_\tau).$$

In other words, period-$(\tau+1)$ player ignores a period-τ player' effort and replicates the latter player's equilibrium strategy (*not his bid*). Denote players' expected payoffs under the constructed strategy profile by $\boldsymbol{b}^\dagger := \left\{b_1^*(a;\gamma_1), \ldots, b_\tau^*(a;\gamma_\tau), b_{\tau+1}^\dagger(a;\gamma_{\tau+1}), b_{\tau+2}^*(a;\gamma_{\tau+2}), \ldots, b_T^*(a;\gamma_T)\right\}$ by $(\Pi_1^\dagger, \ldots, \Pi_T^\dagger)$.

The key step of the proof is to show that a period-τ player would be strictly better off once period-$(\tau+1)$ player changes his equilibrium strategy from $b_{\tau+1}^*(a;\gamma_{\tau+1})$ to $b_{\tau+1}^\dagger(a;\gamma_{\tau+1})$, i.e., $\Pi_\tau^* < \Pi_\tau^\dagger$. The intuition is as follows. A later mover, ceteris paribus, tends to be more aggressive in competition than an earlier mover because the former needs to beat a smaller number of future opponents than the latter, which encourages a later mover to strive for a win. Thus, when the period-$(\tau+1)$ player deviates from his equilibrium strategy and

replicates his immediate predecessor's strategy, the former player would be less likely to outperform the latter. This obviously benefits the period-τ player and leads to an increase in his expected payoff.

To fix ideas, consider a period-τ player, with ability $a_\tau > a_\tau^{**}(\gamma_\tau)$, for a given γ_τ. By Theorem 1, he actively anticipates the contest and exerts an effort that is strictly above γ_τ. When the period-$(\tau+1)$ player replicates period-τ player's strategy, he can defeat the period-τ player if and only if he is of a higher type, which occurs with probability $1-F(a_\tau)$. Under the equilibrium strategy profile, in contrast, the period-$(\tau+1)$ player outperforms the period-τ player as long as the former chooses to stay active; he does so whenever his ability exceeds the threshold $a_{\tau+1}^*(\gamma_{\tau+1}) = a_{\tau+1}^*(b_\tau^*(a_\tau;\gamma_\tau))$, which occurs with a probability of $1-F(a_{\tau+1}^*(b_\tau^*(a_\tau;\gamma_\tau)))$. We formally show in the Appendix that $a_\tau > a_{\tau+1}^*(b_\tau^*(a_\tau;\gamma_\tau))$:[19] In other words, the period-$(\tau+1)$ player behaves less aggressively when he replicates his immediate predecessor's equilibrium strategy.

To complete the proof, first note that $\Pi_{\tau+1}^\dagger \leq \Pi_{\tau+1}^*$ by the definition of PBE. We further have $\Pi_\tau^\dagger \leq \Pi_{\tau+1}^\dagger$ by the construction of $b_{\tau+1}^\dagger = b_\tau^*$. Combining these inequalities yields $\Pi_\tau^* < \Pi_\tau^\dagger \leq \Pi_{\tau+1}^\dagger \leq \Pi_{\tau+1}^*$, which concludes that period-$(\tau+1)$ player receives a higher equilibrium payoff than period-τ player.

5 Further Discussion of First-price Models

In this section, we discuss three variations to our baseline model within the framework of first-price contest models. We first consider a symmetric tie-breaking rule. We then examine an ex post complete-information model in which players' abilities are publicly revealed upon realization. Finally, we introduce a model with a hybrid payment rule that allows losers to pay a fraction of their effort costs. We show that the later-mover advantage established in the baseline model remains intact.

5.1 Symmetric Tie-breaking Rule

The tie-breaking rule adopted in the winner-selection mechanism (1) guarantees the existence of a symmetric PBE. As discussed in Remark 1, the rule literally "favors" later movers, which may cast doubt on the robustness of our result, as this sparks the concern that the later-mover advantage established in Theorem 2 is an artifact of the asymmetric tie-breaking rule.

Next, we consider a symmetric tie-breaking rule and demonstrate that Theorem 1 is robust to this variation. Before we proceed, it deserves to highlight that the unique symmetric

[19] We show in the proof of Theorem 2 that $a_\tau > a_{\tau+1}^*(b_\tau^*(a_\tau;\gamma_\tau))$ holds for an arbitrary contest architecture.

PBE characterized in Theorem 1 is entirely independent of the prevailing tie-breaking rule if the sequential contest involves at least two or more players in each period—i.e., $n_t \geq 2$ for all $t \in \mathcal{T}$.

Specifically, we assume that a contestant i's probability of winning the contest given the effort profile $\boldsymbol{b} \equiv (b^1, \ldots, b^N)$ is

$$p^{i,s}(\boldsymbol{b}) := \begin{cases} 1, & \text{if } b^i > \max_{\mathcal{N}\setminus\{i\}}\{b^j\}, \\ 1/m, & \text{if } b^i \text{ is among the } m \text{ highest of } \{b^j\}_{j\in\mathcal{N}} \text{ with a tie}, \\ 0, & \text{if } b^i < \max_{\mathcal{N}\setminus\{i\}}\{b^j\}, \end{cases} \tag{9}$$

where the superscript s indicates "symmetric tie-breaking rule." By (9), when multiple players exert the same highest effort, they win the contest with equal probabilities regardless of their arrival time. A PBE may fail to exist in the sequential contest game if there exists some $t \in \mathcal{T}$ with $n_t = 1$, which calls for an alternative equilibrium notion. We adopt the notion of ϵ-equilibrium firstly proposed by Radner (1980).[20] Intuitively, it requires that a player deviate only when this deviation leads to a substantially higher expected payoff; and the parameter $\epsilon > 0$ captures how much of payoff improvement is required to trigger a deviation.

Mailath, Postlewaite, and Samuelson (2005) introduce the notion of contemporaneous perfect ϵ-equilibrium in dynamic games with complete information, which is stronger than Radner's equilibrium notion; Jackson, Rodriguez-Barraquer, and Tan (2012) define interim and ex-ante ϵ-equilibrium in static games with incomplete information. Given that the contest we consider is a dynamic game with incomplete information, their definitions can naturally be combined and extended to define an ϵ-equilibrium in our setting.

Fix a player $i \in \mathcal{N}_t$ and an effort profile $(b^j)_{j\in\cup_{k=1}^{t-1}\mathcal{N}_k}$. With a symmetric tie-breaking rule (9), the player's strategy depends not only on his ability and the highest prior effort $\gamma_t \equiv \max_{j\in\cup_{k=1}^{t-1}\mathcal{N}_k}\{b^j\}$, but also on the number of players who chose γ_t, which we denote by z_t. For notational convenience, denote, respectively, by $b^{i,s}(a;\gamma_t,z_t)$ a player i's strategy, with $i \in \mathcal{N}_t$, and $\pi^{i,s}(b,a;\gamma_t,z_t)$ his interim expected payoff when choosing effort b for a given ability a and his opponents' strategy profile $\left\{b^{j,s}(a;\gamma_t,z_t)\right\}_{j\in\mathcal{N}_t, t\in\mathcal{T}, j\neq i}$.

Definition 1 ***(ϵ-equilibrium)*** *Fix $\epsilon > 0$. A strategy profile $\boldsymbol{b}^s \equiv \left\{b^{i,s}(a;\gamma_t,z_t)\right\}_{i\in\mathcal{N}_t, t\in\mathcal{T}}$ constitutes an ϵ-equilibrium of a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ under a symmetric tie-breaking rule* (9)*, if for all $i \in \mathcal{N}_t$, with $t \in \mathcal{T}$, and any history associated with $\gamma_t \in [0,1]$ and $z_t \geq 1$,*

$$\pi^{i,s}(b^{i,s}\left(a;\gamma_t,z_t\right), a;\gamma_t,z_t) \geq \pi^{i,s}\left(b', a;\gamma_t,z_t\right) - \epsilon,$$

[20] For applications of ϵ-optimality see, e.g., Baye and Morgan (2004), Milgrom (2010), and Martin and Schlag (2020).

for all $a \in (0,1]$ *and* $b' \geq 0$.

Definition 1 requires that given any history summarized by (γ_t, z_t), a player's equilibrium strategy be an ϵ-best response after his ability realizes.

Proposition 4 ***(ϵ-equilibrium in Sequential Contests with a Symmetric Tie-Breaking Rule)*** *Consider a sequential contest* $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$. *Fix an arbitrary* $\epsilon > 0$ *and let*

$$b^{i,s}(a; \gamma_t, z_t) := \begin{cases} b_t^*(a; \gamma_t) + a\epsilon \mathbb{1}_{\{a_t^*(\gamma_t) < a \leq a_t^{**}(\gamma_t)\}}, & \text{if } n_t = 1, \\ b_t^*(a; \gamma_t), & \text{if } n_t \geq 2, \end{cases} \tag{10}$$

for all $i \in \mathcal{N}_t$, *with* $t \in \mathcal{T}$. *The strategy profile* $\boldsymbol{b}^s \equiv \left\{b^{i,s}(a; \gamma_t, z_t)\right\}_{i \in \mathcal{N}_t, t \in \mathcal{T}}$ *constitutes a symmetric* ϵ*-equilibrium of the contest game under a symmetric tie-breaking rule as specified in* (9).

The equilibrium construction in Proposition 4 is intuitive. For periods that have multiple players—i.e., $n_t \geq 2$—the alternative tie-breaking rule does not play a critical role because players, by Theorem 1(ii), would never match the highest prior effort. As a result, players' strategies do not depart from the counterpart in the PBE. For periods that have a unique player—i.e., $n_t = 1$—we slightly increase a player's effort whenever he wants to match the highest effort from previous periods under (1) to ensure that he can beat all his predecessors and will not share the prize with them.

By Proposition 4, the ϵ-equilibrium we construct under (9) largely reinstates the prediction of Theorem 2 for sufficiently small ϵ. As a result, all our results obtained under the original tie-breaking rule remain qualitatively intact. Let $\Pi_t^{*,s}$ denote a period-t player's equilibrium expected payoff in the symmetric ϵ-equilibrium described in Proposition 4, with $t \in \mathcal{T}$. The following can formally be stated.

Theorem 3 ***(Later-mover Advantage with a Symmetric Tie-Breaking Rule)*** *Consider a sequential contest* $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ *under a symmetric tie-breaking rule as specified in* (9). *For sufficiently small* $\epsilon > 0$, *a player's expected payoff is higher than that of all earlier movers in the symmetric* ϵ*-equilibrium* $\boldsymbol{b}^s \equiv \left\{b^{i,s}(a; \gamma_t, z_t)\right\}_{i \in \mathcal{N}_t, t \in \mathcal{T}}$, *where* $b^{i,s}(a; \gamma_t, z_t)$ *is defined in* (10). *Formally,* $0 = \Pi_1^{*,s} = \cdots = \Pi_{t_0}^{*,s} < \Pi_{t_0+1}^{*,s} < \cdots < \Pi_T^{*,s}$.

5.2 Contests with Ex Post Complete Information

The later-mover advantage in Theorem 2 is established under the assumption that a player's ability is privately known. Next, we show that Theorem 2 is robust to the prevailing information structure. Specifically, we consider an ex post complete-information variant to

the baseline model: The ability profile $\boldsymbol{a} := (a^1, \ldots, a^N)$ is publicized upon its realization and thus becomes commonly known before all players act.

We first characterize the subgame perfect equilibrium (SPE) of the dynamic contest with complete information given an arbitrary ability profile $\boldsymbol{a} \equiv (a^1, \ldots, a^N)$. Without loss of generality, we focus on a case in which players' abilities all differ from each other, i.e., $a^j \neq a^k$ for $j \neq k$.[21] For the ease of notation, let a_t^{max} and a_t^{sec}—with $t \in \mathcal{T}$—denote the highest and second highest ability levels in period t, respectively.[22] Further, define $\underline{b}_t := \max\{\max_{j \in \cup_{\ell > t} \mathcal{N}_\ell}\{a^j\}, \gamma_t\}$, $\mu_t^{max} := \underline{b}_t / a_t^{sec}$, and $\mu_t^{sec} := (a_t^{max} - a_t^{sec} + \underline{b}_t)/a_t^{max}$.[23] The following result can be obtained.

Proposition 5 ***(Equilibrium in Sequential All-pay Auctions with Complete Information)*** *Consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ under a tie-breaking rule as specified in* (1) *and fix an arbitrary ability profile $\boldsymbol{a} \equiv (a^1, \ldots, a^N)$, with $a^j \neq a^k$ for $j \neq k$.*[24,25] *There exists a unique Subgame Perfect Equilibrium (SPE) of the contest game. The equilibrium strategy of player $i \in \mathcal{N}_t$, with $t \in \mathcal{T}$ is given as follows.*

(i) Suppose that $a^i \leq \max\{a_t^{sec}, \underline{b}_t\}$. The player always bids zero effort regardless of previous maximum bid $\gamma_t \geq 0$.

(ii) Suppose that $a^i > \underline{b}_t$ and $a^i = a_t^{max}$. Player i bids $\underline{b}_t$ if $a_t^{sec} \leq \underline{b}_t$. If $a_t^{sec} > \underline{b}_t$, he bids $\underline{b}_t$ with probability μ_t^{max} and mixes uniformly over $[\underline{b}_t, a_t^{sec}]$ with complementary probability $1 - \mu_t^{max}$.

(iii) Suppose that $a^i > \underline{b}_t$ and $a^i = a_t^{sec}$. Player i bids 0 with probability μ_t^{sec} and mixes uniformly over $[\underline{b}_t, a_t^{sec}]$ with complementary probability $1 - \mu_t^{sec}$.

Baye, Kovenock, and De Vries (1993, 1996) characterize the equilibria of simultaneous-move first-price all-pay auctions. They predict that only the two strongest players would remain active with positive probabilities in equilibrium and each adopts a mixed strategy. Proposition 5 generalizes their results to accommodate the possibility of sequential moves. In line with Baye, Kovenock, and De Vries (1993, 1996), at most two players remain active, and randomization may arise in equilibrium. The strongest player always stay active, as in a simultaneous-move contest. The equilibrium in the sequential setting, however, also differs from its simultaneous-move counterpart in several respects, which deserve to be highlighted.

[21] The condition holds almost surely given that the distribution $F(\cdot)$ is assumed to be atomless.

[22] Define $a_t^{sec} = 0$ if $n_t = 1$.

[23] Two remarks are in order. First, we define $\max_{j \in \cup_{\ell > t} \mathcal{N}_\ell}\{a^j\} = 0$ for $t = T$. Second, μ_t^{max} will not be used in Proposition 5 when $a_t^{sec} = 0$, or equivalently, when $n_t = 1$.

[24] To be more rigorous, we need to specify the ability profile in each period when we define a sequential contest.

[25] It is useful to point out that the equilibrium characterization in Proposition 5 does not depend on the tie-breaking rule and constitutes the unique SPE under the symmetric tie-breaking rule (9).

First, the second strongest player may not be active. By Proposition 5, all players stay inactive with probability one if they act before the strongest player does—i.e., bidding zero effort if $a^i \leq \max\{a_t^{sec}, \underline{b}_t\}$—on the equilibrium path. Instead, the player of the highest ability competes head to head with his most competent *contemporaneous or future* opponent.

Second, by Proposition 5(ii), the strongest player may adopt a pure strategy—i.e., exerting an effort $\underline{b}_t$—in the sequential contest, whereas he always randomizes in the simultaneous-move counterpart. He does so whenever his only effective opponent—i.e., his most competent *contemporaneous or future* opponent—appears in one of the later periods, i.e., $a_t^{sec} \leq \underline{b}_t$.

Third, players in a simultaneous-move contest randomizes over an interval starting from zero. In contrast, Proposition 5(ii) and (iii) predict that when mixed strategies are played, their efforts are randomized over an interval from $\underline{b}_t$ to preempt future competitors. The variable $\underline{b}_t$—shaped jointly by the observable earlier efforts summarized by the parameter γ_t and the threat from future opponents encapsulated by the maximum of their abilities—is reminiscent of a reserve price in a static auction.

Fixing an ability profile $\boldsymbol{a} \equiv (a^1, \ldots, a^N)$, denote by $\pi^{i,c}(\boldsymbol{a})$ player i's equilibrium payoff in the unique SPE, where the superscript c indicates "complete information." By Proposition 5, we can obtain the following result.

Proposition 6 ***(Ex Post Equilibrium Payoff with Complete Information)*** *Consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ under a tie-breaking rule as specified in* (1)*. Fix an arbitrary ability profile $\boldsymbol{a} \equiv (a^1, \ldots, a^N)$, with $a^j \neq a^k$ for $j \neq k$. For player $i \in \mathcal{N}_t$, $\pi^{i,c}(\boldsymbol{a}) = 0$ if $a^i \neq \max_{j \in \mathcal{N}}\{a^j\}$ and $\pi^{i,c}(\boldsymbol{a}) = 1 - (\max_{j \in \cup_{\ell \geq t} \mathcal{N}_\ell \setminus \{i\}}\{a^j\})/a^i$ if $a^i = \max_{j \in \mathcal{N}}\{a^j\}$.*

By Proposition 6, only the strongest player ends up with a positive payoff in the equilibrium, and his payoff depends on the ability of his strongest opponent in the current and future periods. The proposition enables us to derive and compare players' ex ante expected equilibrium payoffs across different periods, denoted by $(\Pi_1^{*,c}, \ldots, \Pi_T^{*,c})$.

Theorem 4 ***(Late-mover Advantage in Complete-information All-pay Auctions)*** *Consider a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ under a tie-breaking rule as specified in* (1)*. Suppose that players' ability is publicized before the contest begins. Then a player's expected payoff is strictly higher than that of all earlier movers in equilibrium, i.e., $\Pi_1^{*,c} < \cdots < \Pi_T^{*,c}$.*

Theorem 4 predicts a strict later-mover advantage under complete information. The intuition is as follows. Consider a player of ability $a \in (0, 1)$. He obtains a positive payoff if and only if his ability turns out to be the highest among all. Note that the probability of this event amounts to $F^{N-1}(a)$ regardless of his timing position. Further, by Proposition 6, the player's expected payoff depends crucially on the expected ability of his strongest opponent

in the current and future periods; the later he moves, the fewer opponents he needs to outperform, and thus the lower the expected highest ability of his contemporaneous and future competitors. A later-mover advantage ensues.

5.3 Hybrid Payment Rule for Losers

In this part, we show that the later-mover advantage result does not rely on the all-pay feature and would hold for a broad class of contest games. Specifically, we consider a sequential contest, in which a contestant's winning probability is again given by (1); the associated payment rule is specified as follows: The winner in the contest is obliged to pay the full cost of his own effort, while a loser only needs to pay a fraction $\theta \in [0,1]$ of it.[26,27] To put this formally, fixing a contestant i's ability a^i and the effort profile $\boldsymbol{b} \equiv (b^1, \ldots, b^N)$, his ex post expected payoff is

$$p^i(\boldsymbol{b})(1 - b^i/a^i) - [1 - p^i(\boldsymbol{b})]\theta b^i/a^i, \text{for all } i \in \mathcal{N}.$$

The above expression degenerates to (2) in the baseline setting as $\theta = 1$, and the contest game turns into a *first-price auction* in which players mover sequentially as $\theta = 0$. In words, this contest involves a hybrid payment rule that involves both winner-pay and all-pay elements.

Fixing a contest architecture $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ and $\theta \in [0,1]$, a sequence of functions $\{H_t(b;\theta), a_t^*(\gamma;\theta), \tilde{\pi}_t(b,a;\theta)\}_{t=1}^T$ in parallel with (3), (4), and (5) can be defined recursively as follows:[28]

$$H_T(b;\theta) \equiv 1, H_{t-1}(b;\theta) := H_t(b;\theta)F^{n_t}\left(a_t^*(b;\theta)\right), \forall b \in [0,1], \tag{11}$$

$$a_t^*(\gamma;\theta) := \max\{0 < a \leq 1 : \tilde{\pi}_t(b,a;\theta) \leq 0, \forall b \geq \gamma\}, \tag{12}$$

$$\tilde{\pi}_t(b,a;\theta) := H_t(b;\theta)F^{n_t-1}(a)\left[1 - (1-\theta)b/a\right] - \theta b/a. \tag{13}$$

With slight abuse of notation, let $\mathcal{T}_0(\theta) := \{t \in \mathcal{T} : H_t(b) \leq \frac{\theta b}{1-b+\theta b}, \forall b \in [0,1]\}$ and define $t_0(\theta) := \max \mathcal{T}_0(\theta)$. Again, we can obtain $0 \leq t_0(\theta) \leq T-1$. Further, let $\Pi_t^*(\theta)$ denote a period-t player's equilibrium expected payoff, with $t \in \mathcal{T}$.

Analogous to Section 3.2, a unique symmetric PBE exists, which can be fully characterized as in Theorem 1. Further, the later-mover advantage established in Theorem 2 persists

[26] See Amann and Leininger (1996) and Baye, Kovenock, and De Vries (2005, 2012) for similar parameterization.

[27] Note that a bid $b \in (0, \gamma_t)$ always leads to a loss and is suboptimal to a period-t player for $\theta > 0$. In contrast, when $\theta = 0$, bidding $b \in (0, \gamma_t)$ is strategically equivalent to bidding zero because a bid does not incur a cost to a loser. In this case, we impose the restriction that period-t players bid 0 or weakly above γ_t without any loss of generality when characterizing the symmetric PBE.

[28] We add θ into $\{H_t(b), a_t^*(\gamma), \tilde{\pi}_t(b,a)\}_{t=1}^T$ to highlight the fact that the defined sequence of functions depends on θ.

under this generalized setting. These results are formally summarized as follows.

Theorem 5 ***(Later-mover Advantage with a Generalized Payment Rule for Losers)*** *Fix $\theta \in [0,1]$ and consider a generalized sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ under a tie-breaking rule as specified in* (1)*. There exists a unique symmetric PBE of the contest game. Moreover, a player's expected payoff is higher than that of all earlier movers in equilibrium. Formally,* $0 = \Pi_1^*(\theta) = \cdots = \Pi_{t_0(\theta)}^*(\theta) < \Pi_{t_0(\theta)+1}^*(\theta) < \cdots < \Pi_T^*(\theta)$.

We again establish a later-mover advantage in the contest with a hybrid payment rule that departs from a standard all-pay contest.

6 Extension: Second-price Models

Thus far, we have assumed a first-price contest setting, such that the winner bears the cost of his own effort. Next, we examine an alternative model in which the winner pays for the cost of the second-highest effort, while losers pay for their own, which are similar in spirit to the war-of-attrition model in the literature (e.g., Krishna and Morgan, 1997; Bulow and Klemperer (1999); Hafer, 2006; Bergemann, Brooks, and Morris, 2019). We demonstrate that winner's payment rule plays a significant role for earlier- or later-mover advantage.

6.1 Equilibrium Characterization in Second-price Models

We first characterize the equilibrium of the contest game. Following the literature, we restrict our attention to symmetric PBE (e.g., Bulow and Klemperer, 1999).[29] We focus on a tractable setting to ease our analysis. Define $\hat{t} := \max\{t \leq T-1 : n_t \geq 2\}$ if there exists some period $t \leq T-1$ such that $n_t \geq 2$. Recall that $G_n(\cdot)$ is the inverse function of $aF^{n-1}(a)$. The following assumption is imposed to simplify the analysis.

Assumption 3 *Fix a sequential contest $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$. Either $n_t = 1$ for all $t \leq T-1$ or $F^{n_T}\big(G_{n_T}(a)\big)F^{T-\hat{t}-1}(a) \leq a$ for all $a \in [0,1]$.*

Assumption 3 is satisfied if the ability distribution satisfies $F(a) \leq a$ for all $a \in [0,1]$.[30]

[29] Multiple equilibria arise even in a static setting. Asymmetric equilibria include those in which one player places a bid equal or greater than one and the remaining players stay inactive and bid zero.

[30] To see this, note that $F^{n_T}\left(G_{n_T}(a)\right)F^{T-\hat{t}-1}(a) \leq F^{n_T}\left(G_{n_T}(a)\right) \leq G_{n_T}(a)F^{n_T-1}\left(G_{n_T}(a)\right) = a$ for all $a \in [0,1]$, where the first inequality follows from $F^{T-\hat{t}-1}(a) \leq 1$, the second inequality from $F(a) \leq a$, and the equality from the definition of $G_{n_T}(\cdot)$.

To proceed, we further define $a_t^\star(\gamma_t)$ as

$$a_t^\star(\gamma_t) := \begin{cases} \gamma_t, & \text{if } n_t = 1, \gamma_t < 1, \\ \text{the unique solution to } (a_t^\star - \gamma_t + 1)F^{n_t-1}(a_t^\star) = 1, & \text{if } n_t \geq 2, \gamma_t < 1, t \leq T-1, \\ G_{n_t}(\gamma_t), & \text{if } n_t \geq 2, \gamma_t < 1, t = T, \\ 1, & \text{if } n_t \geq 2 \text{ or } \gamma_t \geq 1. \end{cases} \tag{14}$$

The following result can then be obtained.

Proposition 7 ***(Equilibrium Characterization of Second-price Sequential Contests)*** *Suppose that Assumption 3 is satisfied, and consider a sequential second-price all-pay auction* $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ *under a tie-breaking rule as specified in* (1)*. The sequential contest possesses a unique symmetric PBE—which we denote by* $\{b_t^\star(a;\gamma_t)\}_{t=1}^T$*—and is characterized as follows:*[31]

(i) If $n_t = 1$*, then*

$$b_t^\star(a;\gamma_t) = \begin{cases} 0, & \text{if } a \leq a_t^\star(\gamma_t), \\ 1, & \text{if } a > a_t^\star(\gamma_t). \end{cases}$$

(ii) If $n_t \geq 2$ *and* $t \leq T-1$*, then*

$$b_t^\star(a;\gamma_t) = \begin{cases} 0, & \text{if } a \leq a_t^\star(\gamma_t), \\ 1 + \int_{a_t^\star(\gamma_t)}^a \frac{sdF^{n_t-1}(s)}{1-F^{n_t-1}(s)}, & \text{if } a > a_t^\star(\gamma_t). \end{cases}$$

(iii) If $n_t \geq 2$ *and* $t = T$*, then*

$$b_t^\star(a;\gamma_t) = \begin{cases} 0, & \text{if } a \leq a_t^\star(\gamma_t), \\ \gamma_t + \int_{a_t^\star(\gamma_t)}^a \frac{sdF^{n_t-1}(s)}{1-F^{n_t-1}(s)}, & \text{if } a > a_t^\star(\gamma_t). \end{cases}$$

By Proposition 7, a period-t player's equilibrium bidding strategy $b_t^\star(a;\gamma_t)$ only depends on the number of players in the current period but not those in the others. In period T, player have no future opponents and would behave as if they were placed in a simultaneous-move war of attribution with a reserve price $\gamma_t \geq 0$, as Proposition 7(iii) shows. In other periods, a player either remains inactive or overbids—i.e., exerting a preemptive effort above 1 to deter future opponents—as described in Proposition 7(i) and (ii). The incentive to *overbid* is an artifact of the second-price feature of the contest game: One pays less than

[31] Equilibrium uniqueness is established under the assumption that a period-t player, with $n_t = 1$ is restricted to choose his bid from the interval $[0, 1]$. This is without loss of generality given that any bid above 1 is strictly dominated by a bid of zero if $\gamma_t > 1$ and generates the same payoff to the player if $\gamma_t \leq 1$.

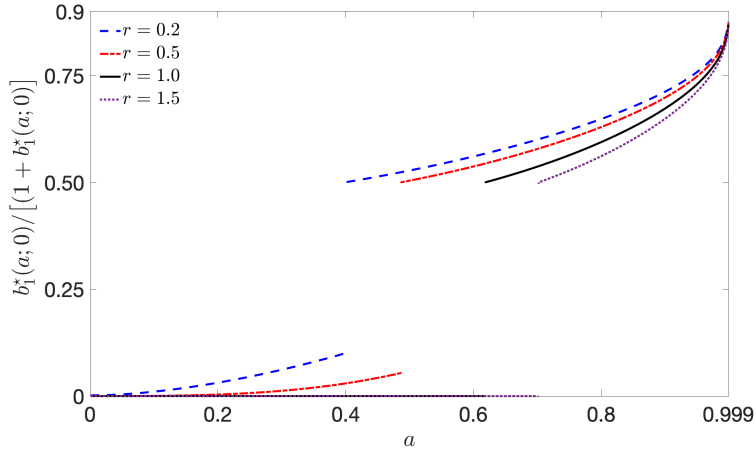


Figure 5: Period-1 Players' Equilibrium Bid in Two-period Second-price All-pay Auctions.

his effort cost as long as he wins, which compels him to step up his effort to avoid being overtaken by future opponents if he is sufficiently confident of a win, i.e., with a sufficiently high ability. Obviously, future players would be entirely discouraged in response to an effort above 1, which leads an active player to ignore the number of opponents in future periods in his bidding strategy.

The polarized bidding strategy of players prior to the last period stems from Assumption 3, as well as the second-price feature of the contest game, and notably simplifies our analysis. To understand the logic, recall that a sufficient condition for the assumption to hold is $F(a) \leq a$ for all $a \in [0,1]$, which implies that high-ability players are more likely to be drawn. This alludes to more intense competition and entices players of relatively low abilities to stay inactive to avoid a loss.

Next, we provide a simple three-player example to demonstrate that equilibrium effort may lie between 0 and 1 in equilibrium if Assumption 3 is violated, i.e., when competition remains mild.

Example 3 *Suppose $F(a) = a^r$ with $r > 0$, and consider a two-stage second-price all-pay auction with $\boldsymbol{n} = (2,1)$. Assumption 3 can be simplified as $r \geq 1$, in which case the equilibrium characterization in Proposition 7 applies. It suffices to focus on the case with $r \in (0,1)$. In the unique symmetric PBE, the period-2 player's equilibrium bidding strategy is $b_2^\star(a;\gamma_2) = \mathbb{1}_{\{a>\gamma_2\}}\gamma_2$, and the period-1 player's equilibrium bidding strategy is*

$$b_1^\star(a;\gamma_1) = \begin{cases} \mathcal{B}(a), & \text{if } 0 < a \leq a_1^\star, \\ 1 + \int_{a_1^\star}^{a} \frac{rs^r}{1-s^r} ds, & \text{if } a_1^\star < a < 1, \end{cases}$$

where $\mathcal{B}(a)$ is governed by the following differential equation:

$$ra^r\left[\mathcal{B}(a)\right]^r + ra^{r+1}\left[\mathcal{B}(a)\right]^{r-1}\mathcal{B}'(a) - r^2\left[\mathcal{B}(a)\right]^{r-1}\mathcal{B}'(a)\int_0^a \mathcal{B}(s)s^{r-1}ds$$
$$-\mathcal{B}'(a) + (r+1)a^r\left[\mathcal{B}(a)\right]^r\mathcal{B}'(a) = 0,$$

with the initial condition $\mathcal{B}(0) = 0$, and the threshold $a_1^\star$ is the unique solution to

$$(a^\star)^{r+1}\left[\mathcal{B}(a^\star)\right]^r - r\left[\mathcal{B}(a^\star)\right]^r\int_0^{a^\star}\mathcal{B}(s)s^{r-1}ds - \left\{1 - \left[a^\star\mathcal{B}(a^\star)\right]^r\right\}\mathcal{B}(a^\star)$$
$$=(a^\star)^{r+1} - r\int_0^{a^\star}\mathcal{B}(s)s^{r-1}ds - (1-a^\star).$$

Figure 5 depicts a period-1 player's equilibrium bidding strategy for $r \in \{0.2, 0.5, 1.0, 1.5\}$. We observe that an effort between 0 and 1 would arise in the equilibrium if $a < a^\star(0)$ for $r \in \{0.2, 0.5\}$.

Two remarks are in order before we move on. First, a period-1 player tends to behave more aggressively—i.e., increasing his effort at all ability levels—when competitions are muted by a smaller r, as Figure 5 shows. A low-ability player is encouraged to take his chance, which leads him to exert effort in the interval $(0, 1)$. A preemptive effort—i.e., $b_1^\star(a; 0) > 1$—is less likely to fall short in contemporaneous and future competitions, which also encourages a high-ability player. Second, as in the equilibrium under Assumption 3, period-1 players' equilibrium bidding strategies are discontinuous in the case of $r \leq 1$, although they may bid between 0 and 1. The discontinuity, as explained above, primarily stems from the second-price payment rule, which gives rise to the incentive to overbid.

6.2 Payoff Comparison in Second-price Models

We are now ready to compare players' equilibrium payoffs across different timing positions, assuming that Assumption 3 is satisfied. Let $\Pi_t^\star$ denote a period-t player's equilibrium expected payoff, with $t \in \mathcal{T}$, in a sequential war of attrition $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$.

Two-player Sequential Competition In parallel with the analysis in Section 4.1, we first consider a two-player sequential-move war of attrition with $\boldsymbol{n} = (1, 1)$. It is straightforward to verify that Assumption 3 is satisfied for all atomless ability distributions $F(\cdot)$. On the equilibrium path, the first mover would place a bid above or equal to 1 regardless of his ability and the second mover would bid 0 by Proposition 7(i). The prize would be distributed to the first mover with certainty; moreover, the first mover's payment is zero. The following result naturally ensues.

Remark 3 ***(Two-player Sequential War of Attrition)*** *Consider a two-player sequential war of attrition with* $\boldsymbol{n} = (1, 1)$*. Then* $\Pi_1^\star = 1 > 0 = \Pi_2^\star$*.*

Remark 3 stands in sharp contrast to Remark 2 and predicts an ambiguous first-mover advantage in a war of attrition. Comparing Remarks 2 and 3 in two-player settings, we can conclude that winner's payment rule plays a decisive role in determining earlier- or later-mover advantage. As shown above, the players in a second-price contest is enticed to overbid, as one is not fully responsible for the cost of his own effort, which compels the first mover to preempt and, in turn, leads the second to concede. In contrast, the players in a first-price model must "underbid"—i.e., exerting effort below his own ability—because he bears his own effort cost regardless of win or loss, which discourages the earlier mover.

Multi-player Sequential Competition We now proceed to the multi-player setting as in Section 4.2 and demonstrate that the rationale laid out above does not fully capture the subtlety in contests with more general architecture.

It is straightforward to infer from Proposition 7 that if $n_\tau = 1$ for some period $\tau \in \mathcal{T}$, then $\Pi_t^\star = 0$ for all $t > \tau$. It thus suffices to restrict our attention to the case with $n_t \geq 2$ for all $t \leq T - 1$. Note that $\hat{t} \equiv \max\{t \leq T - 1 : n_t \geq 2\} = T - 1$ and Assumption 3 is equivalent to $F(a) \leq a$ for all $a \in [0, 1]$. The following can be obtained.

Proposition 8 ***(Earlier-mover and Later-mover Advantage in Two-stage War of Attrition)*** *Suppose that* $F(a) \leq a$ *for all* $a \in [0, 1]$*, and consider a two-stage sequential war of attrition* $\boldsymbol{n} = (n_1, n_2)$*, with* $n_1 \geq 2$*. The following statements hold:*

(i) Fixing n_1*, there exists a threshold* $\bar{n}_2$ *such that* $\Pi_1^\star > \Pi_2^\star$ *if* $n_2 > \bar{n}_2$*.*

(ii) Fixing n_2*, there exists a threshold* $\bar{n}_1$ *such that* $\Pi_2^\star > \Pi_1^\star$ *if* $n_1 > \bar{n}_1$*.*

According to Proposition 8, the payoff comparison between earlier and later movers depends sensitively on the prevailing contest architecture (n_1, n_2). By Proposition 7(ii), a period-1 player's equilibrium strategy depends only on n_1 but not n_2. Note that a period-1 player never exerts an effort in the interval $(0, 1)$. He secures a payoff of zero if he exerts zero effort. He preempts future opponents if he exerts an effort more than 1 regardless of n_2. As a result, his ex ante expected payoff $\Pi_1^\star$ only depends on his contemporaneous competitions—i.e., n_1—and is independent of n_2. An increase in n_2, however, intensifies period-2 competition and tends to decrease $\Pi_2^\star$, despite its irrelevance to $\Pi_1^\star$. An earlier-mover advantage thus ensues when n_2 is sufficiently large for a given n_1, as stated in Proposition 8(i).

Next, consider the implications of an increase in n_1. Intuitively, a period-1 players' ex ante payoff would decrease as n_1 ascends, which is bounded from above by $1/n_1$ and

approaches zero as n_1 goes to infinity. By Proposition 7(iii), a period-2 player's expected payoff is 0 if $\gamma_2 > 1$ and is a positive constant that depends on n_2 if $\gamma_2 = 0$. Therefore, holding fixed n_2, a period-2 player's equilibrium payoff is proportional to the probability of all period-1 players' choosing to remain inactive—i.e., $F^{n_1}(a_1^\star(0))$—in which case the period-2 competition boils down to a standard all-pay auction. By (14), $a_1^\star(0)$ is the unique solution to $\left[a_1^\star(0)+1\right]F^{n_1-1}(a_1^\star(0)) = 1$, from which we can obtain $\lim_{n_1\nearrow\infty} a_1^\star(0) = 1$ and $\lim_{n_1\nearrow\infty} F^{n_1}(a_1^\star(0)) = \lim_{n_1\nearrow\infty} F(a_1^\star(0))/\left[a_1^\star(0)+1\right] = 1/2$. To summarize, although an increase n_1 affects a period-2 player's equilibrium payoff, its impact is bounded and would diminish as n_1 continues to increase. A later-mover advantage is thus expected when n_1 becomes sufficiently large, as predicted in Proposition 8(ii).

The above analysis indicates that players' equilibrium payoffs in a sequential second-price all-pay auction (i.e., war of attrition) may not be *monotone* with respect to their timing positions when we allow for three or more stages. The conjecture is formalized and confirmed by the following result. For notational convenience, we write the period-t player's equilibrium payoff as $\Pi_t^\star(\boldsymbol{n})$ to highlight its dependence on the contest architecture $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$.

Proposition 9 ***(Nonmonotonicity of Equilibrium Payoffs in Multi-stage War of Attrition)*** *Suppose that $F(a) \leq a$ for all $a \in [0,1]$ and consider a multi-stage sequential war of attrition $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$, with $T \geq 3$ and $n_t \geq 2$ for $t \leq T-1$. Fixing a period $\tau \in \mathcal{T}$, $\Pi_\tau^\star(\boldsymbol{n}) < \min_{t\neq\tau} \Pi_t^\star(\boldsymbol{n})$ if n_τ is sufficiently large.*

7 Concluding Remarks

In this paper, we conduct a general analysis of an incomplete-information sequential contest in the form of (first-price) all-pay auctions. Our model allows for a flexible architecture, such that multiple players can be clustered in a single period: They move simultaneously within the period, while observing earlier efforts and anticipating future competitions. Our analysis fully characterizes the equilibrium, assuming a general ability distribution, and demonstrates how the curvature of the distribution functions may shape equilibrium strategies. The analysis adds to the contest literature, as a general analysis of sequential contests remains scarce.

Based on our equilibrium analysis, we examine players' expected payoffs with respect to their timing positions in the sequential game. Despite the complications in the multi-player contest settings, we formally establish a later-mover advantage in the sequential contest game. Namely, a player secures a strictly higher ex ante expected payoff when he is assigned to a later timing position vis-à-vis an earlier one, despite that players' ex ante expected efforts can be nonmonotone with respect to the order of moves. We demonstrate the result

is robust to the variations in terms of tie-breaking rule, information structure, and losers' payment rule. We further investigate a second-price model such that the winner pays the cost of the second-highest effort. We demonstrate that players' ex ante expected payoffs can be nonmonotone with respect to the order of moves. Our analysis sheds light on the voluminous literature regarding earlier-/later-mover advantage in sequential games and provides an alternative perspective to understand the nuance.

A large room for extensions remains. In our setting, players' timing positions are fixed. It would be interesting to extend the analysis to allow endogenously formed timing structure. In addition, we consider a winner-take-all setting. A model of multiple prizes with sequential moves deserves serious scholarly effort.

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Appendix: Proofs

Proof of Lemmas 1 and 2

Proof. It is useful to prove several intermediate results.

Lemma 5 *The following statements hold:*

(i) $a_t^*(\gamma)$ *is continuous, piecewise differentiable, and weakly increasing on* $[0,1]$*, satisfying* $a_t^*(\gamma) \geq \gamma$ *for all* $t \in \mathcal{T}$*; and*

(ii) $H_t(b)$ *is continuous, piecewise differentiable, and strictly increasing on* $[0,1]$*, with* $H_t(0) = 0$ *and* $H_t(1) = 1$ *for all* $t \in \mathcal{T} \setminus \{T\}$*.*

Proof. We prove the lemma by induction. Note that piecewise analyticity implies piecewise differentiability. Therefore, to show that $a_t^*(\gamma)$ and $H_t(b)$ are piecewise differentiable, it suffices to show that they are piecewise analytic. Recall from the main text that $G_n(\cdot)$ is defined as the inverse function of $aF^{n-1}(a)$ for an arbitrary positive integer $n \in \mathbb{N}_+$.

Base case: Note that $G_n(\cdot)$ is strictly increasing, piecewise analytic, and differentiable on $[0,1]$, with $G_n(0) = 0$ and $G_n(1) = 1$. By definition, $H_T(b) = 1$. Therefore, $\tilde{\pi}_T(b,a) \equiv H_T(b)F^{n_T-1}(a) - b/a = F^{n_T-1}(a) - b/a$. These facts, together with (4), imply that

$$a_T^*(\gamma) := \max\left\{0 < a \leq 1 : \tilde{\pi}_T(b,a) \leq 0, \forall b \in [\gamma, 1]\right\} = G_{n_T}(\gamma)$$

and $H_{T-1}(b) = F^{n_T}\big(G_{n_T}(b)\big)$. It is straightforward to verify that $a_T^*(\gamma)$ satisfies part (i) of the lemma and $H_{T-1}(b)$ satisfies part (ii).

Inductive step: Suppose that $H_t(b)$ satisfies part (ii) of the lemma for some $t \leq T-1$. It suffices to show that $a_t^*(\gamma)$ satisfies part (i) of the lemma and $H_{t-1}(b)$ satisfies part (ii).

Fixing $b \in (0,1]$, $\tilde{\pi}_t(b,a)$ strictly increases with $a \in (0,1)$. Define $\breve{a}_t(b)$ as follows:

$$\breve{a}_t(b) := \begin{cases} G^{n_t}\left(\min\left\{\frac{1}{H_t'(0)}, 1\right\}\right), & \text{if } b = 0, \\ 1, & \text{if } b \in (0,1] \text{ and } \tilde{\pi}_t(b,1) < 0, \\ \text{the unique solution to } \tilde{\pi}_t(b,a) = 0, & \text{otherwise.} \end{cases}$$

It can be verified that $a_t^*(\gamma) = \min_{b \geq \gamma} \breve{a}_t(b)$ and $\breve{a}_t(b)$ is continuous on $[0,1]$. This in turn implies that $a_t^*(\gamma)$ is continuous, piecewise analytic, and weakly increasing on $[0,1]$. Further, for $b \geq \gamma$, we have

$$\tilde{\pi}_t(b,\gamma) = F^{n_t-1}(\gamma)H_t(b) - b/\gamma \leq 0,$$

which indicates that $\gamma \in \{0 < a \leq 1 : \tilde{\pi}_t(b, a) \leq 0, \forall b \geq \gamma\}$ and thus $a_t^*(\gamma) \geq \gamma$. To summarize, $a_t^*(\gamma)$ satisfies part (i) of the lemma.

Because $H_t(b)$ satisfies part (ii) of the lemma by assumption and $a_t^*(\gamma)$ satisfies part (i), we can conclude that $H_{t-1}(b) = H_t(b)F^{n_t}\big(a_t^*(b)\big)$ satisfies part (ii). This completes the inductive step.

Conclusion: By the principle of induction, $a_t^*(\gamma)$ satisfies part (i) of Lemma 5 for all $t \in \mathcal{T}$ and $H_t(b)$ satisfies part (ii) for all $t \in \mathcal{T} \setminus \{T\}$. This concludes the proof. ■

Lemma 6 *The following statements hold for all $t \in \mathcal{T}$.*

(i) If $a_t^*(\gamma) < a < 1$, then there exists $b \in [\gamma, 1]$ such that $\tilde{\pi}_t(b, a) > 0$.

(ii) If $0 < a < a_t^*(\gamma)$, then $\tilde{\pi}_t(b, a) < 0$ for all $b \in [\gamma, 1] \setminus \{0\}$.

Proof. Part (i) of the lemma is obvious and it remains to prove part (ii). Fix $0 < a < a_t^*(\gamma)$. Suppose, to the contrary, that $\tilde{\pi}_t(b_0, a) \geq 0$ for some $b_0 \in [\gamma, 1] \setminus \{0\}$. It follows immediately that $a \geq b_0$. Further, we have that $\tilde{\pi}_t(b_0, a_t^*(\gamma)) > \tilde{\pi}_t(b_0, a) \geq 0$, which contradicts with the fact that $\tilde{\pi}_t(b_0, a_t^*(\gamma)) \leq 0$ for all $b \in [\gamma, 1]$. This completes the proof. ■

Now we can prove Lemmas 1 and 2 by induction.

Base case: Consider the last period, i.e., $t = T$. It is evident that the realized highest effort by the end of period T wins the contest with certainty. By definition, $H_T(b) = 1$. Therefore, Lemma 2 holds for $t = T$ and it remains to show that Lemma 1 holds for the last period. We consider the following two cases:

(a) Suppose $n_T = 1$. Then the optimal bidding strategy of the unique period-T player is to bid γ_T if $a > \gamma_T$ and bid 0 otherwise. Therefore, Lemma 1(i) and (ii) hold.

(b) Suppose $n_T \geq 2$. We first show that $b_T^*(a; \gamma_T)$ is increasing in a. Suppose, to the contrary, that there exists an ability pair (a', a''), with $0 < a' < a'' < 1$, such that $b'' := b_T^*(a''; \gamma_T) < b' := b_T^*(a'; \gamma_T)$. Denote the equilibrium winning probability of bidding b by $\mathcal{WP}_T^*(b)$. It is obvious that $\mathcal{WP}_T^*(b'') < \mathcal{WP}_T^*(b')$; otherwise, a type-$a'$ player has a strict incentive to bid b''. Moreover, from players' incentive compatibility constraints, we have that

$$\mathcal{WP}_T^*(b')a' - b' \geq \mathcal{WP}_T^*(b'')a' - b'', \text{ and } \mathcal{WP}_T^*(b'')a'' - b'' \geq \mathcal{WP}_T^*(b')a'' - b',$$

which is equivalent to

$$a'\left[\mathcal{WP}_T^*(b') - \mathcal{WP}_T^*(b'')\right] \geq b' - b'', \text{ and } a''\left[\mathcal{WP}_T^*(b') - \mathcal{WP}_T^*(b'')\right] \leq b' - b''.$$

Combining the above inequalities yields

$$(a' - a'') \times \left[\mathcal{WP}_T^*(b') - \mathcal{WP}_T^*(b'')\right] \geq 0,$$

which is a contradiction given that $\mathcal{WP}_T^*(b'') < \mathcal{WP}_T^*(b')$ and the postulated $a' < a''$.

Let $\bar{a}_T := \inf\{a : b_T^*(a;\gamma_T) > 0\}$. We first show that $b_T^*(a;\gamma_T)$ strictly increases with a for $a > \bar{a}_T$. Suppose, to the contrary, that $\bar{a}_T < a' < a'' < 1$ and $b' := b_T^*(a';\gamma_T) = b'' := b_T^*(a'';\gamma_T)$. It follows immediately that $b_T^*(a;\gamma_T) = b'$ for $a \in [a', a'']$. Then a type-a' player has an incentive to deviate from bidding b'. Specifically, he can raise his effort by an infinitesimal amount to substantially increase his winning probability, which leads to an increase in his interim expected payoff. A contradiction.

Further, note that $b_T^*(a;\gamma_T) \geq \gamma_T$ for $a > \bar{a}_T$ and $b_T^*(a;\gamma_T) = 0$ for $a \leq \bar{a}_T$, and it thus remains to prove that $\bar{a}_T = a_T^*(\gamma_T) \equiv \max\left\{0 < a \leq 1 : \tilde{\pi}_T(b,a) \leq 0, \forall b \in [\gamma_T, 1]\right\}$, where $\tilde{\pi}_T(b,a) \equiv H_T(b)F^{n_T-1}(a) - b/a$. We consider the following two cases:

(i) Suppose that $\bar{a}_T < a_T^*(\gamma_T)$. Consider a type-$a'$ player, with $\bar{a}_T < a' < a_T^*(\gamma_T)$. Recall that $b_T^*(a;\gamma_T)$ strictly increases with a for $a > \bar{a}_T$. Therefore, we have $b_T^*(a';\gamma_T) > 0$. His equilibrium expected payoff is

$$F^{n_T-1}(a') - \frac{b_T^*(a';\gamma_T)}{a'} = \tilde{\pi}_T\left(b_T^*(a';\gamma_T), a'\right) < 0,$$

where the strict inequality follows from $b_T^*(a';\gamma_T) \neq 0$, $b_T^*(a';\gamma_T) \geq \gamma_T$, and Lemma 6(ii). However, he can secure a nonnegative expected payoff by bidding zero. A contradiction.

(ii) Suppose that $\bar{a}_T > a_T^*(\gamma_T)$. Fix $a' \in \left(a_T^*(\gamma_T), \bar{a}_T\right)$. It follows immediately from $a' < \bar{a}_T$ that $b_T^*(a';\gamma_T) = 0$. Note that bidding zero must generate zero expected payoff to a type-a' player. Otherwise, we must have $\gamma_T = 0$; together with $\bar{a}_T > 0$, we can conclude that a player whose type falls below $\bar{a}_T$ can strictly increase his expected payoff by exerting an infinitesimal amount of effort. A contradiction.

By Lemma 6(i), there exists some $b' \in [\gamma_T, 1]$ such that $\tilde{\pi}_T(b', a') > 0$. Then type-$a'$ player's expected payoff of bidding b' is bounded from below by

$$F^{n_T-1}(a') - \frac{b'}{a'} = \tilde{\pi}_T(b', a') > 0.$$

Therefore, a type-a' player has a strict incentive to deviate from exerting zero effort, which is a contradiction.

Inductive step: Suppose that the equilibrium bidding strategy $b_t^*(a;\gamma_t)$ satisfies the properties stated in Lemma 1 and $H_t(b)$ gives the probability of the effort b's exceeding all subsequent efforts in equilibrium, as predicted in Lemma 2, for some $t \leq T$. We show that the same holds for period $t-1$.

Suppose that the realized highest effort by the end of period $t-1$ is b. Then the probability of the effort b's exceeding all subsequent efforts in equilibrium is $H_t(b)F^{n_t}\big(a_t^*(b)\big)$, which is exactly $H_{t-1}(b)$ from (3).

For the case of $n_{t-1} = 1$, note that the problem of the only period-$(t-1)$ player with ability a is $\max_{b \geq \gamma_{t-1}}[H_{t-1}(b) - b/a]$. It is then straightforward to verify that (i) $b_{t-1}^*(a;\gamma_{t-1})$ is increasing in a on $(0,1)$; and (ii) $b_{t-1}^*(a;\gamma_{t-1}) = 0$ for $a \leq a_{t-1}^*(\gamma_{t-1})$ and $b_t^*(a;\gamma_{t-1}) \geq \gamma_{t-1}$ for $a > a_{t-1}^*(\gamma_{t-1})$. For the case of $n_{t-1} \geq 2$, by the same argument as in the base case, we can show that $b_{t-1}^*(a,\gamma_{t-1})$ satisfies all properties stated in Lemma 1. This completes the inductive step.

Conclusion: By the principle of induction, $b_t^*(a;\gamma_t)$ satisfies all properties stated in Lemma 1 for all $t \in \mathcal{T}$. Moreover, $H_t(b)$ gives the probability of the effort b's exceeding all subsequent efforts in equilibrium for all $t \in \mathcal{T}$, as predicted in Lemma 2. This concludes the proof. ■

Proof of Lemma 3

Proof. It is useful to prove the following intermediate result.

Lemma 7 *Suppose that* $a_t^*(\gamma_t) < 1$. *Then* $\tilde{\pi}_t\big(s_t^1(a_t^*(\gamma_t);\gamma_t), a_t^*(\gamma_t)\big) = 0$.

Proof. Evidently, $s_t^1(a_t^*(\gamma_t);\gamma_t) \geq \gamma_t$; together with the definition of $a_t^*(\gamma_t)$, we can obtain $\tilde{\pi}_t\big(s_t^1(a_t^*(\gamma_t);\gamma_t), a_t^*(\gamma_t)\big) \leq 0$. Suppose, to the contrary, that $\tilde{\pi}_t\big(s_t^1(a_t^*(\gamma_t);\gamma_t), a_t^*(\gamma_t)\big) \neq 0$. Then we must have

$$\tilde{\pi}_t\big(s_t^1(a_t^*(\gamma_t);\gamma_t), a_t^*(\gamma_t)\big) < 0.$$

The above inequality, together with the fact that $s_t^1\big(a_t^*(\gamma_t);\gamma_t\big) \in S_t(a;\gamma_t)$, implies that

$$\tilde{\pi}_t\big(b', a_t^*(\gamma_t)\big) < \tilde{\pi}_t\big(s_t^1(a_t^*(\gamma_t);\gamma_t), a_t^*(\gamma_t)\big) < 0, \text{ for all } s_t^1\big(a_t^*(\gamma_t);\gamma_t\big) < b' \leq 1. \tag{15}$$

Next, note that by definition, $s_t^1\big(a_t^*(\gamma_t);\gamma_t\big)$ is the smallest element in the set $S_t(a;\gamma_t)$. Therefore, we have that

$$\tilde{\pi}_t\big(b', a_t^*(\gamma_t)\big) \leq \tilde{\pi}_t\big(s_t^1(a_t^*(\gamma_t);\gamma_t), a_t^*(\gamma_t)\big) < 0, \text{ for all } \gamma_t \leq b' \leq s_t^1\big(a_t^*(\gamma_t);\gamma_t\big). \tag{16}$$

Combining (15) and (16), $\tilde{\pi}_t(b', a_t^*(\gamma_t) + \epsilon) < 0$ for all $b' \in [\gamma_t, 1]$ for sufficiently small $\epsilon > 0$, which contradicts the definition of $a_t^*(\gamma_t)$ and concludes the proof. ■

Now we can prove Lemma 3. Suppose, to the contrary, that $n_t \geq 2$, $a_t^*(\gamma_t) < 1$, and $\lim_{a \searrow a_t^*(\gamma_t)} b_t^*(a;\gamma_t) \neq s_t^1(a_t^*(\gamma_t);\gamma_t)$. We consider the following two cases:

(a) Suppose that $\lim_{a \searrow a_t^*(\gamma_t)} b_t^*(a;\gamma_t) < s_t^1(a_t^*(\gamma_t);\gamma_t)$. Then for sufficiently small $\epsilon > 0$, we have $b_t^*(a;\gamma_t) < s_t^1\big(a_t^*(\gamma_t);\gamma_t\big)$ for all $a < a_t^*(\gamma_t) + \epsilon$. Consider a type-$a_t^*(\gamma_t)$ player. His expected payoff of bidding $s_t^1\big(a_t^*(\gamma_t);\gamma_t\big)$ is at least

$$\begin{aligned}
&H_t\left(s_t^1(a_t^*(\gamma_t);\gamma_t)\right) F^{n_t-1}\left(a_t^*(\gamma_t)+\epsilon\right) - \frac{s_t^1(a_t^*(\gamma_t);\gamma_t)}{a_t^*(\gamma_t)} \\
>&H_t\left(s_t^1(a_t^*(\gamma_t);\gamma_t)\right) F^{n_t-1}\left(a_t^*(\gamma_t)\right) - \frac{s_t^1(a_t^*(\gamma_t);\gamma_t)}{a_t^*(\gamma_t)} \\
=&\tilde{\pi}_t\left(s_t^1\big(a_t^*(\gamma_t);\gamma_t\big), a_t^*(\gamma_t)\right) = 0,
\end{aligned}$$

where the equality follows from Lemma 7. Meanwhile, it follows from Lemma 1(ii) that a type-$a_t^*(\gamma_t)$ player would bid 0 and thus earns zero expected payoff in equilibrium. A contradiction.

(b) Suppose that $\lim_{a \searrow a_t^*(\gamma_t)} b_t^*(a;\gamma_t) > s_t^1(a_t^*(\gamma_t);\gamma_t)$. Consider a player whose type is $a' = a_t^*(\gamma_t) + \epsilon$ for sufficiently small $\epsilon > 0$. His expected payoff of bidding $b' = b_t^*(a';\gamma_t)$ is $\tilde{\pi}_t(b', a')$. It follows from the postulated $\lim_{a \searrow a_t^*(\gamma_t)} b_t^*(a;\gamma_t) > s_t^1(a_t^*(\gamma_t);\gamma_t)$ and the definition of $S_t(a;\gamma)$ that

$$\tilde{\pi}_t\left(\lim_{a \searrow a_t^*(\gamma_t)} b_t^*(a;\gamma_t), a_t^*(\gamma_t)\right) < \tilde{\pi}_t\left(s_t^1\big(a_t^*(\gamma_t);\gamma_t\big), a_t^*(\gamma_t)\right) = 0,$$

where the equality again follows from Lemma 7. By continuity, $\tilde{\pi}_t(b', a') < 0$ for sufficiently small $\epsilon > 0$. Therefore, a type-a' player can secure a strictly higher expected payoff by exerting zero effort, which is a contradiction. This concludes the proof.

■

Proof of Lemma 4

Proof. As mentioned in the text, we focus on the case where $S_t(a;\gamma)$ can be expressed as the union of finitely many disjoint intervals. Fix some type $\tilde{a} \in (a_t^*(\gamma_t), 1]$ such that $b_t^*(\tilde{a} - 0;\gamma_t) = e_t^m(\tilde{a};\gamma_t)$ for some $1 \leq m \leq m_t(\tilde{a};\gamma_t) - 1$. Then we have that (see Figure 1)

$$\tilde{\pi}_t\left(e_t^m(\tilde{a};\gamma_t), \tilde{a}\right) = \tilde{\pi}_t\left(s_t^{m+1}(\tilde{a};\gamma_t), \tilde{a}\right). \tag{17}$$

Suppose, to the contrary, that $b_t^*(\tilde{a} + 0;\gamma_t) \neq s_t^{m+1}(\tilde{a};\gamma_t)$. We consider the following two cases:

(a) Suppose that $b_t^*(\tilde{a}+0;\gamma_t) < s_t^{m+1}(\tilde{a};\gamma_t)$. Then there exists $\epsilon > 0$ such that $b_t^*(a;\gamma_t) < s_t^{m+1}(\tilde{a};\gamma_t)$ for all $a < \tilde{a}+\epsilon$. Consider a player whose ability is $\tilde{a}-\epsilon'$ for sufficiently small $\epsilon' > 0$. His expected payoff of bidding $s_t^{m+1}(\tilde{a};\gamma_t)$ is no less than

$$\begin{aligned} F^{n_t-1}(\tilde{a}+\epsilon)H_t\left(s_t^{m+1}(\tilde{a};\gamma_t)\right) - \frac{s_t^{m+1}(\tilde{a};\gamma_t)}{\tilde{a}-\epsilon'} >& F^{n_t-1}(\tilde{a})H_t\left(s_t^{m+1}(\tilde{a};\gamma_t)\right) - \frac{s_t^{m+1}(\tilde{a};\gamma_t)}{\tilde{a}} \\ =& \tilde{\pi}_t\left(s_t^{m+1}(\tilde{a};\gamma_t),\tilde{a}\right) \\ =& \tilde{\pi}_t\left(e_t^m(\tilde{a};\gamma_t),\tilde{a}\right), \end{aligned}$$

where the second equality follows from (17). Note that $b_t^*(\tilde{a}-0;\gamma_t) = e_t^m(\tilde{a};\gamma_t)$ and thus $\tilde{\pi}_t(e_t^m(\tilde{a};\gamma_t),\tilde{a})$ is the limit of player's equilibrium expected payoff as ϵ' approaches 0. Therefore, the player can obtain a strictly higher payoff by bidding $s_t^{m+1}(\tilde{a};\gamma_t)$, which is a contradiction.

(b) Suppose that $b_t^*(\tilde{a}+0;\gamma_t) > s_t^{m+1}(\tilde{a};\gamma_t)$. Consider a player whose type is $\tilde{a}+\epsilon'$ for sufficiently small $\epsilon' > 0$. Note that his equilibrium expected payoff of bidding $b_t^*(\tilde{a}+\epsilon';\gamma_t)$ can then be bounded from above by

$$\tilde{\pi}_t\left(b_t^*(\tilde{a}+\epsilon';\gamma_t),\tilde{a}+\epsilon'\right) < \tilde{\pi}_t\left(s_t^{m+1}(\tilde{a};\gamma_t),\tilde{a}\right) = \tilde{\pi}_t\left(e_t^m(\tilde{a};\gamma_t),\tilde{a}\right),$$

where the inequality follows from the definition of $S_t(a;\gamma)$ and the equality from (17). Meanwhile, his expected payoff of bidding $b_t^*(\tilde{a}-0;\gamma_t)$ is

$$F^{n_t-1}(\tilde{a})H_t\left(e_t^m(\tilde{a};\gamma_t)\right) - \frac{e_t^m(\tilde{a};\gamma_t)}{\tilde{a}+\epsilon'} \geq \tilde{\pi}_t\left(e_t^m(\tilde{a};\gamma_t),\tilde{a}\right).$$

Therefore, the player has a strict incentive to deviate from his equilibrium bid $b_t^*(\tilde{a}+\epsilon';\gamma_t)$, which is a contradiction.

■

Proof of Theorem 1

Proof. We consider the following two cases:

(a) Suppose $n_t = 1$. It is evident that $b_t^*(a;\gamma_t) = 0$ for $a \leq a_t^*(\gamma_t)$ and $b_t^*(a;\gamma_t)$ solves

$$\max_{b\geq\gamma_t}\left[H_t(b) - b/a\right]$$

for $a > a_t^*(\gamma_t)$. Further, let $a_t^{**}(\gamma_t) := \sup_{a_t^*(\gamma_t) \le a \le 1} \{a : H_t(\gamma_t) - \gamma_t/a > H_t(b') - b'/a, \forall b' \in (\gamma_t, 1]\}$. It follows immediately that

$$\frac{b' - \gamma_t}{a} > \frac{b' - \gamma_t}{a_t^{**}(\gamma_t)} \ge H_t(b') - H_t(\gamma_t), \text{ for all } a_t^*(\gamma_t) \le a < a_t^{**}(\gamma_t) \text{ and } b' \in (\gamma_t, 1],$$

which in turn implies that $b_t^*(a; \gamma_t) = \gamma_t$ for when the player's ability a lies between $a^*(\gamma_t)$ and $a^{**}(\gamma_t)$. To summarize, period-t player's equilibrium bidding strategy for the case of $n_t = 1$ is characterized by (7) in part (i) of the theorem.

(b) Suppose that $n_t \ge 2$. For $a \le a_t^*(\gamma_t)$, it follows immediately from Lemma 1(ii) that $b_t^*(a; \gamma_t) = 0$. For $a > a_t^*(\gamma_t)$, we have that

$$b_t^*(a; \gamma_t) \in \arg\max_{b > \gamma_t} \left[H_t(b) F^{n_t - 1}\left((b_t^*)^{-1}(b; \gamma_t) \right) - b/a \right].$$

This implies that

$$a \in \arg\max_{\check{a} > a_t^*(\gamma_t)} \check{\pi}_t(\check{a}, a; \gamma_t) := H_t\left(b_t^*(\check{a}; \gamma_t) \right) F^{n_t - 1}(\check{a}) - b_t^*(\check{a}; \gamma_t)/a.$$

Suppose that $b_t^*(a; \gamma_t)$ is continuous in some interval $\mathcal{U}_{\tilde{a}} = (\tilde{a}, \tilde{a} + \epsilon)$. In the equilibrium, the following first-order condition should be satisfied:

$$\left. \frac{\partial \check{\pi}_t(\check{a}, a; \gamma_t)}{\partial \check{a}} \right|_{\check{a} = a} = 0, \text{ for } a \in \mathcal{U}_{\tilde{a}},$$

which is equivalent to

$$(n_t - 1) H_t\left(b_t^*(a; \gamma_t) \right) F^{n_t - 2}(a) f(a) + (b_t^*)'(a; \gamma_t) \times \left. \frac{\partial \tilde{\pi}_t(b, a)}{\partial b} \right|_{b = b_t^*(a; \gamma_t)} = 0, \text{ for } a \in \mathcal{U}_{\tilde{a}}, \tag{18}$$

and can be further simplified as (8) in the text. Condition (18), together with Lemma 1(ii), Lemma 3, and Lemma 4, indicates that the equilibrium bidding strategy $b_t^*(a; \gamma_t)$, if a PBE exists, is fully characterized as in Theorem 1(i).

It remains to verify that $b_t^*(a; \gamma_t)$ as described in Theorem 1(ii) indeed constitutes a PBE of the contest game. We first verify the monotonicity of $b_t^*(a; \gamma_t)$. Evidently, the first term on the left-hand side of (18) always remains positive, indicating that $(b_t^*)'(a; \gamma_t) \ne 0$; moreover, from the definition of $s_t^1(a_t^*(\gamma_t); \gamma_t)$, we have that

$$\left. \frac{\partial \tilde{\pi}_t\left(b, a_t^*(\gamma_t) \right)}{\partial b} \right|_{b = s_t^1\left(a_t^*(\gamma_t); \gamma_t \right)} < 0.$$

Therefore, $(b_t^*)'(a;\gamma_t) > 0$ at $a = a_t^*(\gamma_t)$; otherwise, (18) cannot be satisfied. We can thus conclude from these facts that $b_t^*(a;\gamma_t)$ strictly increases with a whenever $b_t^*(a;\gamma_t)$ is continuous and is governed by (18). It remains to verify the monotonicity of $b_t^*(a;\gamma_t)$ at discontinuity points. Suppose that there exists $\tilde{a}$ such that $b_t^*(\tilde{a}-0;\gamma_t) = e_t^m(\tilde{a};\gamma_t)$. By Lemma 4, we have $b_t^*(\tilde{a}-0;\gamma_t) = e_t^m(\tilde{a};\gamma_t) < s_t^{m+1}(\tilde{a};\gamma_t) = b_t^*(\tilde{a}+0;\gamma_t)$.

Simple algebra would verify that

$$\frac{\partial^2 \check{\pi}_t(\check{a}, a;\gamma_t)}{\partial \check{a} \partial a} > 0,$$

which implies that $\frac{\partial \check{\pi}_t(\check{a},a;\gamma_t)}{\partial \check{a}}$ is increasing in a. Therefore, we have

$$\frac{\partial \check{\pi}_t(\check{a}, a;\gamma_t)}{\partial \check{a}} > \left.\frac{\partial \check{\pi}_t(\check{a}, a;\gamma_t)}{\partial \check{a}}\right|_{\check{a}=a} = 0, \text{ for } \check{a} < a,$$

and

$$\frac{\partial \check{\pi}_t(\check{a}, a;\gamma_t)}{\partial \check{a}} < \left.\frac{\partial \check{\pi}_t(\check{a}, a;\gamma_t)}{\partial \check{a}}\right|_{\check{a}=a} = 0, \text{ for } \check{a} > a.$$

That is, the necessary first-order condition $\left.\frac{\partial \check{\pi}_t(\check{a},a;\gamma_t)}{\partial \check{a}}\right|_{\check{a}=a} = 0$ is also a sufficient condition for global maximizer. This concludes the proof.

■

Proof of Proposition 1

Proof. See main text. ■

Proof of Proposition 2

Proof. It is useful to prove the following intermediate result.

Lemma 8 *Suppose that Assumption 1 is satisfied. Then $H_t(b)$ is continuous, twice differentiable, weakly increasing, strictly concave on $[0,1]$, and satisfies $\lim_{b\searrow 0}[bh_t(b)] = 0$ for all $t \leq T-1$.*

Proof. We prove the lemma by induction.

Base case: Consider the penultimate period, i.e., $t = T-1$. It follows from (4), (5), and Assumption 1 that $a_T^*(\gamma) = G_{n_T}(\gamma)$; together with (3), we have $H_{T-1}(b) = F^{n_{T-1}}\left(G_{n_{T-1}}(b)\right)$. Evidently, $H_{T-1}(b)$ is continuous, twice differentiable, and weakly increasing, and it remains to show that $H_{T-1}(b)$ is strictly concave on $[0,1]$ and satisfies $\lim_{b\searrow 0}[bh_{T-1}(b)] = 0$.

For notational convenience, define $\widehat{G}_t(b) := F^{n_t}\left(G_{n_t}(b)\right)$, $\forall t \in \{2, \ldots, T\}$. It follows from Assumption 1 that $\widehat{G}_t(b)$ is continuous, twice differentiable, and weakly increasing. We first show that $\widehat{G}_t(b)$ is strictly concave. Carrying out the algebra, we have that

$$\widehat{G}_t'(b) = \frac{n_t F\left(G_{n_t}(b)\right) f\left(G_{n_t}(b)\right)}{F\left(G_{n_t}(b)\right) + (n_t - 1) G_{n_t}(b) f\left(G_{n_t}(b)\right)}.$$

Because $G_{n_t}(b)$ is strictly increasing in b, it suffices to show that for all $x \in (0, 1)$,

$$\frac{d}{dx} \frac{F(x) f(x)}{F(x) + (n_t - 1) x f(x)} < 0 \iff \frac{d}{dx}\left[\frac{1}{f(x)} + (n_t - 1)\frac{x}{F(x)}\right] > 0.$$

The strict concavity of $F(x)$ implies that both $\frac{1}{f(x)}$ and $\frac{x}{F(x)}$ are strictly increasing in x. Therefore, $\widehat{G}_t(b)$ is strictly concave in b.

Next, we show that $\lim_{b \searrow 0}[b\widehat{G}_t'(b)] = 0$. The analysis is straightforward for $n_t = 1$, and it suffices to consider the case of $n_t \geq 2$. Carrying out the algebra, we have that

$$\lim_{b \searrow 0} \frac{b}{G_{n_t}(b)} = \lim_{b \searrow 0}\left[F^{n_t - 1}\left(G_{n_t}(b)\right)\right] = 0,$$

and

$$\lim_{b \searrow 0}\left[G_{n_t}(b)\widehat{G}_t'(b)\right] = \lim_{b \searrow 0} \frac{n_t b F(b) f(b)}{F(b) + (n_t - 1) b f(b)} = \lim_{b \searrow 0} \frac{n_t}{\frac{1}{bf(b)} + (n_t - 1)\frac{1}{F(b)}} = 0.$$

Therefore,

$$\lim_{b \searrow 0}[b\widehat{G}_t'(b)] = \lim_{b \searrow 0} \frac{b}{G_{n_t}(b)} \times \lim_{b \searrow 0}[G_{n_t}(b)\widehat{G}_t'(b)] = 0.$$

Note that $H_{T-1}(b) = \widehat{G}_{T-1}(b)$. The above analyses indicate that $H_{T-1}(b) = \widehat{G}_{T-1}(b)$ is strictly concave and $\lim_{b \searrow 0}[bh_{T-1}(b)] = \lim_{b \searrow 0}[b\widehat{G}_{T-1}'(b)] = 0$.

Inductive step: Suppose that $H_t(b)$ is continuous, twice differentiable, weakly increasing, strictly concave on $[0, 1]$, and satisfies $\lim_{b \searrow 0}[bh_t(b)] = 0$ for some $t \leq T - 1$. Next, we show that $H_{t-1}(b)$ has the same properties. Before we proceed, note that $H_t(0) = 0$ for all $t \leq T - 1$.

It is straightforward to verify that $H_{t-1}(b)$ is continuous and twice differentiable from its definition. Further, it can be verified from the concavity of $H_t(b)$ that $b/H_t(b)$ is strictly increasing in b. Because $H_t(b)$, $\widehat{G}_t(b)$, and $b/H_t(b)$ are all increasing, $H_{t-1}(b)$ is an increasing function.

Next, we prove the strict concavity of $H_{t-1}(b)$. Carrying out the algebra, we can obtain

that

$$h_{t-1}(b) = h_t(b)\left[\widehat{G}_t\left(\frac{b}{H_t(b)}\right) - \widehat{G}_t'\left(\frac{b}{H_t(b)}\right)\frac{b}{H_t(b)}\right] + \widehat{G}_t'\left(\frac{b}{H_t(b)}\right),$$

and

$$h_{t-1}'(b) = \underbrace{h_t'(b)}_{<0}\underbrace{\left[\widehat{G}_t\left(\frac{b}{H_t(b)}\right) - \widehat{G}_t'\left(\frac{b}{H_t(b)}\right)\frac{b}{H_t(b)}\right]}_{>0} + \underbrace{\widehat{G}_t''\left(\frac{b}{H_t(b)}\right)}_{<0}\underbrace{\left[\frac{H_t(b)-bh_t(b)}{H_t(b)}\right]}_{>0}\underbrace{\left[\frac{b}{H_t(b)}\right]'}_{>0}.$$

From the previous analysis, $\widehat{G}_t(b)$ is strictly concave and $\lim_{b\searrow 0}[b\widehat{G}_t'(b)] = 0$, which implies that $\widehat{G}_t''\left(\frac{b}{H_t(b)}\right) < 0$ and

$$\widehat{G}_t\left(\frac{b}{H_t(b)}\right) - \widehat{G}_t'\left(\frac{b}{H_t(b)}\right)\frac{b}{H_t(b)} > 0, \forall b \in (0,1).$$

The monotonicity of $b/H_t(b)$ implies that $\frac{H_t(b)-bh_t(b)}{H_t(b)} > 0$ and $\left[\frac{b}{H_t(b)}\right]' > 0$. Further, the strict concavity of $H_t(b) < 0$ implies that $h_t'(b) < 0$. Therefore, $h_{t-1}'(b) < 0$ and thus $H_{t-1}(b)$ is strictly concave.

Finally, we have

$$\lim_{b\searrow 0}\left[bh_{t-1}(b)\right] = \lim_{b\searrow 0}\left\{bh_t(b)\left[\widehat{G}_t\left(\frac{b}{H_t(b)}\right) - \widehat{G}_t'\left(\frac{b}{H_t(b)}\right)\frac{b}{H_t(b)}\right] + H_t(b)\frac{b}{H_t(b)}\widehat{G}_t'\left(\frac{b}{H_t(b)}\right)\right\}. \tag{19}$$

Note that

$$0 \le \widehat{G}_t\left(\frac{b}{H_t(b)}\right) - \widehat{G}_t'\left(\frac{b}{H_t(b)}\right)\frac{b}{H_t(b)} \le \widehat{G}_t\left(\frac{b}{H_t(b)}\right) \le 1, \tag{20}$$

$$0 \le \frac{b}{H_t(b)}\widehat{G}_t'\left(\frac{b}{H_t(b)}\right) \le \widehat{G}_t\left(\frac{b}{H_t(b)}\right) \le 1, \tag{21}$$

Equations (19) to (21), together with $H_t(0) = 0$ and the postulated $\lim_{b\searrow 0}[bh_t(b)] = 0$, imply that $\lim_{b\searrow 0}[bh_{t-1}(b)] = 0$. This completes the inductive step.

Conclusion: By the principle of induction, $H_t(b)$ is continuous, twice differentiable, weakly increasing, strictly concave on $[0,1]$, and satisfies $\lim_{b\searrow 0}[bh_t(b)] = 0$ for all $t \le T-1$. This concludes the proof. ■

Now we can prove Proposition 2. By Lemma 8, $H_t(b)$ is concave. Therefore, $\tilde{\pi}_t(b,a)$ is concave in b and $S_t(a;\gamma)$ reduces to a single interval. The proposition then follows immediately from Theorem 1. ■

Proof of Proposition 3

Proof. It suffices to show that every period-t player, with $t \leq T-1$, would exert zero effort independent of γ_t. It can be verified that $H_t(b) \leq b$ for all $b \in [0,1]$ and $t \leq T-1$ if Assumption 2 is satisfied, which implies that $\mathcal{T}_0 = \{1, \ldots, T-1\}$ and thus $t_0 = T-1$. The proposition follows immediately from Proposition 1, which concludes the proof. ■

Proof of Example 2

Proof. The analysis for the case of $N = 2$ is laid out in the main text and it suffices to consider the case of $N \geq 3$. We first derive players' equilibrium expected efforts in fully sequential contests, then conduct a comparison.

Derivation of Players' Expected Efforts We first derive $\mathbb{E}[b_t^*|\gamma_t]$. According to the equilibrium bidding strategy characterized in Proposition 2(i), we have that

$$\begin{aligned}
\mathbb{E}[b_t^*|\gamma_t] &= \gamma_t \left\{ F\left[\frac{1}{h_t(\gamma_t)}\right] - F\left[\frac{\gamma_t}{H_t(\gamma_t)}\right] \right\} + \int_{h_t(1/\gamma_t)}^{1} \frac{1}{h_t^{-1}(a)} dF(a) \\
&= \gamma_t \left\{ F\left[\frac{1}{h_t(\gamma_t)}\right] - F\left[\frac{\gamma_t}{H_t(\gamma_t)}\right] \right\} + \int_{\gamma_t}^{\bar{\gamma}_t} x dF\left[\frac{1}{h_t(x)}\right] \\
&= \begin{cases} \bar{\gamma}_t - \gamma_t F\left[\frac{\gamma_t}{H_t(\gamma_t)}\right] - \int_{\gamma_t}^{\bar{\gamma}_t} F\left[\frac{1}{h_t(x)}\right] dx, & \text{if } \gamma_t \leq \bar{\gamma}_t, \\ \gamma_t \left\{ 1 - F\left[\frac{\gamma_t}{H_t(\gamma_t)}\right] \right\}, & \text{if } \gamma_t > \bar{\gamma}_t, \end{cases}
\end{aligned} \tag{22}$$

where $\bar{\gamma}_t := h_t^{-1}(1)$ for $t < T$ and $\bar{\gamma}_T := 0$. Given that $F(a) = a^r$, $r \in (0,1)$, it can be verified that $H_t(b) = b^{1-(1-r)^{T-t}}$, which implies that

$$F\left[\frac{\gamma}{H_t(\gamma)}\right] = \gamma^{r(1-r)^{T-t}}, \tag{23}$$

and

$$F\left[\frac{1}{h_t(\gamma)}\right] = \min\left\{1, \frac{\gamma^{r(1-r)^{T-t}}}{[1-(1-r)^{T-t}]^r}\right\}. \tag{24}$$

Combining (22), (23), and (24), for $\gamma \leq \bar{\gamma}_t$, we have

$$\begin{aligned}
\mathbb{E}[b_t^*|\gamma_t = \gamma] &= \bar{\gamma}_t - \gamma^{1+r(1-r)^{T-t}} - \frac{\bar{\gamma}_t^{1+r(1-r)^{T-t}} - \gamma^{1+r(1-r)^{T-t}}}{[1+r(1-r)^{T-t}][1-(1-r)^{T-t}]^r} \\
&= \frac{r(1-r)^{T-t}}{1+r(1-r)^{T-t}} \bar{\gamma}_t + \frac{1-[1+r(1-r)^{T-t}][1-(1-r)^{T-t}]^r}{[1+r(1-r)^{T-t}][1-(1-r)^{T-t}]^r} \gamma^{1+r(1-r)^{T-t}},
\end{aligned}$$

where the second equality follows from $\frac{\bar{\gamma}_t^{r(1-r)^{T-t}}}{[1-(1-r)^{T-t}]^r} = 1$; for $\gamma > \bar{\gamma}_t$, we have

$$\mathbb{E}[b_t^*|\gamma_t = \gamma] = \gamma \left[1 - \gamma^{r(1-r)^{T-t}}\right].$$

Next, we derive the CDF of γ_t. Note that

$$\gamma_t = \max_{\tau<t} \left\{ h_\tau^{-1}\left(\frac{1}{a_\tau}\right)\right\}, \text{ for } t \geq 2. \tag{25}$$

The above expression can be obtained by induction. Evidently, (25) holds for $t = 2$. Suppose that (25) holds for $t = \ell \geq 2$. It suffices to show that (25) holds for $t = \ell + 1$. Consider the period-ℓ player. From Proposition 2(i), we have that

$$\gamma_{\ell+1} = \begin{cases} \gamma_\ell, & \text{if } a_\ell \leq 1/h_\ell(\gamma_\ell), \\ h_\ell^{-1}\left(\frac{1}{a_\ell}\right), & \text{if } a_\ell > 1/h_\ell(\gamma_\ell). \end{cases}$$

Note that $h_\ell^{-1}\left(\frac{1}{a_\ell}\right)$ is increasing in a_ℓ. The above expression can be rewritten as

$$\gamma_{\ell+1} = \max\left\{\gamma_\ell, h_\ell^{-1}\left(\frac{1}{a_\ell}\right)\right\};$$

together with the postulated $\gamma_\ell = \max_{\tau<\ell}\left\{h_\tau^{-1}\left(\frac{1}{a_\tau}\right)\right\}$, we can obtain

$$\gamma_{\ell+1} = \max\left\{\max_{\tau<\ell}\left\{h_\tau^{-1}\left(\frac{1}{a_\tau}\right)\right\}, h_\ell^{-1}\left(\frac{1}{a_\ell}\right)\right\} = \max_{\tau<\ell+1}\left\{h_\tau^{-1}\left(\frac{1}{a_\tau}\right)\right\}.$$

By (25), the CDF of γ_t, with $t \geq 2$, is given by

$$\Gamma_t(\gamma) := \Pi_{\tau<t} F\left[\frac{1}{h_\tau(\gamma)}\right].$$

Note that $\bar{\gamma}_{\tau+1} < \bar{\gamma}_\tau$ for $1 \leq \tau \leq T-2$. Therefore, for $t \geq 2$, we have that

$$\Gamma_t(\gamma) = \begin{cases} \Pi_{\tau<t} F\left[\frac{1}{h_\tau(\gamma)}\right] = \frac{\gamma^{r\sum_{\tau<t}(1-r)^{T-\tau}}}{\Pi_{\tau<t}[1-(1-r)^{T-\tau}]^r}, & \text{if } \gamma \leq \bar{\gamma}_t, \\ \Pi_{\tau\leq k} F\left[\frac{1}{h_\tau(\gamma)}\right] = \frac{\gamma^{r\sum_{\tau\leq k}(1-r)^{T-\tau}}}{\Pi_{\tau\leq k}[1-(1-r)^{T-\tau}]^r}, & \text{if } \bar{\gamma}_{k+1} < \gamma \leq \bar{\gamma}_k, \text{ for } 1 \leq k \leq t-1. \end{cases} \tag{26}$$

Now we can calculate $\mathbb{E}[b_t^*]$. Note that

$$\mathbb{E}[b_t^*] = \int_0^{\bar{\gamma}_1} \mathbb{E}[b_t^*|\gamma] d\Gamma_t(\gamma) = \int_0^{\bar{\gamma}_t} \mathbb{E}[b_t^*|\gamma] d\Gamma_t(\gamma) + \int_{\bar{\gamma}_t}^{\bar{\gamma}_1} \mathbb{E}[b_t^*|\gamma] d\Gamma_t(\gamma). \tag{27}$$

Combining (22) and (26), for $t \geq 2$, we can obtain that

$$\begin{aligned}
\int_0^{\bar{\gamma}_t} \mathbb{E}[b_t^*|\gamma] d\Gamma_t(\gamma) &= \frac{r(1-r)^{T-t}}{1+r(1-r)^{T-t}} \frac{\bar{\gamma}_t^{1+r\sum_{\tau<t}(1-r)^{T-\tau}}}{\Pi_{\tau<t}[1-(1-r)^{T-\tau}]^r} \\
&\quad + \frac{r\sum_{\tau<t}(1-r)^{T-\tau}}{\Pi_{\tau<t}[1-(1-r)^{T-\tau}]^r} \left(\frac{1}{A_{T-t}} - 1\right) \int_0^{\bar{\gamma}_t} \gamma^{r\sum_{\tau\leq t}(1-r)^{T-\tau}} dx \\
&= \frac{r(1-r)^{T-t}}{A_{T-t}} \frac{\bar{\gamma}_t^{1+r\sum_{\tau\leq t}(1-r)^{T-\tau}}}{\Pi_{\tau<t}[1-(1-r)^{T-\tau}]^r} + \frac{r\sum_{\tau<t}(1-r)^{T-\tau}}{\Pi_{\tau<t}[1-(1-r)^{T-\tau}]^r} \left(\frac{1}{A_{T-t}} - 1\right) \frac{\bar{\gamma}_t^{1+r\sum_{\tau\leq t}(1-r)^{T-\tau}}}{1+r\sum_{\tau\leq t}(1-r)^{T-\tau}} \\
&= \frac{\bar{\gamma}_t^{1+r\sum_{\tau\leq t}(1-r)^{T-\tau}}}{A_{T-t}\Pi_{\tau<t}[1-(1-r)^{T-\tau}]^r} \left[r(1-r)^{T-t} + (1-A_{T-t})\frac{r\sum_{\tau<t}(1-r)^{T-\tau}}{1+r\sum_{\tau\leq t}(1-r)^{T-\tau}}\right],
\end{aligned} \tag{28}$$

where $A_{T-t} := [1+r(1-r)^{T-t}][1-(1-r)^{T-t}]^r$; and that

$$\begin{aligned}
\int_{\bar{\gamma}_t}^{\bar{\gamma}_1} \mathbb{E}[b_t^*|\gamma] d\Gamma_t(\gamma) &= \sum_{k=1}^{t-1} \int_{\bar{\gamma}_{k+1}}^{\bar{\gamma}_k} \gamma \left\{1 - F\left[\frac{\gamma}{H_t(\gamma)}\right]\right\} d\Pi_{\tau\leq k} F\left[\frac{1}{h_\tau(\gamma)}\right] \\
&= \sum_{k=1}^{t-1} \frac{r\sum_{\tau\leq k}(1-r)^{T-\tau}}{\Pi_{\tau\leq k}[1-(1-r)^{T-\tau}]^r} \left(\frac{\bar{\gamma}_k^{1+r\sum_{\tau\leq k}(1-r)^{T-\tau}} - \bar{\gamma}_{k+1}^{1+r\sum_{\tau\leq k}(1-r)^{T-\tau}}}{1+r\sum_{\tau\leq k}(1-r)^{T-\tau}}\right. \\
&\qquad \left. - \frac{\bar{\gamma}_k^{1+r(1-r)^{T-t}+r\sum_{\tau\leq k}(1-r)^{T-\tau}} - \bar{\gamma}_{k+1}^{1+r(1-r)^{T-t}+r\sum_{\tau\leq k}(1-r)^{T-\tau}}}{1+r(1-r)^{T-t}+r\sum_{\tau\leq k}(1-r)^{T-\tau}}\right).
\end{aligned} \tag{29}$$

Therefore, for $t \geq 2$, $\mathbb{E}[b_t^*]$ can be calculated through (27), (28), and (29). For $t = 1$, $\mathbb{E}[b_t^*]$ can again be calculated through (27), (28), and (29) by setting $\Pi_{\tau<1}[1-(1-r)^{T-\tau}]^r$ to 1 and $\sum_{\tau<t}(1-r)^{T-\tau}$ to 0.

Effort Comparison Carrying out the algebra, it can be verified that

$$\mathbb{E}[b_t^*] = (T-t)r^2 + o(r^2),$$

as $r \searrow 0$, from which we can conclude that $\mathbb{E}[b_1^*] > \mathbb{E}[b_2^*] > \cdots > \mathbb{E}[b_T^*]$ when r is sufficiently close to 0.

Similarly, as $r \nearrow 1$, we have

$$\mathbb{E}[b_t^*] = \begin{cases} \frac{1}{e}(1-r)^{T-t} + o((1-r)^{T-t}), & \text{if } t < T, \\ \frac{2e-1}{2e^2}(1-r) + o(1-r), & \text{if } t = T, \end{cases}$$

from which we can conclude that $\mathbb{E}[b_{T-1}^*] > \mathbb{E}[b_T^*] > \mathbb{E}[b_{T-2}^*] > \cdots > \mathbb{E}[b_1^*]$ when r is sufficiently close to 1. This concludes the proof. ■

Proof of Theorem 2

Proof. Recall the unique symmetric PBE is denoted by $\boldsymbol{b}^* := \{b_1^*(a;\gamma_1), \ldots, b_T^*(a;\gamma_T)\}$. Fix an arbitrary period $\tau \in \{t_0+1, \ldots, T-1\}$ and a player i in period $\tau+1$, i.e., $i \in \mathcal{N}_{\tau+1}$. We conduct the following thought experiment. Holding fixed all other players' strategies—including those in period $\tau+1$, if any—we modify player i's equilibrium bidding strategy from $b_{\tau+1}^*(a;\gamma_{\tau+1})$ to

$$b_{\tau+1}^{i\dagger}(a;\gamma_{\tau+1}) := b_\tau^*(a;\gamma_\tau).$$

For ease of exposition, denote the constructed profile of bidding strategies by $\boldsymbol{b}^\dagger$. Further, denote player i's expected payoff and a period-τ player's under $\boldsymbol{b}^\dagger$ by $\Pi_{\tau+1}^{i\dagger}$ and $\Pi_\tau^\dagger$, respectively.

We first show that $\Pi_\tau^* < \Pi_\tau^\dagger$. Consider an indicative period-τ player j, $j \in \mathcal{N}_\tau$, whose ability we denote by a^j. Denote his interim expected payoff under $\boldsymbol{b}^*$ and that under $\boldsymbol{b}^\dagger$ by $\pi_\tau^*(a^j;\gamma_\tau)$ and $\pi_\tau^\dagger(a^j;\gamma_\tau)$, respectively.

If $b^j := b_\tau^*(a^j;\gamma_\tau) = 0$, then the period-$\tau$ player loses under both $\boldsymbol{b}^*$ and $\boldsymbol{b}^\dagger$, indicating $\pi_\tau^*(a^j;\gamma_\tau) = \pi_\tau^\dagger(a^j;\gamma_\tau) = 0$. If $b^j > 0$, then we have $b^j \geq \gamma_\tau$ and

$$\begin{aligned} \pi_\tau^*(a^j;\gamma_\tau) &= F^{n_\tau-1}(a^j)H_\tau(b^j) - b^j/a^j \\ &= F^{n_\tau-1}(a^j)H_{\tau+1}(b^j)F^{n_{\tau+1}}\left(a_{\tau+1}^*(b^j)\right) - b^j/a^j, \end{aligned}$$

where the second equality follows from (3). Similarly, we have

$$\pi_\tau^\dagger(a^j;\gamma_\tau) = F(\acute{a}^j)F^{n_\tau-1}(a^j)H_{\tau+1}(b^j)F^{n_{\tau+1}-1}\left(a_{\tau+1}^*(b^j)\right) - b^j/a^j,$$

where $\acute{a}^j$ is defined as

$$\acute{a}^j := \begin{cases} a_\tau^*(\gamma_\tau), & \text{if } n_\tau = 1 \text{ and } a_\tau^*(\gamma_\tau) < a^j \leq a_\tau^{**}(\gamma_\tau), \\ a^j, & \text{otherwise}, \end{cases}$$

and satisfies $b^*_\tau(\acute{a}^j;\gamma_\tau) = b^*_\tau(a^j;\gamma_\tau) > 0$. It is straightforward to verify that

$$\pi^*_\tau(a^j;\gamma_\tau) < \pi^\dagger_\tau(a^j;\gamma_\tau) \iff F\left(a^*_{\tau+1}(b^j)\right) < F(\acute{a}^j) \iff a^*_{\tau+1}(b^j) < \acute{a}^j. \tag{30}$$

It follows immediately from $b^*_\tau(\acute{a}^j;\gamma_\tau) = b^*_\tau(a^j;\gamma_\tau) > 0$ that $\pi^*_\tau(\acute{a}^j;\gamma_\tau) > 0$, from which we can conclude

$$H_{\tau+1}(b^j)F^{n_{\tau+1}-1}\left(a^*_{\tau+1}(b^j)\right) - b^j/\acute{a}^j > 0. \tag{31}$$

Further, it follows from the definition of $a^*_{\tau+1}(\cdot)$ [see Equation (4)] that

$$\tilde{\pi}_{\tau+1}\left(b^j, a^*_{\tau+1}(b^j)\right) = H_{\tau+1}(b^j)F^{n_{\tau+1}-1}\left(a^*_{\tau+1}(b^j)\right) - b^j/a^*_{\tau+1}(b^j) \le 0. \tag{32}$$

Combining (31) and (32) yields

$$a^*_{\tau+1}(b^j) \le \frac{b^j}{H_{\tau+1}(b^j)F^{n_{\tau+1}-1}\left(a^*_{\tau+1}(b^j)\right)} < \acute{a}^j.$$

The above condition, together with (30), implies that $\pi^*_\tau(a^j;\gamma_\tau) < \pi^\dagger_\tau(a^j;\gamma_\tau)$ and

$$\Pi^*_\tau = \mathbb{E}\left[\pi^*_\tau(a^j;\gamma_\tau)\right] < \mathbb{E}\left[\pi^\dagger_\tau(a^j;\gamma_\tau)\right] = \Pi^\dagger_\tau, \tag{33}$$

where the expectation is taken with respect to both a^j and γ_τ.

To complete the proof, first note that $\Pi^*_{\tau+1} \ge \Pi^{i\dagger}_{\tau+1}$ by the definition of PBE. Moreover, it follows immediately from the construction $b^{i\dagger}_{\tau+1}(a;\gamma_{\tau+1}) := b^*_\tau(a;\gamma_\tau)$ that $\Pi^{i\dagger}_{\tau+1} \ge \Pi^\dagger_\tau$. These inequalities, together with (33), imply that $\Pi^*_{\tau+1} \ge \Pi^{i\dagger}_{\tau+1} \ge \Pi^\dagger_\tau > \Pi^*_\tau$. ■

Proof of Proposition 4

Proof. Consider the strategy profile $\boldsymbol{b}^s \equiv \left\{b^{i,s}(a;\gamma_t,z_t)\right\}_{i\in\mathcal{N}_t, t\in\mathcal{T}}$ as defined in (10). Fix $t \in \mathcal{T}$ and consider an indicative player $i \in \mathcal{N}_t$. Note that a tie occurs with zero probability under $\boldsymbol{b}^s$. We consider the following two cases:

(a) Suppose that $n_t = 1$. Then player i's interim expected payoff of bidding b^i under (9), given that all other players bid according to $\boldsymbol{b}^s$, is

$$\pi^{i,s}(b^i,a^i;\gamma_t,z_t) = \begin{cases} -b^i/a^i, & \text{if } b^i < \gamma_t, \\ \frac{1}{z_t+1}\Pr\left(a^l \le a^*_k(b^i), \forall k \in \{t+1,\dots,T\}, \forall l \in \mathcal{N}_k\right) - b^i/a^i, & \text{if } b^i = \gamma_t, \\ \Pr\left(a^l \le a^*_k(b^i), \forall k \in \{t+1,\dots,T\}, \forall l \in \mathcal{N}_k\right) - b^i/a^i, & \text{if } b^i > \gamma_t. \end{cases}$$

With slight abuse of notation, denote player i's interim expected payoff under (1) and

$\boldsymbol{b}^* \equiv \left\{b_t^*(a;\gamma_t)\right\}_{t\in\mathcal{T}}$ by $\pi^i(b^i, a^i; \gamma_t)$. Note that if $b^i \neq \gamma_t$, $\pi^{i,s}(b^i, a^i; \gamma_t, z_t) = \pi^i(b^i, a^i; \gamma_t)$; if $b^i = \gamma_t$, $\pi^{i,s}(b^i, a^i; \gamma_t, z_t) \leq \pi^i(b^i, a^i; \gamma_t)$.

We consider the following two subcases:

(i) Suppose that $b_t^*(a^i; \gamma_t) \neq \gamma_t$, which occurs when $a^i \leq a_t^*(\gamma_t)$ or $a^i > a_t^{**}(\gamma_t)$. By (10), we have $b^{i,s}(a^i; \gamma_t, z_t) = b_t^*(a^i; \gamma_t)$. Note that $b_t^*(a^i; \gamma_t)$ solves $\max_{b^i \geq 0} \pi^i(b^i, a^i; \gamma_t)$ in the unique symmetric PBE under (1). Moreover, it can be verified that $\pi^{i,s}(b^i, a^i; \gamma_t, z_t) = \pi^i(b^i, a^i; \gamma_t)$ except for the case of $b^i = \gamma_t$. Therefore, $b_t^*(a^i; \gamma_t)$ also solves $\max_{b^i \geq 0} \pi^{i,s}(b^i, a^i; \gamma_t, z_t)$ and is thus optimal for player i.

(ii) Suppose that $b_t^*(a^i; \gamma_t) = \gamma_t$, which occurs when $a_t^*(\gamma_t) < a^i \leq a_t^{**}(\gamma_t)$. It follows from (10) that $b^{i,s}(a^i; \gamma_t, z_t) = b_t^*(a^i; \gamma_t) + a^i\epsilon = \gamma_t + a^i\epsilon$. Note that for all $b^i \geq 0$, we have that

$$\pi^{i,s}(b^i, a^i; \gamma_t, z_t) \leq \pi^i(b^i, a^i; \gamma_t) \leq \pi^i(b_t^*(a^i; \gamma_t), a^i; \gamma_t) = \pi^i(\gamma_t, a^i; \gamma_t).$$

That is, player i's interim expected payoff is bounded from above by $\pi^i(\gamma_t, a^i; \gamma_t)$. Moreover, bidding $\gamma_t + a^i\epsilon$ would result in an expected payoff of $\pi^i(\gamma_t + a^i\epsilon, a^i; \gamma_t)$, which is no less than $\pi^i(\gamma_t, a^i; \gamma_t) - \epsilon$. Therefore, bidding $b^{i,s}(a; \gamma_t, z_t)$ is ϵ-optimal for player i.

(b) Suppose that $n_t \geq 2$. Then

$$\pi^{i,s}(b^i, a^i; \gamma_t, z_t) = \begin{cases} -b^i/a^i, & \text{if } b^i < \gamma_t, \\ \frac{1}{z_t+1} \Pr\left(\begin{array}{c} a^l \leq a_k^*(b^i), \forall k \in \{t, \ldots, T\}, \\ \forall l \in \mathcal{N}_k \setminus \{i\} \end{array} \right) - b^i/a^i, & \text{if } b^i = \gamma_t, \\ \Pr\left(\begin{array}{c} a^\jmath \leq (b_t^*)^{-1}(b^i; \gamma_t), \forall \jmath \in \mathcal{N}_t \setminus \{i\}, \text{ and} \\ a^l \leq a_k^*(b^i), \forall k \in \{t+1, \ldots, T\}, \forall l \in \mathcal{N}_k \end{array} \right) - b^i/a^i, & \text{if } b^i > \gamma_t. \end{cases}$$

Again, it can be verified that $\pi^{i,s}(b^i, a^i; \gamma_t, z_t) = \pi^i(b^i, a^i; \gamma_t)$ if $b_i \neq \gamma_t$ and $\pi^{i,s}(b^i, a^i; \gamma_t, z_t) \leq \pi^i(b^i, a^i; \gamma_t)$ if $b^i = \gamma_t$. By (10), we have $b^{i,s}(a^i; \gamma_t, z_t) = b_t^*(a^i; \gamma_t)$. Moreover, it follows from Lemma 1(ii) and (iii) that $b_t^*(a^i; \gamma_t) = 0$ for $a^i \leq a_t^*(\gamma_t)$ and $b_t^*(a^i; \gamma_t) > \gamma_t$ for $a^i > a_t^*(\gamma_t)$. Therefore, $b_t^*(a^i; \gamma_t) \neq \gamma_t$ for all $a^i \in (0, 1)$. Further, for all $a^i \in (0, 1)$ and $b^i \geq 0$, we have that

$$\pi^{i,s}\left(b^i, a^i; \gamma_t, z_t\right) \leq \pi^i\left(b^i, a^i; \gamma_t\right) \leq \pi^i\left(b_t^*(a^i; \gamma_t), a^i; \gamma_t\right) = \pi^{i,s}\left(b^{i,s}(a^i; \gamma_t, z_t), a^i; \gamma_t, z_t\right).$$

Therefore, $b^{i,s}(a^i; \gamma_t, z_t)$ solves $\max_{b^i \geq 0} \pi^{i,s}(b^i, a^i; \gamma_t, z_t)$ and is optimal for player i.

To summarize, $\pi^{i,s}(b^{i,s}(a;\gamma_t,z_t),a;\gamma_t,z_t) \geq \pi^{i,s}(b',a;\gamma_t,z_t)-\epsilon$ for all $a \in (0,1]$ and $b' \geq 0$, and thus $\boldsymbol{b}^s \equiv \left(b^{i,s}(a;\gamma_t,z_t)\right)_{i\in\mathcal{N}_t,t\in\mathcal{T}}$ constitutes a symmetric ϵ-equilibrium of the contest game under (9). This concludes the proof. ■

Proof of Proposition 5

Proof. Consider an indicative player $i \in \mathcal{N}_t$. We start from the last period T. By definition, $\max_{j\in\cup_{\ell>t}\mathcal{N}_\ell}\{a^j\} = 0$ for $t = T$ and hence $\underline{b}_T = \gamma_T$. If $n_T = 1$, then $a_t^{sec} = 0$ and it is optimal for the unique period-T player to choose γ_T if $a^i > \gamma_T$ and exert zero effort otherwise. If $n_T \geq 2$, the stage game is a simultaneous all-pay auction with reserve price $\underline{b}_T = \gamma_T$. It can be verified that every player bidding according to the strategy described in Proposition 5 constitutes an equilibrium; equilibrium uniqueness follows from Bertoletti (2016).

Suppose that the statement holds for periods $t+1$ through T and consider period t. We focus on the case of $n_t \geq 2$; the analysis for the case of $n_t = 1$ is similar. Denote player i's bid by b^i. If $b^i < \gamma_t$, then he loses with certainty. If $b^i < \max_{j\in\cup_{\ell>t}\mathcal{N}_\ell}\{a^j\}$, then his bid will be exceeded by some later mover in equilibrium and will lose again. If $b^i \geq \underline{b}_t$, then all players after period t would remain inactive in equilibrium.

The above analysis indicate that fixing γ_t, the stage game in period t is essentially a simultaneous all-pay auction with a reserve price $\underline{b}_t \equiv \max\{\max_{j\in\cup_{\ell>t}\mathcal{N}_\ell}\{a^j\},\gamma_t\}$. It can then be verified that the strategy profile described in the proposition constitutes the unique equilibrium. This concludes the proof. ■

Proof of Proposition 6

Proof. Fix an indicative player $i \in \mathcal{N}_t$ and consider the following two cases:

(a) Suppose that $a^i \neq \max_{j\in\mathcal{N}}\{a^j\}$. By Proposition 5, player i would bid 0 with a positive probability in equilibrium, from which we can conclude that $\pi^{i,c}(\boldsymbol{a}) = 0$.

(b) Suppose that $a^i = \max_{j\in\mathcal{N}}\{a^j\}$. By Proposition 5, all the players that move before period t bids 0 in equilibrium, so $\gamma_t = 0$ in equilibrium and thus $\underline{b}_t = \max_{j\in\cup_{\ell>t}\mathcal{N}_\ell}\{a^j\}$. We consider the following two subcases:

   (i) If $a_t^{sec} \leq \underline{b}_t$, then player i bids $\underline{b}_t$ and wins the contest. His expected payoff amounts to $1-(\max_{j\in\cup_{\ell>t}\mathcal{N}_\ell}\{a^j\})/a^i$, which is equivalent to $1-(\max_{j\in\cup_{\ell\geq t}\mathcal{N}_\ell\setminus\{i\}}\{a^j\})/a^i$.

   (ii) If $a_t^{sec} > \underline{b}_t$, then player i would use a mixed strategy in the equilibrium and his expected payoff is equal to that when he bids $\underline{b}_t$, which is $1-a_t^{sec}/a^i$ and is equivalent to $1-(\max_{j\in\cup_{\ell\geq t}\mathcal{N}_\ell\setminus\{i\}}\{a^j\})/a^i$.

To summarize, $\pi^{i,c}(\boldsymbol{a}) = 1 - (\max_{j \in \cup_{\ell \geq t} \mathcal{N}_\ell \setminus \{i\}} \{a^j\})/a^i$ in the case of $a^i = \max_{j \in \mathcal{N}} \{a^j\}$. This concludes the proof.

■

Proof of Theorem 4

Proof. Fix $i \in \mathcal{N}_t$ and $j \in \mathcal{N}_{t+1}$, with $t \in \{1, \ldots, T-1\}$. By Proposition 6, the interim expected payoff of player i with ability $a \in [0,1]$, which we denote by $\pi_t^{*,c}(a)$, is

$$\begin{aligned}\pi_t^{*,c}(a) &= \Pr\left[a > a^\imath, \forall \imath \in \mathcal{N} \setminus \{i\}\right] \mathbb{E}\left[1 - \left(\max_{\jmath \in \cup_{\ell \geq t} \mathcal{N}_\ell \setminus \{i\}} \{a^\jmath\}\right)/a \middle| a > a^\imath, \forall \imath \in \mathcal{N} \setminus \{i\}\right] \\ &= \Pr\left[a > a^\imath, \forall \imath \in \mathcal{N} \setminus \{i\}\right] \mathbb{E}\left[1 - \left(\max_{\jmath \in \cup_{\ell \geq t} \mathcal{N}_\ell \setminus \{i\}} \{a^\jmath\}\right)/a \middle| a > a^\jmath, \forall \jmath \in \cup_{\ell \geq t} \mathcal{N}_\ell \setminus \{i\}\right].\end{aligned} \tag{34}$$

Similarly, the expected payoff of player j with type a, which we denote by $\pi_{t+1}^{*,c}(a)$, is

$$\pi_{t+1}^{*,c}(a) = \Pr\left[a > a^\imath, \forall \imath \in \mathcal{N} \setminus \{j\}\right] \mathbb{E}\left[1 - \left(\max_{\jmath \in \cup_{\ell \geq t+1} \mathcal{N}_\ell \setminus \{j\}} \{a^\jmath\}\right)/a \middle| a > a^\jmath, \forall \jmath \in \cup_{\ell \geq t+1} \mathcal{N}_\ell \setminus \{j\}\right]. \tag{35}$$

It is evident that

$$\Pr\left[a > a^\imath, \forall \imath \in \mathcal{N} \setminus \{i\}\right] = F^{N-1}(a) = \Pr\left[a > a^\imath, \forall \imath \in \mathcal{N} \setminus \{j\}\right]. \tag{36}$$

Moreover, we have that

$$\mathbb{E}\left[\max_{\jmath \in \cup_{\ell \geq t} \mathcal{N}_\ell \setminus \{i\}} \{a^\jmath\} \middle| a > a^\jmath, \forall \jmath \in \cup_{\ell \geq t} \mathcal{N}_\ell \setminus \{i\}\right] > \mathbb{E}\left[\max_{\jmath \in \cup_{\ell \geq t+1} \mathcal{N}_\ell \setminus \{j\}} \{a^\jmath\} \middle| a > a^\jmath, \forall \jmath \in \cup_{\ell \geq t+1} \mathcal{N}_\ell \setminus \{j\}\right]. \tag{37}$$

Combining (34), (35), (36), and (37) yields $\pi_t^{*,c}(a) < \pi_{t+1}^{*,c}(a)$, from which we can conclude that

$$\Pi_t^{*,c} = \int_0^1 \pi_t^{*,c}(a) dF(a) < \int_0^1 \pi_{t+1}^{*,c}(a) dF(a) = \Pi_{t+1}^{*,c}.$$

This completes the proof. ■

Proof of Theorem 5

Proof. Fixing a contest architecture $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ and $\theta \in [0,1]$, denote a symmetric PBE of the contest game, if it exists, by $\{b_t^*(a;\gamma_t)\}_{t=1}^T$, with slight abuse of notation. Recall that the sequence of functions $\{H_t(b;\theta), a_t^*(\gamma;\theta), \tilde{\pi}_t(b,a;\theta)\}_{t=1}^T$ is defined by (11), (12), and (13). By similar arguments as for the case of $\theta = 1$, we can show that Lemmas 1 to 4 extend to $\theta \in [0,1]$. The proof of existence and uniqueness of symmetric PBE resembles that of Theorem 1, except that $\check{\pi}_t(\check{a}, a; \gamma)$ is now defined as

$$\check{\pi}_t(\check{a}, a; \gamma) := H_t\left(b_t^*(\check{a};\gamma_t);\theta\right) F^{n_t-1}(\check{a}) \left[a - (1-\theta) b_t^*(\check{a};\gamma_t)\right] - \theta b_t^*(\check{a};\gamma_t),$$

and the differential equation that governs a period-t player's bidding strategy $b_t^*(a;\gamma_t)$—given that $n_t \geq 2$ and $b_t^*(a;\gamma_t)$ is continuous in some interval $\mathcal{U}_{\tilde{a}} = (\tilde{a}, \tilde{a}+\epsilon)$—is

$$(n_t - 1) H_t\left(b_t^*(a;\gamma_t);\theta\right) F^{n_t-2}(a) f(a) \left[a - (1-\theta) b_t^*(a;\gamma_t)\right] + (b_t^*)'(a;\gamma_t) a \frac{\partial \tilde{\pi}_t(b,a;\theta)}{\partial b}\bigg|_{b=b_t^*(a;\gamma_t)} = 0.$$

The proof of the later-mover advantage is similar to that in Theorem 2 and omitted for brevity. ■

Proof of Proposition 7

Proof. Recall that $a_t^\star(\cdot)$ is defined in (14). It is useful to prove an intermediate result.

Lemma 9 *Consider a sequential second-price all-pay auction $\boldsymbol{n} \equiv (n_1, \ldots, n_T)$ under a tie-breaking rule as specified in* (1)*. Moreover, suppose that there exists a symmetric PBE and $n_t \geq 2$. Then a period-t player's equilibrium bidding strategy $b_t^\star(a;\gamma_t)$ is equal to* 0 *for $a \leq a_t^\star(\gamma_t)$ and strictly increases with a on $\left(a_t^\star(\gamma_t), 1\right)$. Moreover, $b_t^\star\left(a_t^\star(\gamma_t)+0;\gamma_t\right) = 1$ if $t \in \mathcal{T} \setminus \{T\}$ and $b_t^\star\left(a_t^\star(\gamma_t)+0;\gamma_t\right) = \gamma_t$ if $t = T$.*

Proof. Fix $t \in \mathcal{T}$, if $\gamma_t \geq 1$, then the optimal strategy for a period-t player is $b_t^\star(a;\gamma_t) = 0$ for all $a \in (0,1)$. We have $a_t^\star(1) = 1$ from (14), and thus the properties stated in Lemma 9 are satisfied. In what follows, we focus on the case of $\gamma_t \in [0,1)$, with $t \in \mathcal{T}$.

We first show that $b_t^\star(a;\gamma_t)$ is weakly increasing in a on $(0,1)$. Denote the equilibrium winning probability of bidding b and the corresponding expected second highest effort by $\mathcal{WP}_t^\star(b)$ and $\mathcal{C}_t^\star(b)$, respectively. Suppose, to the contrary, that there exists an ability pair (a', a''), with $0 < a' < a'' < 1$, such that $b'' := b_t^\star(a'';\gamma_t) < b' := b_t^\star(a';\gamma_t)$. It is obvious that $\mathcal{WP}_t^\star(b'') \leq \mathcal{WP}_t^\star(b')$; otherwise, type-$a'$ player has a strict incentive to bid b''. We consider the following two cases:

(a) Suppose that $\mathcal{WP}_t^\star(b'') < \mathcal{WP}_t^\star(b')$. From players' incentive compatibility constraints, we have that

$$a'\mathcal{WP}_t^\star(b') - \mathcal{C}_t^\star(b') \geq a'\mathcal{WP}_t^\star(b'') - \mathcal{C}_t^\star(b''), \text{ and } a''\mathcal{WP}_t^\star(b'') - \mathcal{C}_t^\star(b'') \geq a''\mathcal{WP}_t^\star(b') - \mathcal{C}_t^\star(b'),$$

which is equivalent to

$$a' \left[\mathcal{WP}_t^\star(b') - \mathcal{WP}_t^\star(b'')\right] \geq \mathcal{C}_t^\star(b') - \mathcal{C}_t^\star(b''), \text{ and } a'' \left[\mathcal{WP}_t^\star(b') - \mathcal{WP}_t^\star(b'')\right] \leq \mathcal{C}_t^\star(b') - \mathcal{C}_t^\star(b'').$$

Combining the above inequalities yields

$$(a' - a'') \times \left[\mathcal{WP}_t^\star(b') - \mathcal{WP}_t^\star(b'')\right] \geq 0,$$

which is a contradiction given that $\mathcal{WP}_t^\star(b'') < \mathcal{WP}_t^\star(b')$ and the postulated $a' < a''$.

(b) Suppose that $\mathcal{WP}_t^\star(b'') = \mathcal{WP}_t^\star(b')$. Consider the following three sets:

$$\begin{aligned}
\mathcal{A}_1 &:= \{a \in (a', a'') : b_t^\star(a; \gamma_t) > b'\}, \\
\mathcal{A}_2 &:= \{a \in (a', a'') : b_t^\star(a; \gamma_t) < b''\}, \\
\mathcal{A}_3 &:= \{a \in (a', a'') : b'' \leq b_t^\star(a; \gamma_t) \leq b'\}.
\end{aligned}$$

It follows from the postulated $\mathcal{WP}_t^\star(b'') = \mathcal{WP}_t^\star(b')$ that $\mathcal{A}_3$ has F-measure zero. Next, note that $\mathcal{A}_1$ also has F-measure zero. To see this, suppose, to the contrary, that $\mathcal{A}_1$ has positive F-measure. Then there exists some type $a''' \in \mathcal{A}_1$ such that the set $\{a \in (a', a'') : b''' > b_t^\star(a; \gamma_t) > b'\}$ has positive F-measure, where $b''' := b_t^\star(a'''; \gamma_t)$. This in turn implies that $b''' > b' > b''$ and $\mathcal{WP}_t^\star(b''') > \mathcal{WP}_t^\star(b') = \mathcal{WP}_t^\star(b'')$, which would lead to a contradiction by the same argument as in case (a). Similarly, we can show that $\mathcal{A}_2$ has F-measure zero. However, by definition, at least one of the three sets $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$ has positive F-measure, which is a contradiction.

The weak monotonicity of the equilibrium bidding strategy implies that the measure of the set of discontinuity points for each bidding strategy is zero in the equilibrium. Fix a profile of equilibrium bidding strategies $\{b_t(a; \gamma_t)\}_{t=1}^T$ and define $\{b_t^\star(a; \gamma_t)\}_{t=1}^T$ as follows: $b_t^\star(a; \gamma_t) = \lim_{\tilde{a} \nearrow a} b_t(\tilde{a}; \gamma_t)$ for all $t \in \mathcal{T}$. By construction, $b_t^\star(a; \gamma_t)$ is left-continuous in a. Moreover, it can be verified that if $\{b_t(a; \gamma_t)\}_{t=1}^T$ constitutes a symmetric PBE of the contest game, then $\{b_t^\star(a; \gamma_t)\}_{t=1}^T$ also constitutes a symmetric PBE. In what follows, we assume that the equilibrium bidding strategy is left-continuous without loss of generality.

Let $\hat{a}_t := \inf\{a : b_t^\star(a; \gamma_t) > 0\}$. We can show that $b_t^\star(a; \gamma_t)$ strictly increases with a for $a > \hat{a}_t$. To see this, suppose, to the contrary, that $\hat{a}_t < a' < a'' < 1$ and $b' := b_t^\star(a'; \gamma_t) = b'' := b_t^\star(a''; \gamma_t)$. It follows immediately that $b_t^\star(a; \gamma_T) = b''$ for $a \in [a', a'']$. Then a type-a'' player has an incentive to deviate from bidding b''. Specifically, suppose that he raises his effort by an infinitesimal amount. Such a deviation would then (i) slightly increase his expected effort cost but have no impact on his expected winning probability conditional on he loses or becomes the sole winner under the equilibrium bid b''; and (ii) significantly increase his

expected winning probability but have no impact on his effort cost conditional on he ties with some player. Therefore, this deviation would lead to an increase in his interim expected payoff, which is a contradiction.

Next, consider the following two cases:

(a) Suppose that $t = T$. From (14), we have $a_T^\star(\gamma_T) = G_{n_T}(\gamma_t)$. We first show that $b_T^\star(\hat{a}_T + 0) = \gamma_T$. Evidently, bidding $b \in (0, \gamma_T)$ is strictly dominated by bidding 0 and we must have $b_T^\star(\hat{a}_T + 0; \gamma_T) \geq \gamma_T$. Suppose, to the contrary, that $b_T^\star(\hat{a}_T + 0; \gamma_T) > \gamma_T$. Consider a type-$(\hat{a}_T + \epsilon)$, for sufficiently small $\epsilon > 0$. In the equilibrium, he would bid $b_T^\star(\hat{a}_T + \epsilon; \gamma_T)$ and the corresponding equilibrium expected payoff is bounded from above by

$$\varpi_T^\star(\epsilon) := \frac{F^{n_T-1}(\hat{a}_T + \epsilon)(\hat{a}_T + \epsilon - \gamma_T) - \left[1 - F^{n_T-1}(\hat{a}_T + \epsilon)\right] b_T^\star(\hat{a}_T + \epsilon; \gamma_T)}{\hat{a}_T + \epsilon}.$$

Suppose that the player deviates and bids $b' \in \left(\gamma_T, b_T^\star(\hat{a}_T + 0; \gamma_T)\right)$. His expected payoff amounts to

$$\varpi_T^\dagger(\epsilon) := \frac{F^{n_T-1}(\hat{a}_T)(\hat{a}_T + \epsilon - \gamma_T) - \left[1 - F^{n_T-1}(\hat{a}_T)\right] b'}{\hat{a}_T + \epsilon}.$$

It is straightforward to verify that $\varpi_T^\star(\epsilon) < \varpi_T^\dagger(\epsilon)$ for sufficiently small ϵ, which is a contradiction.

It remains to show $a_T^\star(\gamma_T) = \hat{a}_T$. Note that the equilibrium expected payoff of a type-$\hat{a}_T$ player is 0. As a result, his expected payoff of bidding γ_T must be non-positive, i.e.,

$$\frac{F^{n_T-1}(\hat{a}_T)(\hat{a}_T - \gamma_T) - \left[1 - F^{n_T-1}(\hat{a}_T)\right] \gamma_T}{\hat{a}_T} \leq 0. \tag{38}$$

Meanwhile, the equilibrium expected payoff of a type-$(\hat{a}_T + \epsilon)$ player is non-negative, i.e., $\varpi_T^\star(\epsilon) \geq 0$. Together with $b_T^\star(\hat{a}_T + 0; \gamma_T) = \gamma_T$, we can obtain

$$\lim_{\epsilon \searrow 0} \varpi_T^\star(\epsilon) = \frac{F^{n_T-1}(\hat{a}_T)(\hat{a}_T - \gamma_T) - \left[1 - F^{n_T-1}(\hat{a}_T)\right] \gamma_T}{\hat{a}_T} \geq 0. \tag{39}$$

Combining (38) and (39) yields that $F^{n_T-1}(\hat{a}_T)(\hat{a}_T - \gamma_T) - \left[1 - F^{n_T-1}(\hat{a}_T)\right] \gamma_T = 0$, which is equivalent to $F^{n_T-1}(\hat{a}_T)\hat{a}_T = \gamma_T$. Therefore, $\hat{a}_T = G_{n_T}(\gamma_t) = a_T^\star(\gamma_T)$.

(b) Suppose that $t \in \{1, \ldots, T-1\}$. From (14), $a_t^\star(\gamma_t)$ is the unique solution to $(a_t^\star - \gamma_t + 1)F^{n_t-1}(a_t^\star) = 1$. First, note that $\hat{a}_t < 1$ and thus the right neighborhood of $\hat{a}_t$ is contained in $(0, 1)$; otherwise, all period-t players bid 0 regardless of their types and bidding 1 is a profitable deviation for a player whose ability exceeds γ_t.

In what follows, we show that $b_t^\star(\hat{a}_t + 0; \gamma_t) = 1$. We first rule out the possibility of $b_t^\star(\hat{a}_t + 0; \gamma_t) > 1$. Note that if a player bids $b \geq 1$, all subsequent players would remain inactive. By the same argument as in the case of $t = T$, we can show that if $b_t^\star(\hat{a}_t + 0; \gamma_t) > 1$, then a type-$(\hat{a}_t + \epsilon)$ player has a strict incentive to deviate to some bid $b' \in \big(1, b_t^\star(\hat{a}_t + 0; \gamma_t)\big)$ for sufficiently small ϵ .

Next, we rule out the possibility of $\hat{b} := b_t^\star(\hat{a}_t + 0; \gamma_t) \in (0, 1)$. Consider the type-$\hat{a}_t$ player. Because his equilibrium expected payoff is 0, his expected payoff of bidding $\hat{b}$ must be non-positive. Further, note that a type-$(\hat{a}_t + \epsilon)$ player's expected payoff of bidding $\hat{b}$ is non-negative for sufficiently small $\epsilon > 0$. These altogether indicate that the type-$\hat{a}_t$ player's expected payoff of bidding $\hat{b}$ is zero, i.e.,

$$\mathcal{WP}_t^\star(\hat{b})(\hat{a}_t - \gamma_t) - \big[1 - \mathcal{WP}_t^\star(\hat{b})\big]\hat{b} = 0. \tag{40}$$

Similarly, bidding 1 also yields a non-positive expected payoff to the type-$\hat{a}_t$ player. Note that the corresponding winning probability is at least $F^{n_t-1}(\hat{a}_t)$. Fixing $\hat{a}_t > 0$, the player's expected payoff of bidding 1 is no less than

$$\frac{F^{n_t-1}(\hat{a}_t)(\hat{a}_t - \gamma_t) - \big[1 - F^{n_t-1}(\hat{a}_t)\big]}{\hat{a}_t}.$$

Therefore, we must have $\frac{F^{n_t-1}(\hat{a}_t)(\hat{a}_t-\gamma_t)-\big[1-F^{n_t-1}(\hat{a}_t)\big]}{\hat{a}_t} \leq 0$, which can be simplified as

$$(\hat{a}_t - \gamma_t + 1)F^{n_t-1}(\hat{a}_t) \leq 1, \text{ for } \hat{a}_t > 0. \tag{41}$$

Recall $\hat{t} := \max\{t \leq T-1 : n_t \geq 2\}$, indicating that $n_\ell = 1$ for all $\ell \in \{\hat{t}+1 \ldots, T-1\}$. Note that a period-$\ell$ player bids below $\hat{b}$ if and only if his ability is below $\hat{b}$, which occurs with probability $F(\hat{b})$. Further, the equilibrium probability that no bid in period T exceeds $\hat{b}$ is $F^{n_T}(a_T^\star(\hat{b})) = F^{n_T}(G_{n_T}(\hat{b}))$. Therefore, we have that

$$\Pr\left(\begin{array}{c}\text{No bid after period } \hat{t} \\ \text{exceeds } \hat{b}\end{array}\middle|\begin{array}{c}\text{No bid in period } \ell \\ \text{exceeds } \hat{b} \text{ for all } t < \ell \leq \hat{t}\end{array}\right) = F^{T-\hat{t}-1}(\hat{b})F^{n_T}\left(G_{n_T}(\hat{b})\right). \tag{42}$$

The equilibrium winning probability $\mathcal{WP}_t^\star(\hat{b})$ can then be bounded from above by

$$\begin{aligned}
\mathcal{WP}_t^\star(\hat{b}) =& F^{n_t-1}(\hat{a}_t)\Pr\Big(\text{No bid after period } t \text{ exceeds } \hat{b}\Big) \\
=& F^{n_t-1}(\hat{a}_t) \times \Pr\left(\begin{array}{c|c} \text{No bid after period } \hat{t} & \text{No bid in period } \ell \\ \text{exceeds } \hat{b} & \text{exceeds } \hat{b} \text{ for all } t < \ell \leq \hat{t} \end{array}\right) \\
& \times \Pr(\text{No bid in period } \ell \text{ exceeds } \hat{b} \text{ for all } t < \ell \leq \hat{t}) \\
\leq& F^{n_t-1}(\hat{a}_t) \times \Pr\left(\begin{array}{c|c} \text{No bid after period } \hat{t} & \text{No bid in period } \ell \\ \text{exceeds } \hat{b} & \text{exceeds } \hat{b} \text{ for all } t < \ell \leq \hat{t} \end{array}\right) \\
=& F^{n_t-1}(\hat{a}_t) F^{T-\hat{t}-1}(\hat{b}) F^{n_T}\Big(G_{n_T}(\hat{b})\Big) \\
\leq& F^{n_t-1}(\hat{a}_t)\hat{b}, \qquad (43)
\end{aligned}$$

where the last equality follows from (42) and the last inequality from Assumption 3. Combining (40) and (43) yields

$$(\hat{a}_t - \gamma_t + \hat{b})F^{n_t-1}(\hat{a}_t)\hat{b} - \hat{b} \geq \mathcal{WP}_t^\star(\hat{b})(\hat{a}_t - \gamma_t) - \big[1 - \mathcal{WP}_t^\star(\hat{b})\big]\hat{b} = 0. \qquad (44)$$

For $\hat{a}_t = 0$, we have $(\hat{a}_t - \gamma_t + \hat{b})F^{n_t-1}(\hat{a}_t)\hat{b} - \hat{b} = -\hat{b} < 0$, which contradicts (44). Moreover, for $\hat{a}_t > 0$, we have $(\hat{a}_t - \gamma_t + \hat{b})F^{n_t-1}(\hat{a}_t)\hat{b} - \hat{b} < (\hat{a}_t - \gamma_t + 1)F^{n_t-1}(\hat{a}_t)\hat{b} - \hat{b} \leq \hat{b} - \hat{b} = 0$, where the strict inequality follows from the postulated $\hat{b} \in (0, 1)$ and the second inequality from (41). A contradiction again.

Last, we rule out the possibility of $\hat{b} := b_t^\star(\hat{a}_t + 0; \gamma_t) = 0$. Suppose that $\hat{b} = 0$. It follows immediately that $\gamma_t = 0$. Fix $\epsilon > 0$. Consider a type-a' player, with $a' \in (\hat{a}_t, 1)$, such that $b' := b_t^\star(a'; \gamma_t) < \epsilon$. By the same argument of proving (43), we can obtain $\mathcal{WP}_t^\star(b') \leq F^{n_t-1}(a')b'$. Therefore, the type-$a'$ player's equilibrium expected payoff is at most

$$\begin{aligned}
\frac{\mathcal{WP}_t^\star(b')a' - \big[1 - \mathcal{WP}_t^\star(b')\big]\, b'}{a'} &= \frac{\mathcal{WP}_t^\star(b')(a' + b') - b'}{a'} \\
&\leq \frac{\big[F^{n_t-1}(a')(a' + \epsilon) - 1\big]\,\epsilon}{a'} < 0,
\end{aligned}$$

for sufficiently small ϵ. A contradiction.

From the above analyses, we have $\hat{b} \equiv b_t^\star(\hat{a}_t + 0; \gamma_t) = 1$. Plugging $\hat{b} = 1$ and $\mathcal{WP}_t^\star(1) = F^{n_t-1}(\hat{a}_t)$ into (40) yields $\hat{a}_t = a_t^\star(\gamma_t)$. This concludes the proof.

■

Now we can prove Proposition 7. Parts (i) and (iii) are obvious and it remains to prove

part (ii). Fix $t \in \mathcal{T} \setminus \{T\}$ and suppose $n_t \geq 2$. By Lemma 9, $b_t^\star(a;\gamma_t) = 0$ for $a \leq a_t^\star(\gamma_t)$. Further, $b_t^\star\left(a_t^\star(\gamma_t)+0;\gamma_t\right) = 1$ given that $t \in \mathcal{T} \setminus \{T\}$. Therefore, for $a > a_t^\star(\gamma_t)$, $b_t^\star(a;\gamma_t)$ solves the following maximization problem:

$$\max_{b > \mathbb{1}_{\{t<T\}} + \gamma_T \mathbb{1}_{\{t=T\}}} \Bigg\{ F^{n_t-1}\left((b_t^\star)^{-1}(b;\gamma_t)\right) - \frac{\gamma_t}{a} F^{n_t-1}\left(a_t^\star(\gamma_t)\right) \\ - \int_{a_t^\star(\gamma_t)}^{(b_t^\star)^{-1}(b;\gamma_t)} \frac{b_t^\star(a';\gamma_t)}{a} dF^{n_t-1}(a') - \left[1 - F^{n_t-1}\left((b_t^\star)^{-1}(b;\gamma_t)\right)\right] \frac{b}{a} \Bigg\}.$$

Equivalently, a solves $\max_{\check{a} > a_t^\star(\gamma_t)} \psi_t(\check{a}, a; \gamma_t)$, where $\psi_t(\check{a}, a; \gamma_t)$ is defined as

$$\psi_t(\check{a}, a; \gamma_t) := F^{n_t-1}(\check{a}) - \frac{\gamma_t}{a} F^{n_t-1}\left(a_t^\star(\gamma_t)\right) - \int_{a_t^\star(\gamma_t)}^{\check{a}} \frac{b_t^\star(a';\gamma_t)}{a} dF^{n_t-1}(a') - \left[1 - F^{n_t-1}(\check{a})\right] \frac{b_t^\star(\check{a};\gamma_t)}{a}.$$

The equilibrium bidding strategy is then governed by the following first-order condition:

$$\left.\frac{\partial \psi_t(\check{a}, a; \gamma_t)}{\partial \check{a}}\right|_{\check{a}=a} = \frac{dF^{n_t-1}(a)}{da} - \frac{1}{a}\left[1 - F^{n_t-1}(a)\right] \frac{db_t^\star(a;\gamma_T)}{da} = 0.$$

The above equation, together with Lemma 9(ii), yields that

$$b_t^\star(a;\gamma_t) = \mathbb{1}_{\{t<T\}} + \gamma_t \mathbb{1}_{\{t=T\}} + \int_{a_t^\star(\gamma_t)}^{a} \frac{a'}{1 - F^{n_t-1}(a')} dF^{n_t-1}(a'), \text{ if } a > a_t^\star(\gamma_t).$$

Simple algebra would verify that

$$\frac{\partial^2 \psi_t(\check{a}, a; \gamma_t)}{\partial \check{a} \partial a} > 0,$$

which implies that $\frac{\partial \psi_t(\check{a}, a; \gamma_t)}{\partial \check{a}}$ is increasing in a. Therefore, we have

$$\frac{\partial \psi_t(\check{a}, a; \gamma_t)}{\partial \check{a}} > \left.\frac{\partial \psi_t(\check{a}, a; \gamma_t)}{\partial \check{a}}\right|_{\check{a}=a} = 0, \text{ for } \check{a} < a,$$

and

$$\frac{\partial \psi_t(\check{a}, a; \gamma_t)}{\partial \check{a}} < \left.\frac{\partial \psi_t(\check{a}, a; \gamma_t)}{\partial \check{a}}\right|_{\check{a}=a} = 0, \text{ for } \check{a} > a.$$

That is, the necessary first-order condition $\left.\frac{\partial \psi_t(\check{a}, a; \gamma_t)}{\partial \check{a}}\right|_{\check{a}=a} = 0$ is also a sufficient condition for global maximizer, and thus the strategy profile characterized above does constitute a PBE. This completes the proof. ■

Proof of Example 3

Proof. Fix $\boldsymbol{n} = (2, 1)$ and suppose $F(a) = a^r$ with $r \in (0, 1)$. The derivation of a period-2 player's equilibrium strategy is obvious and it suffices to derive $b_1^\star(a; 0)$. By the same argument as in the proof of Lemma 9, we can show that $b_1^\star(a; 0)$ is weakly increasing in a. Therefore, there exists a threshold $\hat{a}_1$ such that $b_1^\star(a; 0) = 0$ if and only if $a \leq \hat{a}_1$. Moreover, $b_1^\star(a; 0)$ is strictly increasing in $a \in (\hat{a}_1, 1)$.

Note that $\hat{a}_1 = 0$. Otherwise, a type-$\hat{a}_1$ player can earn a strictly positive expected payoff by bidding $b \in \left(0, \hat{a}_1^{\frac{1+r}{1-r}}\right)$. To see this, note that the player's winning probability of bidding b is at least $F(\hat{a}_1)F(b) = \hat{a}_1^r b^r$, and thus his expected payoff is at least

$$\frac{\hat{a}_1^{r+1} b^r - (1 - b^r \hat{a}_1^r) b}{\hat{a}_1} > \frac{\hat{a}_1^{r+1} b^r - b}{\hat{a}_1} > 0,$$

which contradicts the fact that the player's equilibrium payoff is 0.

From the above analysis, $b_1^\star(a; 0)$ solves

$$\max_{b>0} \Big\{ aF\left(\min\{b, 1\}\right) F\left((b_1^\star)^{-1}(b; 0)\right) - F\left(\min\{b, 1\}\right) \int_0^{(b_1^\star)^{-1}(b;0)} b_1^\star(a'; 0) dF(a') \\ - \left[1 - F\left(\min\{b, 1\}\right) F\left((b_1^\star)^{-1}(b; 0)\right)\right] b \Big\}.$$

Equivalently, a solves the following maximization problem:

$$\max_{\check{a}>0} \Big\{ aF\left(\min\left\{b_1^\star(\check{a}; 0), 1\right\}\right) F(\check{a}) - F\left(\min\left\{b_1^\star(\check{a}; 0), 1\right\}\right) \int_0^{\check{a}} b_1^\star(a'; 0) dF(a') \\ - \left[1 - F\left(\min\left\{b_1^\star(\check{a}; 0), 1\right\}\right) F(\check{a})\right] b_1^\star(\check{a}; 0) \Big\}. \tag{45}$$

If $b_1^\star(a; 0) < 1$, we denote $b_1^\star(a; 0)$ by $\mathcal{B}(a)$. From the first-order condition for the maximization problem (45), we have that

$$ra^r \left[\mathcal{B}(a)\right]^r + ra^{r+1} \left[\mathcal{B}(a)\right]^{r-1} \mathcal{B}'(a) - r^2 \left[\mathcal{B}(a)\right]^{r-1} \mathcal{B}'(a) \int_0^a \mathcal{B}(s) s^{r-1} ds \\ - \mathcal{B}'(a) + (r+1) a^r \left[\mathcal{B}(a)\right]^r \mathcal{B}'(a) = 0.$$

Similarly, if $b_1^\star(a; 0) \geq 1$, we can obtain

$$0 = a \frac{dF^{n_t - 1}(a)}{da} - \left[1 - F^{n_t - 1}(a)\right] \frac{db_t^\star(a; \gamma_T)}{da}.$$

By the same argument as in the proof of Proposition 7, the above first-order conditions are also sufficient conditions for global maximizer.

It remains to pin down the cutoff $a_1^\star$. Consider a type-$a_1^\star$ player in the first period. Evidently, his expected payoff of bidding $\mathcal{B}(a_1^\star)$ is equal to that of bidding 1, i.e.,

$$\begin{aligned}&(a_1^\star)^{r+1}\left[\mathcal{B}(a_1^\star)\right]^r - r\left[\mathcal{B}(a_1^\star)\right]^r\int_0^{a_1^\star}\mathcal{B}(s)s^{r-1}ds - \left\{1-\left[a_1^\star\mathcal{B}(a_1^\star)\right]^r\right\}\mathcal{B}(a_1^\star)\\ =&(a_1^\star)^{r+1} - r\int_0^{a_1^\star}\mathcal{B}(s)s^{r-1}ds - (1-a_1^\star),\end{aligned}$$

from which we can solve for $a_1^\star$. This completes the proof. ■

Proof of Remark 3

Proof. By Proposition 7, $b_1^\star(a;\gamma_1)=1$ for all $a\in(0,1]$. As a result, $\gamma_2=1$ and $b_2^\star(a;\gamma_2)\equiv 0$ for all $a\in(0,1]$, which implies that $\Pi_1^\star=\int_0^1 dF(a)=1>0=\Pi_2^\star$. ■

Proof of Proposition 8

Proof. We first prove part (i) of the proposition. By Proposition 7, $\Pi_1^\star$ depends only on n_1 but not n_2. Moreover, individual rationality for the second period implies that $\Pi_2^\star\le 1/n_2$. Therefore, $\Pi_1^\star<\Pi_2^\star$ for sufficiently large n_2.

Next, we prove part (ii). Denote the expected payoff of a period-2 player conditional on $\gamma_2=0$ by $\widehat{\Pi}_2$. By definition, $\widehat{\Pi}_2$ does not depend on n_1. Further, it follows immediately from Proposition 7 that $\Pi_2^\star=\widehat{\Pi}_2F^{n_1}\left(a_1^\star(0)\right)$. Therefore, we have

$$\lim_{n_1\nearrow\infty}\Pi_2^\star=\widehat{\Pi}_2\lim_{n_1\nearrow\infty}F^{n_1}\left(a_1^\star(0)\right). \tag{46}$$

By (14), $a_1^\star(0)$ solves $\left[a_1^\star(0)+1\right]F^{n_1-1}\left(a_1^\star(0)\right)=1$ for $n_1\ge 2$, from which we can conclude $\lim_{n_1\nearrow\infty}a_1^\star(0)=1$, and thus

$$\lim_{n_1\nearrow\infty}F^{n_1}\left(a_1^\star(0)\right)=\lim_{n_1\nearrow\infty}\left\{F\left(a_1^\star(0)\right)/\left[1+a_1^\star(0)\right]\right\}=1/2. \tag{47}$$

Combining (46) and (47) yields $\lim_{n_1\nearrow\infty}\Pi_2^\star=\widehat{\Pi}_2/2$. Next, note that individual rationality for the first period implies that $\Pi_1^\star\le 1/n_1$ and thus $\lim_{n_1\nearrow\infty}\Pi_1^\star=0$. Therefore, $\Pi_2^\star>\Pi_1^\star$ for sufficiently large n_1. This concludes the proof. ■

Proof of Proposition 9

Proof. Denote the expected payoff of a period-t player conditional on $\gamma_t=0$ by $\widehat{\Pi}_t$. By definition, $\widehat{\Pi}_t$ does not depend on $(n_1,\ldots,n_{t-1})$.

Fix a period $\tau \in \mathcal{T}$. By Proposition 7, $\Pi_t^\star(\boldsymbol{n})$ does not depend on n_τ if $t < \tau$ and

$$\Pi_t^\star(\boldsymbol{n}) = \widehat{\Pi}_t \left[\prod_{\ell < t \text{ and } \ell \neq \tau} F^{n_\ell}\left(a_\ell^\star(0)\right) \right] F^{n_\tau}\left(a_\tau^\star(0)\right), \text{ if } t > \tau.$$

By the same argument as in the proof of Proposition 8, we can show that $\lim_{n_\tau \nearrow \infty} F^{n_\tau}(a_\tau^\star(0)) = 1/2$. The above analysis implies that $\lim_{n_\tau \nearrow \infty} \Pi_t^\star(\boldsymbol{n}) > 0$ for all $t \neq \tau$. Further, individual rationality for period τ implies that $\lim_{n_\tau \nearrow \infty} \Pi_\tau^\star(\boldsymbol{n}) = 0$. As a result, $\Pi_\tau^\star(\boldsymbol{n}) < \min_{t \neq \tau} \Pi_t^\star(\boldsymbol{n})$ if n_τ is sufficiently large. This completes the proof. ■